A Handbook

of

Engineering Chemistry



Title of the Book: **"**A Handbook of Engineering Chemistry**"**

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PREFACE

The book "**A Handbook of Engineering Chemistry**" is exclusively dedicated to First Year B. Tech Graduate students under JNTU Kakinada affiliated colleges (CE and ME branches). This book was written in accordance with the syllabus for autonomous colleges at JNTU Kakinada. This book is well written, and the concepts are easy to understand. This book is required reading for students preparing for semester exams. All topics are suggested for the semester end exam of the Engineering Chemistry Paper.

This book contains 5 chapters. 1. Polymer Technology., 2. Electrochemical Cells and Corrosion, 3. Chemistry of Materials, 4. Fuels and 5. Water technology.

After studying **polymer technology**, students will be able to outline the properties of polymers and various additives added, as well as different methods of forming plastic materials. Describe how some plastic materials are made, their properties, and their applications. Determine the mechanism of conduction in conducting polymers. Discuss the differences between natural and synthetic rubbers, as well as their applications.

At the end of **electrochemical cells and corrosion**, students will be able to explain the theory of battery and fuel cells, categorize the causes of corrosion, and study some corrosion methods.

By the end of **chemistry of materials**, students will be able to outline their understanding of materials such as nanomaterials and fullerenes and their applications, explain the techniques used to detect and measure changes in reaction state, and illustrate commonly used industrial materials.

At the end of the **fuels** chapter, students will be able to differentiate between petroleum, gasoline, and synthetic gasoline and understand how they are produced, as well as study alternate fuels and analyze flue gas.

In **water technology**, students will be able to explain the impurities found in raw water, the problems associated with them, and how to avoid them.



Dr. R. S. S. Srikanth Vemuri, an associate professor with eight years of teaching experience, has been working at Vignan's Institute of Engineering for Women since 2014. In 2014, Andhra University awarded him a doctorate. For his contributions to the field of research, he received two awards. One is the VD GOOD organization's Young Scientist Award (2021), and the second is the Sastra Award (2022). He authored 20 research papers that were published in both national and international journals. His research focuses on chemical kinetics.



Dr. Bhargavi Kondapalli, an analytical scientist with research expertise in drug validation, impurity identification and quantification by HPLC and LC-MS/MS, as well as a thorough knowledge of analytical chemistry. Ph.D. award received in 2022 from Andhra University. Currently employed by Aurobindo Pharma Ltd. as a scientist. She has 12 publications in the field of study (9 articles and 3 conferences). Physical chemistry and analytical chemistry are two of her areas of interest.

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Unit 1

Polymer Technology

1.1. Introduction

The words polymer, which mean "many," and meres, which mean "parts," come from classical Greek. A polymer is a long-chain molecule made up of numerous repeating units with the same structure, to put it simply. While some polymers, like proteins, cellulose, and silk, are found in nature, many others, like nylon, polystyrene, and polyethylene, can only be made synthetically. In some circumstances, synthetic production of naturally occurring polymers is also possible.

Natural rubber (from the Hevea tree), also known as polyisoprene in its synthetic form, is a prime example. Elastomers are frequently made of polymers with high extension properties under ambient conditions. Nitrile and butyl rubber are two significant synthetic elastomers that are used in addition to natural rubber. Other polymers may possess properties that make it possible to create long fibers suitable for textile applications from them. Natural fibers like cotton, wool, and silk can be effectively replaced by synthetic fibers, primarily nylon and polyester.

Contrary to popular belief, commercial materials made from synthetic polymers that aren't elastomers and fibers are referred to as plastics instead of polymers. Common commercial plastic resins can include two or more polymers in addition to a range of fillers and additives. These are included to enhance certain characteristics like processability, thermal or environmental stability, or mechanical properties.

The middle of the nineteenth century is when polymer science first emerged. Charles Goodyear invented the vulcanization process in the 1830s, which turned the gummy natural rubber latex into a practical elastomer for tyre use. Nitric acid and cellulose were combined by Christian F. Schönbein in 1847 to create cellulose nitrate. This was the first thermoplastic made by humans and was used in the 1860s as celluloid. Leo Hendrik Baekeland 1 began manufacturing Bakelite (phenol-formaldehyde resin) in 1907. General Electric created Glyptal (unsaturated polyester resin) in 1912 as a coating resin for protective surfaces.

Based on how they respond to thermal processing, all polymers can be divided into two main categories. Thermoplastics are polymers that can be processed into a desired form by heating them up to soften them. By using heat and pressure, used thermoplastics can be recovered and refabricated. An important illustration of a commercial thermoplastic is polystyrene. Poly (vinyl chloride) and polyolefins (such as polyethylene and polypropylene) are two additional noteworthy examples. In contrast, thermosets are polymers that have had their individual chains chemically linked during the polymerization process or after a subsequent chemical or thermal treatment during fabrication. Once created, these crosslinked networks can withstand solvent attack, mechanical deformation caused by heat, and heat softening. However, they cannot be thermally processed. Thermosets are suitable materials for adhesive, coating, and composite applications due to these characteristics. The main thermosets used in the production of glass-reinforced composites like Fiberglas are epoxy, phenol-formaldehyde resins, and unsaturated polyesters.

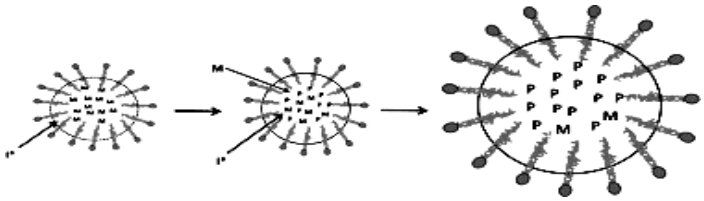
1.2. Polymerization methods

Emulsion Polymerization method

In this method the monomers & water-soluble initiators form the emulsion with water. The surfactants such as soap, detergents etc. are add which hold the monomers & initiators (H_2O_2 or persulphate) in the form of a micelles. Polymerization takes place inside the micelle on exposing to heat or radiation. De-emulsifying agents (3% solution of $Al_2(SO_4)_3$) & filtration can isolate the obtained polymer.

Monomer + **Initiator** + **surfactant** ⟶ **polymer**

(Dissolved in inert solvent) (water soluble) (Emulsion in water)



Advantages

- High molecular weight polymers can be obtained.
- Polymer process is fast
- Final product can directly use for fabrication.
- High purity product is obtained.
- This method is used to prepare synthetic rubbers like BuNa-S, acryl rubber, Teflon, paints or adhesives etc.,

Suspension Polymerization Method (Pearl Polymerization method):

In this method water insoluble monomers are suspended in water as tiny droplets by continuously agitating (shaking). Initiators are soluble in monomers. Monomers and initiator forms heterogeneous mixture with the water. Polymerization is initiator by heating or exposing to radiation.

As the polymerization takes place, the polymer is obtained as spherical pearls or beads; hence it is also called as pearl polymerization methods. Since the product is insoluble in water. It can be easily separated just by filtration & the product needs no further purification.

Monomer + Initiator + suspending agent ⟶ **polymer**

(Suspension in water) (dissolved in monomer) (suspended in water as beads)

Advantages

- Viscosity buildup is negligible
- Better heat transfer can be achieved.
- High purity product is obtained.
- Isolation of the product is easy and it does not require any purification.

Disadvantages

- Few monomers are water soluble.

Uses

- This method is used for preparing PVC, PVA (Poly vinyl acetate), styrene, divinylbenzene (ion exchange)

1.3. Physical Properties of Polymers

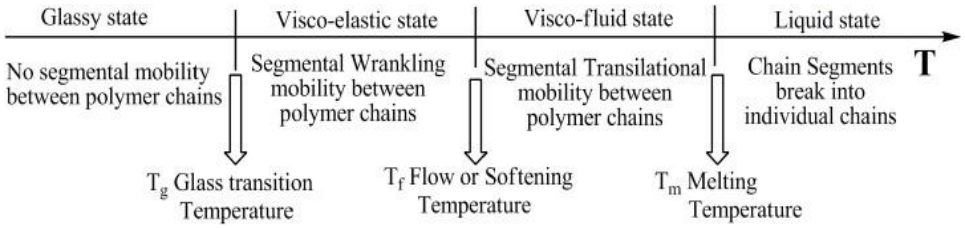
1. **Crystallinity:** Polymeric molecules does not have a definite structure like solids (cubic, ruby, face centered cube, body centered cube) structures. A polymer is a mixture of 60% crystalline and 40% amorphous. As the crystallinity increases, brittleness, strength and chemical resistance increases.

2. **Molecular weight:** The polymer molecule has different number of monomeric units hence different molecular weights. As a result, the molecular weight of polymer should be expressed as average molecular weight $\overline{M}$ here are two types of average molecular weights ($\overline{M_n}$ & $\overline{M_w}$)

   $\overline{M_n} = \frac{\sum M_i N_i}{\sum N_i}$, $\overline{M_w} = \frac{\sum M_i^2 N_i}{\sum M_i N_i}$

3. **Solubility:** When a polymer is dissolved in a suitable solvent, the polymer absorbs water and swells in size. Slowly polymer goes into the solution a viscous polymer solution which is heterogeneous.

4. **Effect of heat on Polymer**



At room temperature all polymers are solids existing in rubbery or viscoelastic state. As the temperature increases the polymer melts and this is called melting point of polymer (T_f). As the temperature further increases to boiling (Tb) the polymer decomposes to monomers which arein vapor phase. When the temperature is lowered below zero degree the polymer becomes hard, brittle and glassy and the polymer behaves like glass.

5. **Glass transition temperature:** The temperature at which the polymertransformsfromrubberystatetoglassystateisknownasglasstransition temperature. The rotation about the inter unit bonds in the polymer is effectively frozen at T_g and below T_g.

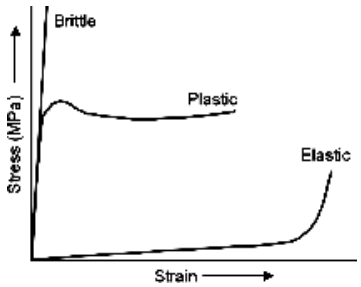
   e.g.: T_g of **LDPE** = -110° C; T_g of **HDPE** = -90° C

6. **Diffusion and permeability:** Crystalline polymers resist diffusion of molecules through voids or other gaps and tend to resist diffusion because of much greater degree of molecular packing. The diffusion rates are higher in rubber.

1.4. Mechanical Properties of Polymers:

1. **Toughness:** Below the Tg temperature the polymer break with a brittle fracture and they become tougher. The size of the crystalline structure also determines the toughness. The smallest spherical crystalline masses increase toughness.

2. **Strength:** The strength of a polymer depends on length of polymer chain, branching or cross-linking which are controlled by elastomeric properties, crystallinity and branching of chain. The strength of a polymer increases with increase in molecular weight. The intermolecular forces of attraction, presence of polar groups and chain length increases the strength of the polymers. A polymer strength is determined by stress-strain test.



3. **Elasticity:** Elasticity is the property of recovering original shape after the removal of deforming stress. Natural rubbers possess high elasticity due to coiled structure.

1.5.Compounding of Plastics

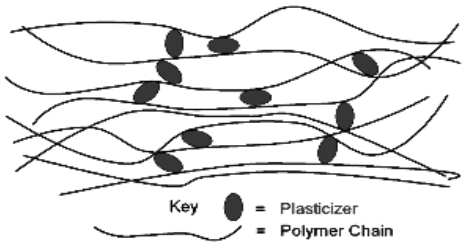
During moulding, different constituents are added to impart the required property. These constituents that are added to plastic during its moulding process are called moulding constituents.

Ingredients used in compounding of plastics

Binder (or) resin: Most important component for compounding. Binder holds different constituents together. Large molecular weight compounds are usually used as binders.
E.g.: resin, glue, gumtec.

Fillers or Extenders: Fillers are generally added to thermosetting plastics to increase tensile strength, hardness and crack resistance. Fillers also reduces the cost of final of the polymers.
e.g.: wood flour, Asbestos, Mica, Cotton, Carbon black, ZnO, PbO etc.

Plasticizers: Plasticizers are substances added to enhance the plasticity and flexibility of the material and to reduce the cracking on the surface. Plasticizers also increase the flow property of the plastics. Plasticizer neutralizes the intermolecular forces of attraction between polymer chains.



e.g.: Dibutyl oxalate, Castor oil and Tricresyl phosphate, vegetable oil, camphor, etc.,

Lubricants: These are added to prevent the plastics from sticking to the moulds and they impart glassy finish to the final product.
e.g.: Oils, Waxes and Soaps, etc.

Stabilizers: They are added during compounding to prevent various adverse effects during processing or during end use. For example, PVC undergoes decomposition and decolorization at moulding temperature. So, during their moulding, stabilizers are used.

a) **Antioxidants:** These are can be used to prevent oxidation. e.g.: di-tertiary butyl-p-cresol.

b) **Heat stabilizers:** To prevent thermal degradation. e.g.: white lead, lead chromate, red lead, etc.

c) **Radiation absorbers**: Most of the polymer degrades quickly when exposed to UV radiation. e.g.: carbon black, titanium oxide.

Colorants: These are added to impart the desired color to the plastics and give decorative effect. e.g.: organic dyes, Carbon black- black; Titanium oxide (white), CdS (yellow) and Pb_3O_4 (red)

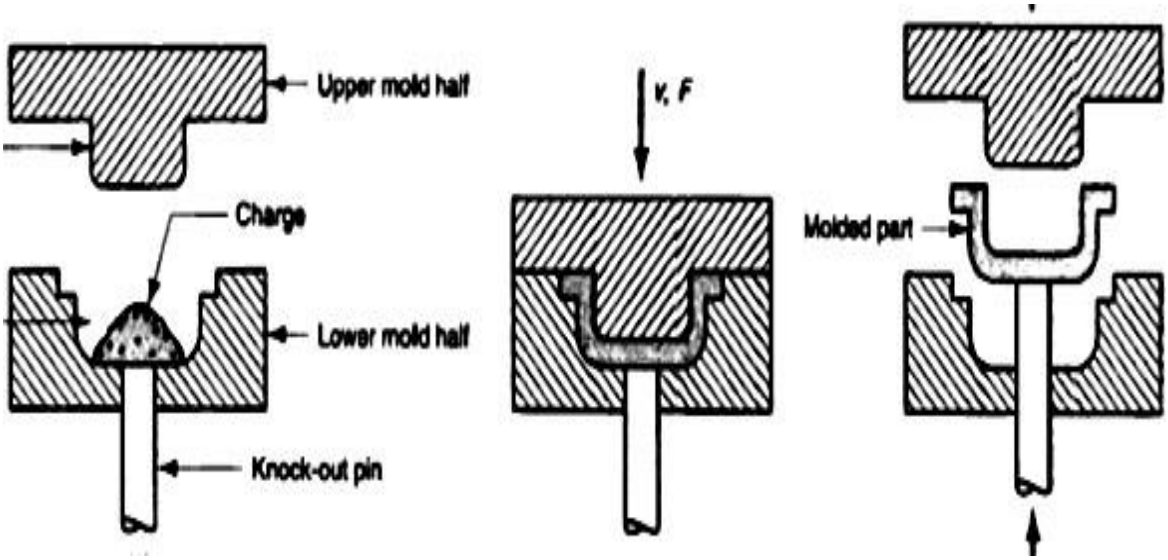
1.6. Fabrication of plastics (or) Moulding of Plastics

Fabrication of plastics is the manufacturing process of plastic articles. The methods, usually depends upon the types of resins used i.e., whether thermosetting or thermoplastic. Different fabrication techniques are described below.

1.6.1. Compression Moulding

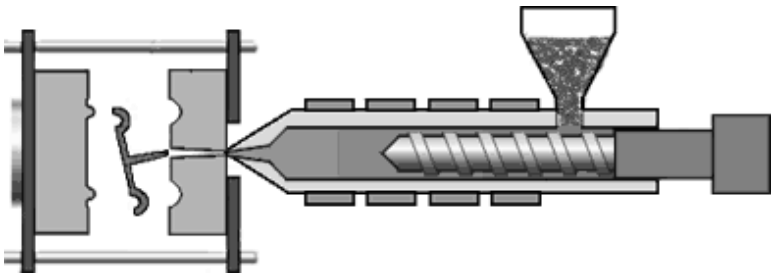
- This method is applied to **thermosetting resins.**
- The predetermined quantity of plastic ingredients in proper proportions is filled between the two half –pieces of mould which are capable of being moved relative to each other.
- The heat and pressure are then applied according to specifications.
- Two halves are closed very slowly.
- Under the influence of heat, the compacted material gets cured and hardened.
- After curing the moulded article is taken out by opening the mould parts.

- It is used for manufacturing rubber tires, shoes, electric plug sockets, pot handles, etc.
- Advantages: Low expensive and low maintenance
- Disadvantages: Longer cycle time.



1.6.2. Injection Moulding

- It is carried out for producing articles made of ***thermos plastics*** *material*
- In this process, a correct amount of plastic material as granules, pellets or powder is added from the hopper into the cylinder.
- The plunger pushes the plastic material in to the heating chamber where the material is converted to viscous liquid.
- Again, with the help of the plunger the molten plastic material is injected through the nozzle into the mould cavity maintained at room temperature.
- When the materials have been cured sufficiently, half of the mould is opened to get the moulded article.



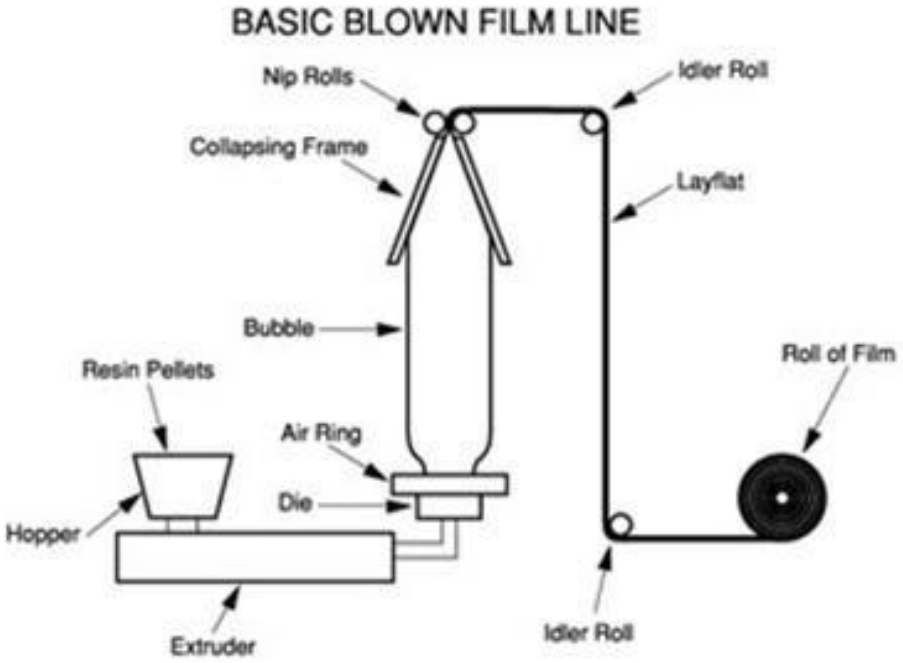
Advantages

- high speed production and low mould cost
- Loss of material will be less.

Disadvantage

- Large number of cavities cannot be filled, so there is a limitation to designed articles.

1.6.3. Blown Film Method



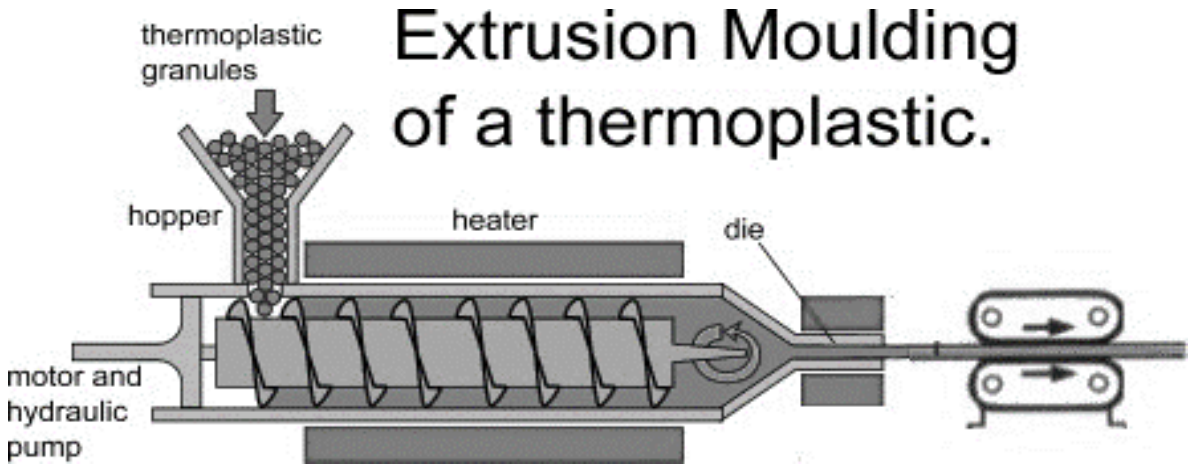
Blown Film is a manufacturing process that produces films of plastic sheeting and tubes. These sheets or tubes can either be fully cylindrical, or have other object inserted in to them so they form useful objects like shopping bags, rubbish bags or cling wrap

- Blowing film is a fairly easy process. Plastic pellets must first be heated until they melt (much like in injection moulding).
- The plastic is then pressed through a cylindrical die to create a bubble with a thin wall. This bubble, which is kept afloat by air, is fed air through a scaffold as the tube rises. The plastic starts to cool after this time period.
- When it has cooled down sufficiently, it folds in on itself to form a flattened tube. This tube is spun onto a spool after being passed through a number of rollers. Before being wound onto a spool, the tube's edges can be trimmed and it can also be punched, cut, or printed on.
- Producing goods that need uniform properties along the length and width of the product is a great application for blown film manufacturing.
- It's also a fantastic high-volume production technique, but buying a blown film machine comes at a high capital expense.
- Up to 130 meters of products can be produced by blown film in a single minute. They can create sheets and films that range in thickness from 250 microns (0.25mm) to as thin as 10-20 microns (0.01mm - 0.02mm).
- High- or low-density polyethylene (HDPE or LDPE) makes up the majority of blown film, though there are some exceptions. Blown film's drawback is that it can only be produced as sheets or tubes.

1.6.4. Extrusion Moulding

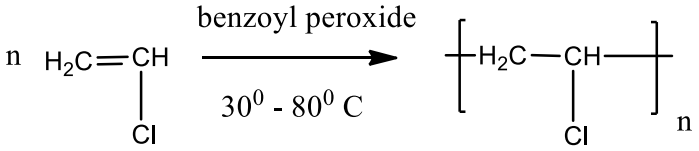
- Extrusion moulding is used mainly for continuous moulding of ***thermoplastic*** materials into articles of uniform cross section like tubes, rods, strips, insulated electric cables.
- In this process, a correct amount of plastic material as granules, pellets or powder is added from the hopper into the cylinder.
- The plunger pushes the plastic material in to the heating chamber where the material is converted to viscous liquid.
- The molten mass pushed by means of a rotating screw into a die, having the required outer shape of the article to be the manufactured.
- The material extruded out of the die is quite hot and carried over the conveyor belt.

- Here the plastic mass gets cooled, due to the atmospheric exposure (or artificially by air jets).
- Advantage: High production rate.



1.7. Poly Vinyl Chloride

Poly Vinyl Chloride is produced by free radical chain polymerization of vinyl chloride in presence of a small amount of benzoyl peroxide or hydrogen peroxide.



{Vinyl chloride, so needed is generally prepared by treating acetylene at 1to1.5 atmospheres with hydrogen chloride at 60° C to 80° C in the presence of metal chloride as catalyst.}

$CH{\equiv}CH + HCl \longrightarrow CH_2{=}CHCl$

Acetylene — Vinylchloride

Properties

- It is a colorless, non-inflammable, chemically inert material.
- It is having low density and high softening point148°C
- It is resistant to light, atmospheric oxygen, inorganic acids and alkalis.
- It is not stable to heat and UV radiation.
- Strong and brittle.

Uses

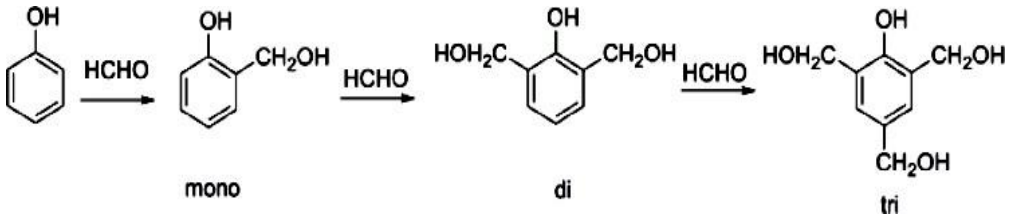
- It is mainly used as cable insulation, tank-lining, packing and toys.
- It is used for manufacturing of safety helmets, refrigerator components and floor covering.
- PVC pipes are used for carrying corrosive chemicals in petrochemical factories.
- Water carrying pipes, pipes for drainage and making frames for doors and windows.

1.8. Bakelite (Phenol Formaldehyde Resin)

It is prepared by condensing phenol with formaldehyde in presence of acidic/alkaline catalyst.

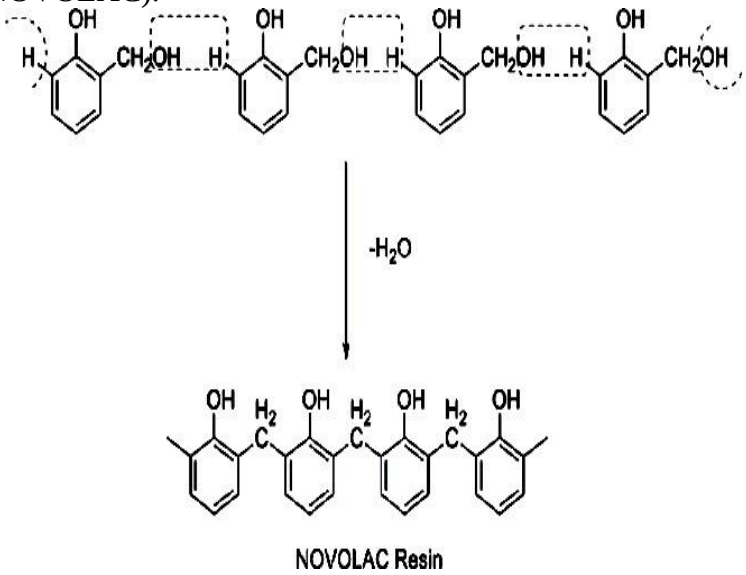
Preparation

Step1: Phenol reacts with formaldehyde in basic condition to form mono, di or tri methanolate Phenol (methylal phenol), depending on the formaldehyde ratio taken (1:1,1:2or1:3)



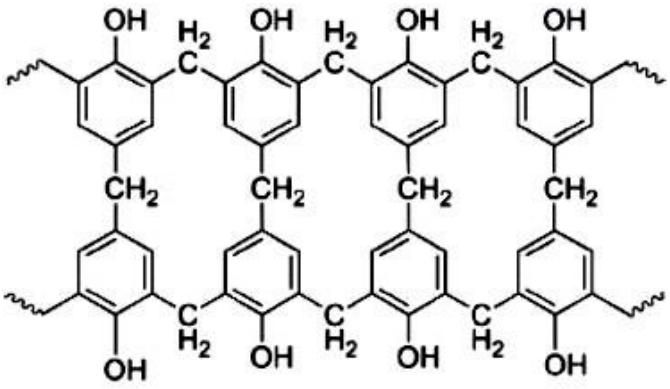
Step2: Preparation of PF resin

Mono-methanolate phenol derivative condense under acid catalysis to give a low molecular weight polymer called resin (NOVOLAC).



Step3: Bakelite formation

This stage is done during moulding. NOVOLAC resin is mixed with additives and curing agent hexamethylenetetramine (HMTA) and heated. Methylene Bridge cross linking occurs between NOVOLAC resins. A hard insoluble material called Bakelite is formed.



Properties and Uses

- It is electrically non-conducting, they are insulators. Used in the manufacture of circuit boards,

sockets, plug points

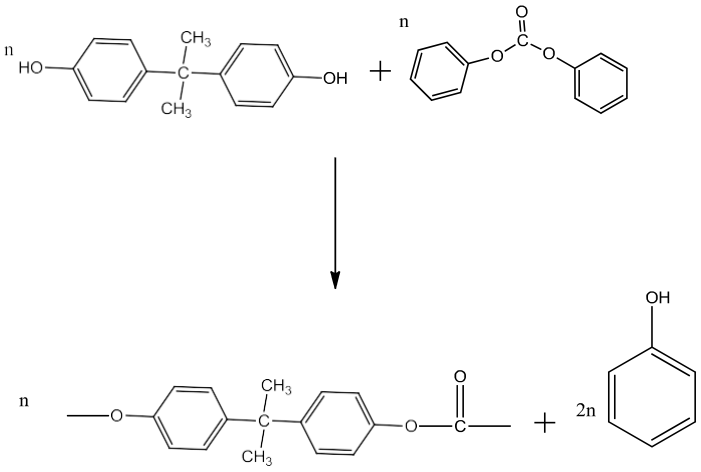
- They are heat resistant. Used in the manufacture of handles for kitchenware, heaters etc.
- They are resistant to oxidizing acids. Used for the manufacture of acid storage cans.
- It is a good adhesive. Used as adhesive for grinding wheels.
- It is a good ion exchanging resins used in water purification by ion exchange method
- It has good chemical and corrosive resistance.

1.9.Polycarbonates

Polycarbonate (PC), known by the trade names Lexan, makrolon, Makro clear. These are a group of thermos plastic resins containing carbonate group in their structure and it is an epoxy resin. They contain [-O–CO–O-] linkage.

Synthesis

They are also prepared by condensation reaction of diphenyl carbonate and Bisphenol-A.



Properties

- Polycarbonate is a transparent and good electrical insulator.
- Polycarbonate has a high melting point, around 265°C.
- It has a high tensile strength and impact resistance.
- It is resistant to water and many organic compounds but alkalis slowly hydrolyze it.

Applications

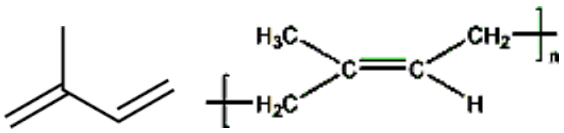
- It is used for making hair driers, electric razors
- It finds application in making many useful articles such as safety goggles, windows in automobiles, bullet proof glasses, police safety shields, Telephone parts and auto mobile light lenses, etc.,
- Polycarbonates are mainly used in CD-ROMs, DVD and baby bottles, syringes are well known examples.

Elastomers

1.10. Natural Rubber

Rubber is a natural elastic polymer of isoprene. It is obtained from the milk of rubber called 'Latex' of Hevea brasilein is type of plants.

- Trans polyisoprene is called gutta-percha



Preparation of Natural Rubber from Latex

- The main composition of natural rubber is polyisoprene.
- Latex is rubber milk containing about 30 to 45% of rubber.
- The rubber milk is diluted with water and allowed to stand for some time.
- The clear liquid from the top is treated with acetic acid or formic acid to precipitate rubber.
- The precipitated rubber is collected and passed through rollers to get sheets of rubber. This is called 'Creep rubber'.
- In the second type of process, the rubber sheets are hung in smoked chambers for 3-4 days at 40-50 °C. This is called smoked rubber.

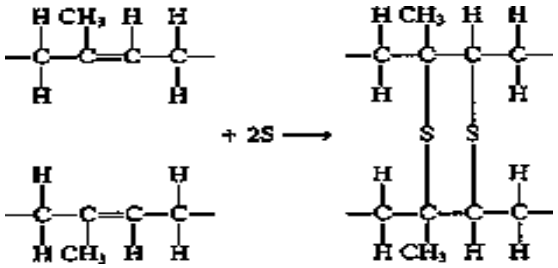
Defects of natural rubber

The natural rubber obtained from latex cannot be used in industries because it has some defects.

- It becomes soft and sticky during summer and it became hard and brittle during winter.
- It is attacked by atmospheric O_2, CO_2, moisture, light and gets less durability
- It swells up in oils.
- It flows plastically due to prolonged stress.
- Chemicals easily affect natural rubber.
- It possesses poor tensile strength

1.11. Vulcanization of Rubber

- Vulcanization is compounding of rubber with Sulphur.
- Vulcanization is done by heating rubber with Sulphur at 140°C in CO_2 atmosphere
- Sulphur adds to the double bonds present in rubber to provide crosslinks between the polymer chains.
- 2 to 4% Sulphur addition gives soft elastic rubber. When Sulphur content is more than 30%, we get hard rubber called 'Ebonite'.



Properties of Vulcanized Rubber

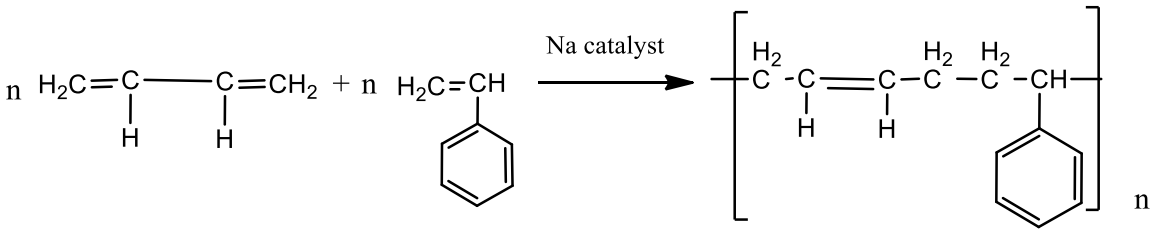
- Vulcanized rubber has good tensile strength.
- It has good resistance to organic solvents.
- It acts as good electrical insulator.
- It has low water absorption tendency.
- It has good impact resistance.

- The elasticity of vulcanized rubber is very low.

1.12. BuNa-S (styrene butadiene rubber)/SBR /GR-S

Buna–S prepared by the copolymerization of butadiene with styrene using sodium as a catalyst.

Bu stands for butadiene-monomer
Na stands for sodium–catalyst
S-Stands for styrene



Properties

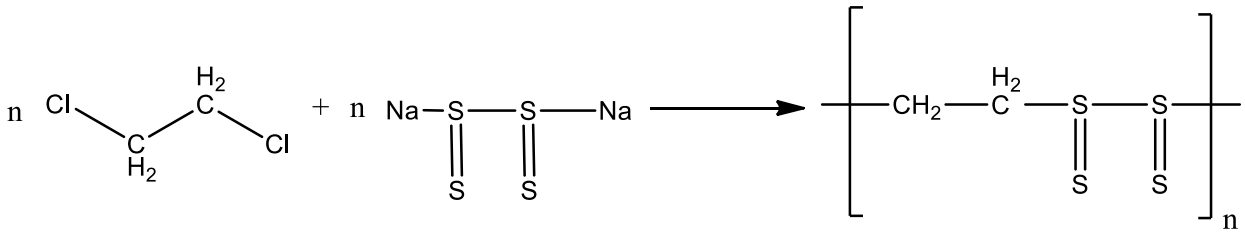
- It is a strong and tough polymer.
- It is vulcanized by sulphur mono chloride or sulphur.
- It resembles natural rubber in processing characteristics and quality off in shed products.
- It has excellent abrasion resistance and also good insulator.
- It is resistant to chemicals but swells in oil sand attacked by even traces of ozone present in atmosphere.
- It has high load bearing capacity.

Applications

- Major applications of styrene rubber are in the manufacture of tires.
- It is used in the foot wear industry for making shoe soles and footwear components.
- It is also used for making wires and cables insulations.
- The other applications of styrene rubber are for the production of floor tiles, tank linings in chemical industries and as an adhesive.

1.13. Thiokol Rubber (or) GRP (or) Poly sulphide Rubber

Thiokol rubber was prepared by co-polymerisation of sodium poly sulphide (Na_2S_4) and ethylene di chloride produces Thiokol rubber.



ethylene di chhloride sodium sulphide Thiokol

Properties

- It has strength and impermeability to gases.
- This rubber cannot be vulcanized and it cannot form hard rubber.
- It possesses extremely good resistance to mineral oils, fuels, oxygen, solvents, ozone and sunlight.
- Thiokol rubber possesses low abrasion resistance.

Applications

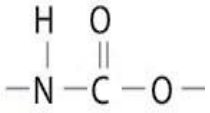
- Fabrics coated with Thiokol are used for barrage balloons, life rafts and jackets which are inflated by

CO_2.

- Thiokol are used for lining hoses for conveying gasoline and oil.
- It is used for making gasket, sand seals for printing rolls.

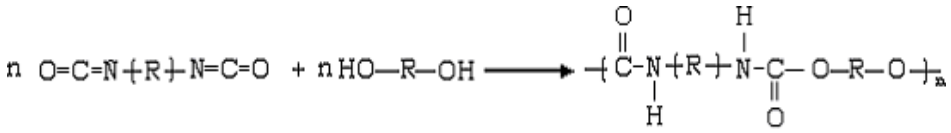
1.14. Polyurethane

Polyurethanes are produced by the polymerization of di-isocyanate and diol or triol (or the addition reaction between p-phenyl di-isocyanate with glycol).



urethane bond

Synthesis



Properties

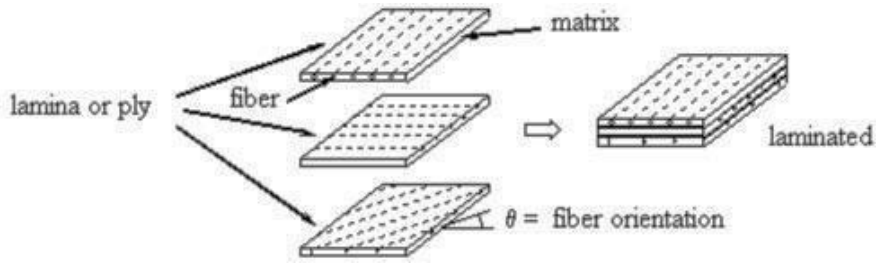
- These can be obtained in the form of foams, fibers, Elastomers, Coatings, etc.
- They have high strength, hardness.
- Good resistance to gas, oil
- Excellent resistance to oxygen and ozone.

Uses

- Flexible foams are used for cushions in automobiles & furniture.
- Rigid foams are used to reinforce hallow structural units.
- Fibers of Polyurethanes are used in light weight garments and swimsuits because of their stretching property.
- These are used to coat gymnasium floor and dance floor.
- These are used in tyre treads and industrial wheels.

1.15. Composite Materials

Combination of two or more structurally different components combine to form a new class of material suitable for structural applications is referred as composite material. When one of the components is a polymer, the resulting composite is called a polymer composite.



Constituents of Composites

Two essential constituents of composites are

1. **Matrix Phase:** It is the continuous body constituents which encloses the composite and give its bulk

form. Matrix phase may be metal, ceramics (or) polymers. Composites using these matrixes are known as metal matrix composites (MMC), Ceramic Matrix Composites (CMC) and Polymer Matrix Composites (PMC) respectively.

Functions of Matrix Phase

- It binds the dispersed phase together.
- Acts as a medium to transmit and distribute an externally applied load to the dispersed phase.
- It protects the dispersed phase from chemical action and keeps in proper position and orientation during application of load.
- It prevents propagation of brittle cracks due to its plasticity and softness.

2. **Dispersed Phase:** It is the structural constituent, which determines the internal structure of composite. Important dispersed phases of composites are fibers, particles, whiskers, or flakes.

Some examples of composite materials

Fiber Reinforced Plastics (FRP)

- FRP are produced by reinforcing a plastic matrix with a high strength fiber as dispersed phase.
- Natural fibers such as sisal and asbestos are used for reinforcement.
- Depending on the desired properties of final composite the nature of fiber used is decided.
- FRP are produced by following methods.
- Filament winding technique.
- Hand lay-up technique.
- Spray lay-up technique.
- Continuous lamination method
- Some important types of fiber reinforced composite are glass fiber reinforced composite, carbon fiber reinforced composite, aramid fiber reinforced composite etc.

Fiberglass (glass fiber reinforced composite)

- Glass fiber is most extensively used reinforced fiber because of durability, acid proof, water proof and fire proof nature of glass.
- Glass is drawn into threads or fibers into the form of filaments
- Then the filaments are woven in the form of mats
- The fiber material is suitably bonded with plastic material to be reinforced
- The common plastic resins use dare polyester, epoxy, silicon, vinyl derivatives.

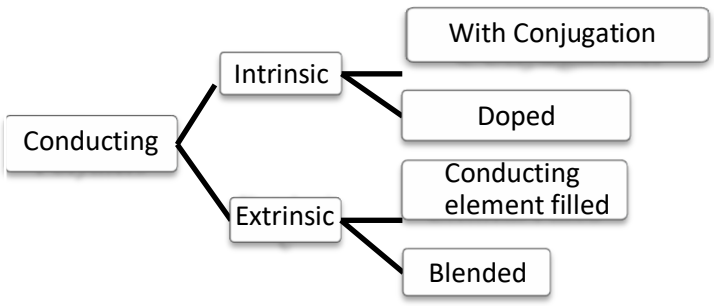
Applications composite materials

- Automobile parts and storage tanks are made from fiber glass
- It is used for making industrial floorings, plastic pipes
- It is used for making transport vehicles to reduce vehicle weight and boost fuel efficiency.

1.16. Conducting Polymers

Those polymers which conduct electricity are called conducting polymers. The conduction of polymers may be due to unsaturation or due to the presence of externally added ingredients in them.

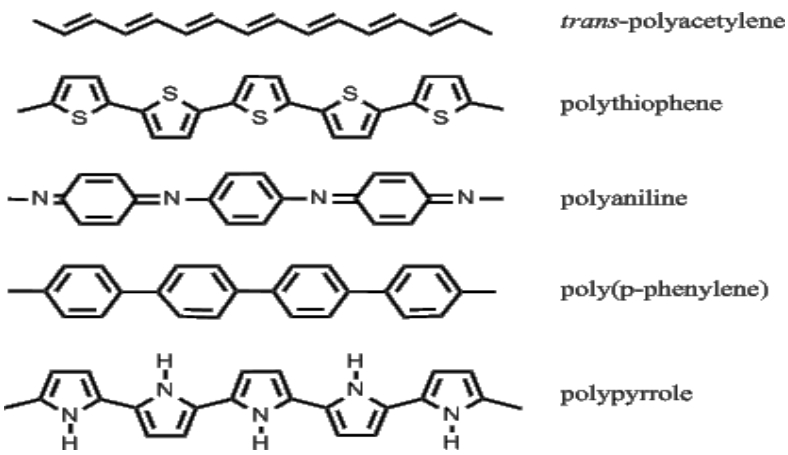
The conducting polymers can be classified in the following ways.



A. Intrinsic Conducting Polymers

Intrinsic conducting polymers have extensive conjugation in the back bone which is responsible for conduction. These are two types.

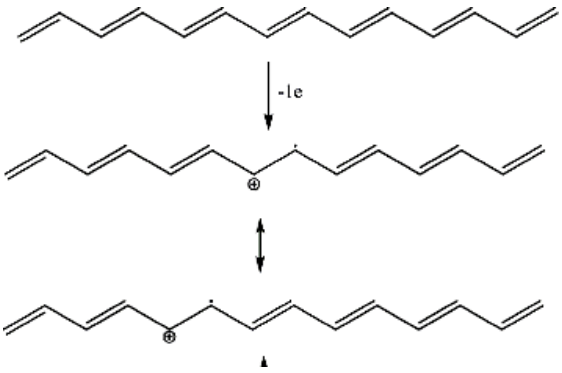
1. **Conducting polymers having conjugation:** These polymers having conjugated double bonds in their backbone. In such case, overlapping of orbitals having conjugated π electrons over the entire backbone forms the valence band as well as conduction band which were separated by a significant Fermi energy gap. The electrical conductivity takes place only after thermal or photolytic activation of the electrons, which give them sufficient energy to jump the electrons from valence band to conduction band.
   e.g.: polyacetylene, polyaniline, polypyrrole and polybutadienylene.



2. **Doped conducting polymers**: Intrinsic conducting polymers can be easily oxidized or reduced because of their low ionization energy and high electron affinities. The conductivity such polymers can be increased by creating either positive or negative charge on the polymer backbone by oxidation or reduction. This process is referred as doping, which is the following two types.

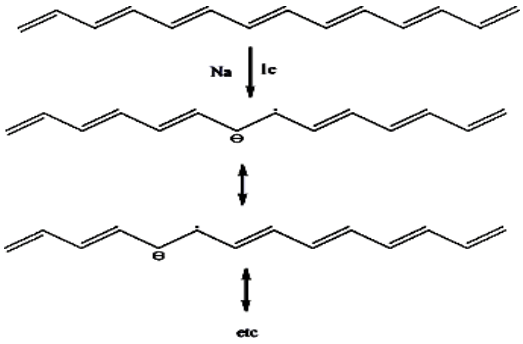
   a. **P-type doping:** Creating positive charge on polymer backbone is called p-doping. It is done by oxidation of a conducting polymer with a suitable oxidizing agent such as a Lewis acid ($AlCl_3$, $FeCl_3$) or iodine vapor.

   $$\text{-(-}CH_2{=}CH_2\text{-)}_n + FeCl_3 \longrightarrow (CH_2{=}CH_2)_n + FeCl_4^- + FeCl_2$$



   b. **n-type doping:** Creating negative charge on polymer backbone is called n-doping. It is done by reduction of a conducting polymer with a suitable reducing agent such as a Lewis base (naphthylamine, sodium napthalide, etc.)

   $$\text{-(-}CH_2{=}CH_2\text{-)}_n\text{-} + Na^+(C_{10}H_8)^- \longrightarrow (CH_2{=}CH_2)^-_nNa^+ + C_{10}H_8$$



B. Extrinsic Conducting Polymers

The conductivity of these polymers is due to the presence of externally added ingredients in them. These are further two types.

a. **Conducting element filled polymers:** In such case, polymer acts as a binder to hold the conducting such as carbon black, metallic fibers and metallic oxides etc. The minimum concentration of conducting filler is added so that the polymer starts conducting. This minimum concentration of conductive filler is called percolation threshold. At this concentration of filler, a conducting path is formed in polymeric material.
Advantages: These polymers possess quite good bulk conductivity, light in weight, mechanically durable, can be easily molded and are of low cost.

b. **Blended conducting polymers:** These are obtained by blending a conventional polymer with a conducting polymer. Such polymers possess better physical, chemical, electrical and mechanical properties.

Factors affecting conductivity

1. **Band gap:** Band gap increases, conductivity decreases.
2. **Doping level:** Conductivity increases with increase in doping level
3. **Temperature:** The conductivity of a conducting polymer decreases with decrease in temperature**.**
4. **Conjugation length:** Conductivity increases with increase in conjugation (or) chain length.

Applications of conducting polymers

- Conducting polymers find their applications in rechargeable batteries.
- Conducting polymers also used for making analytical sensors for pH, O_2, NO_X, SO_2, NH_3 and glucose.
- Conducting polymers also used for making ion exchangers.
- Conducting polymers are used for making electronic displays and optical fibers.
- Conducting polymers are applicable in photo voltaic devices.

1.17. Biodegradable Polymers

Biodegradable plastics are plastics that are capable of being decomposed by bacteria or other microorganisms.
e.g.: Aliphatic polyesters are biodegradable due to their enzymatically hydrolysable ester bond. These are classified as

1.Natural

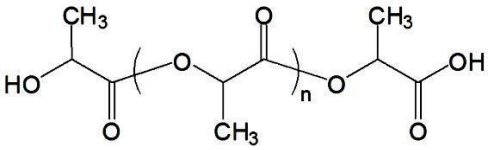
Polysaccharides - **e.g.:** starch and cellulose.
Proteins– **e.g.:** gelatin, casein, silk, wool
Polyesters - **e.g.:** polyhydroxy alkenoates
Others - **e.g.:** lignin, natural rubber.

2.Synthetic

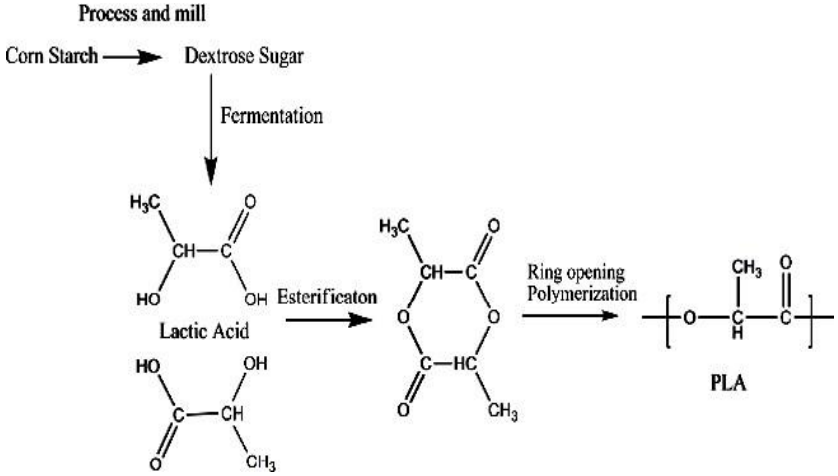
E.g.: PVA, PLA, PHA, PHBetc.,

- **Polylactic acid/ polylactide (PLA):**

- This is a biodegradable thermoplastic polyester belonging to the class of polyhydroxy alkanoates.

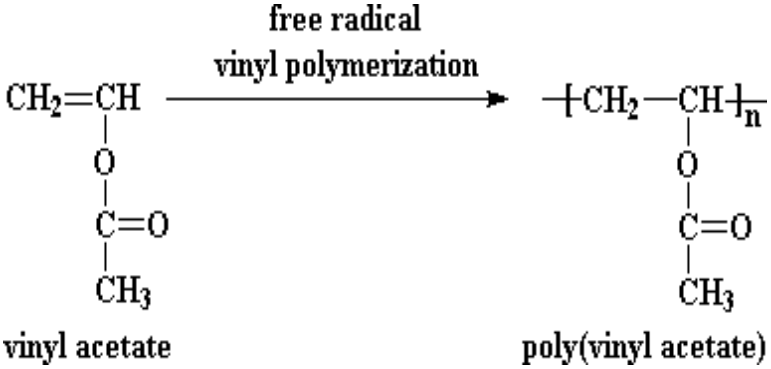


- It is derived from renewable resources (corn starch, tapioca roots, sugar cane, etc.)



- **Poly vinyl acetate (PVA):**

- Poly vinyl acetate is a water soluble and biodegradable polymer, possessing excellent mechanical properties and compatibility with starch.



Properties of biodegradable polymers

- Moisture sensitive
- Sensitive towards Acids and bases
- Dissolves in chlorinated organic solvents
- Good oxygen permeability.

- PLA is biocompatible

Application of biodegradable polymers

- It is used in number of biomedical applications such as drug delivery, stents, dialysis media, medical implants like anchors, screws, pins, mesh, surgical threads etc.,
- Disposable table wares, packages for food and flushable sanitary products, diapers etc.
- Used in food industry as a packing material, food storage and compostable bags, chewing gums.
- Adhesive for wood, paper & cloth for book bindings, envelops, wallpaper, etc.

FAQ

1. Explain emulsion polymerization.
2. Write a note on suspension polymerization.
3. Write preparation, properties and applications of Bakelite.
4. Mention the synthesis, properties and applications of polycarbonate and polyurethane.
5. Explain compression moulding.
6. Discuss injection moulding.
7. Write a note on blown film method with the help of neat sketch.
8. Write the procedure of extrusion moulding technique for fabrication of plastics.
9. Define composite material. Write about fibre reinforced plastics.
10. What is the importance of vulcanization of rubber?
11. What are conducting polymers? Explain.
12. List out applications of biomedical polymers.
13. Write a note on biodegradable polymers.
14. Discuss biopolymers.
15. Mention preparation and properties of Buna-S, Thiokol and PVC.
16. Write a note on conducting polymers.
17. Describe biodegradable polymers.

Electrochemical cells and corrosion

2.1. Introduction

An electrochemical cell is a mechanism that uses the energy released by a spontaneous redox reaction to generate an electric current. The galvanic or voltaic cell, which bears the names of Luigi Galvani and Alessandro Volta, is one example of this type of cell. The late 18th century saw a number of experiments by these scientists on chemical reactions and electric current.

The anode and cathode, two conductive electrodes found in electrochemical cells, work together. The electrode where oxidation takes place is referred to as the anode. The electrode on which reduction occurs is called the cathode. Any sufficiently conductive material, including metals, semiconductors, graphite, and even conductive polymers, can be used to create electrodes. The electrolyte, which contains ions that can move freely, is sandwiched between these electrodes.

Two different metal electrodes, one in an electrolyte solution, are used in the voltaic cell. The cathode will experience reduction, and the anode will experience oxidation. The metal in the anode will oxidise, moving from a solid state with an oxidation state of zero to one with a positive value, and it will then become an ion. The metal ion in the solution will take one or more electrons from the cathode at the cathode, reducing its oxidation state to zero. On the cathode, this results in the formation of a solid metal.

To allow for an electron flow from the metal of the anode to the ions on the surface of the cathode, the two electrodes must be electrically connected to one another. This electron flow is an electrical current that can be used to power devices like motors and lights.

2.2. Single Electrode Potential

Single electrode potential: The potential of half-cell i.e., the potential difference between the metal and its salt solution in which it is dipped is called single electrode potential. It cannot be directly measured.

$$Zn|ZnSO_4 \;\|\; CuSO_4|Cu$$

<half-cell > <half-cell>

< Single electrode> <Single electrode>
(anode) (cathode)

$$E_{cell} = E_{anode} + E_{cathode}$$

$$E_{cell} = E_{Ox} + E_{Red}$$

$$E_{cell} = E_R - E_L$$

E_R of Cu^{2+}/Cu is 0.337 V and E_L of Zn/Zn^{2+} is -0.763 V.
E_{cell} = 0.337 - (-0.763) = 1.1 V

Direction of spontaneous reaction: If $E_{cell} > 0$, the cell reaction has a tendency to go from left to right. If, however the $E_{cell} < 0$ the cell reaction will have a tendency to go from right to left. The reaction is spontaneous or feasible only when E_{cell} is positive for that case the standard free energy of the reaction (ΔG^0) would be negative ($\Delta G^0 = -nFE^0$).

2.3. Electrochemical Series and uses of Series

A series of electrodes or half cells arranged in order of their increasing standard oxidation potentials or in the decreasing of their reduction potentials is called as electromotive force series (EMF series) or electrochemical series.

Characteristics electrical series

1. In this series, all reduction potentials are given on scale and hydrogen E^0 is taken as zero.
2. The standard reduction potential of an element is a measure of thetendency of that element to get reduced.
3. The element which has greater reduction potential gets reduced easily, while the elements with low reduction potential will get easily oxidized.
4. Elements that lose electron more easily have lower reduction potential and those which lose electrons with greater difficulty or instead of losing they accept electrons more easily have a higher reduction potential.
5. In EMF series elements having higher +ve reduction potential is placed at the bottom and while those having lower –ve reduction potential is placed at the top. SHE has the middle position in the electrochemical series.
6. The substances which are stronger oxidising agents than hydrogen is placed below the hydrogen in the series and have +ve standard reduction potential.
7. The substances which are weaker reducing agents than hydrogen isplaced above the hydrogen in the series.

Electrochemical Series:

Oxidized species		Reduced species	E^0 in volts Standard Reduction Potentials at 298K
$Li^+ + e^-$	↔	Li	-3.05
$K^+ + e^-$	↔	K	-2.93
$Ca^{2+} + 2e^-$	↔	Ca	-2.87
$Na^+ + e^-$	↔	Na	-2.71
$Mg^{2+} + 2e^-$	↔	Mg	-2.37
$Al^{3+} + 3e^-$	↔	Al	-1.66
$Mn^{2+} + 2e^-$	↔	Mn	-1.18
$2H_2O$ (l) + $2e^-$	↔	H_2 (g) + $2OH^-$ (aq)	-0.83
$Zn^{2+} + 2e^-$	↔	Zn	-0.76
$Cd^{2+} + 2e^-$	↔	Cd	-0.40
$Ni^{2+} + 2e^-$	↔	Ni	-0.25
$Pb^{2+} + 2e^-$	↔	Pb	-0.13
$2H^+ + 2e^-$	↔	H_2	0.00
$Sn^{2+} + 2e^-$	↔	Sn	0.15
$Cu^{2+} + 2e^-$	↔	Cu	0.34
$Fe^{3+} + e^-$	↔	Fe^{2+}	0.77
$Ag^+ + e^-$	↔	Ag	0.80
$2Hg^{2+} + 2e^-$	↔	Hg_2^{2+}	0.91
$Br_2^{2+} + 2e^-$	↔	$2Br^-$	1.08
$Cl_2 + 2e_-$	↔	$2Cl^-$	1.36
$Au^{3+} + 3e^-$	↔	Au	1.50
$Co^{3+} + e^-$	↔	Co^{2+}	1.82
$F_2 + 2e^-$	↔	$2F^-$	2.87

Applications/Uses

1. **For comparison of the relative oxidizing and reducing power of various metals:** A positive number indicate stronger reductionpotential of the electrode, and it functions as a stronger oxidizing agent. A negative number means weaker reduction potential of the electrode, and hence it is a more powerful reducing agent.

   e.g.: The F_2 electrode with the standard reduction potential of +2.87V isa strong oxidizing agent and Li^+ ion with the standard reduction potential value of -3.05 V is a strong reducing agent.

2. **For comparison of the relative activities of metals towards displacement reactions based on their placement in the activity series:** Any metal that is higher in the activity series can displace other metal that is placed lower in the activity series, from its compound.

   e.g.: $Al + Fe_2O_3 \square Al_2O_3 + Fe$

   Al (-1.66 V) has higher oxidation potential than Fe (0.4 V) to form Al_2O_3

3. **To calculate the standard EMF of any electrochemical cell (E^0_{cell}):** The standard EMF of the cell is an addition product of standard electrode potential of two half-cells: Reduction half-cell and oxidation half cells.

   $E^0_{cell} = E^0_{red} + E^0_{ox}$

   Standard reduction potential (E^0_{red}) = - standard oxidation potential (E^0_{ox})

   e.g.: For a reaction $Zn(s) + Pb^{2+}(aq) \rightarrow Zn^{2+}(aq) + Pb(s)$

   The standard reduction potentials given are

   $Pb^{2+}(aq)|Pb(s) = -0.125$ V& $Zn^{2+}(aq)|Zn(s) = -0.764$V

   $E^0_{cell} = E_{cathode} - E_{anode} = -0.125-(-0.764) = 0.639$V

4. **In predicting the feasibility/occurrence of a redox reaction:** For a redox reaction to be spontaneous or occur readily, the EMF of the cell must be positive. If it comes out to be negative, then the direct, spontaneous reaction cannot take place.

   e.g.: 1M $CuSO_4$ cannot be stored in Nickel vessel.

   Because $E^0\ Ni^{2+}|Ni = -0.25$ V and $E^0\ Cu^{2+}|Cu = +0.35$V.

   $E^0_{cell} = 0.35-(-0.25) = 0.6$V.

   Reaction is spontaneous.

5. **To predict whether a metal reacts with acid to liberate Hydrogen gas:** For a metal to react with acid (H+), the following reaction should take place.

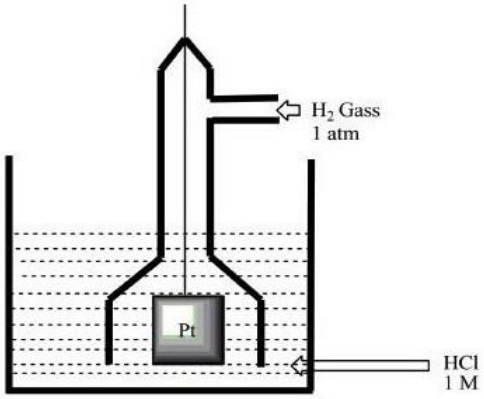
   $M(s) + H^+(aq) \rightarrow M^+(aq) + 1/2H_2(g)$

   The metal undergoes oxidation, and the hydrogen ion by accepting the electron undergoes reduction. If the metal has negative reduction potential and is place above the hydrogen in the electrochemical series, it can react with an acid to liberate hydrogen gas. The metals copper, silver, gold and platinum and placed below the hydrogen are less reactive, and they do not replace hydrogen from water or acid.

2.4. Standard Hydrogen Electrode or Hydrogen Electrode (SHE) or Normal Hydrogen Electrode (NHE):

Standard Hydrogen Electrode (SHE) is the primary reference electrode. Its electrode potential is arbitrarily fixed as zero. SHE is an example of gas electrode. It consists of an inert platinum rod dipped

in 1MHCl solution. H2 gas is bubbled at 1 atm pressure at 298 K temperature.
H^+ (aq)(1M)/H_2(g)(1 atm)/Pt.



Depending on the type of electrode to which it was connected, it could either act as an anode or a cathode. SHE is a reversible electrode as a result. SHE functions as the cathode if the coupled electrode's potential is less than zero. At SHE, reduction takes place.

$$H^+ + e^- \rightarrow H_2$$

If the potential of the coupled electrode is greater than zero, thenSHE acts as anode. Oxidation takes place at SHE.

$$H_2 \rightarrow H^+ + e^-$$

Limitations of SHE

1. SHE is difficult to construct. Bubbling H_2 gas at 1 atm around Ptthrough the solution is difficult. Maintaining activity of H+ at unity is difficult.
2. Pt metal has high adsorption and catalytic activity. Hydrogen on Pt surface is reduction catalyst. Many substances get reduced.

2.5. Calomel Electrode

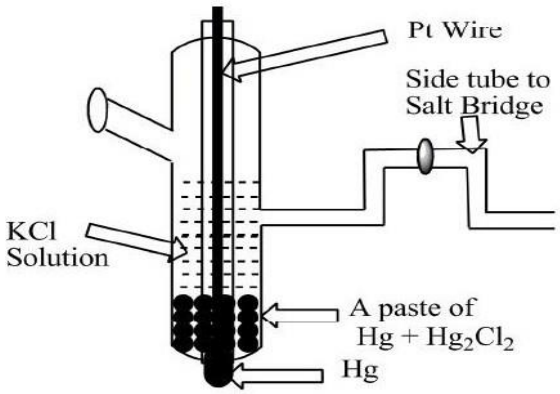
Due to limitations of SHE, secondary reference electrodes are needed. The potential of secondary reference electrodes is calibratedusing SHE.

Calomel Electrode is an example of metal-metal insoluble salt electrode.

Composition: Mercury is the metal. Mercurous chloride (Hg_2Cl_2, calomel)is the insoluble salt. KCl is a soluble salt with a common anion. Electrical contact is made using Pt wire.

Electrode representation: $Cl^-_{(aq)}$ / Hg_2Cl_2 /Hg

Construction: It is made up of glass tube. Mercury is kept at the bottom. Over which a paste of Hg^+ Hg_2Cl_2 kept. A solution of KCl is placed calomel. A platinum wire is used to make electrical contact. The side tube is used to make contact with other electrodes through a salt bridge.



It is a reversible electrode. It can act as cathode or anode depending on the potential of the electrode to which it is coupled. If the potential of coupled electrode is greater than calomel electrode, calomel electrodeundergoes oxidation (Anode). If the potential of coupled electrode is less than calomel electrode, calomel electrode undergoes reduction (Cathode).

The electrode reaction (reduction),

$$Hg_2Cl_2 + 2e^- \rightarrow 2Hg(l) + 2Cl\text{-}(aq).$$

The potential of the electrode is given by,

$$E_{cell}=E^0_{cell}-\frac{2.303RT}{2F}\log[Cl^-]^2$$

$$E_{cell}=E^0_{cell}-\frac{2.303RT}{F}\log[Cl^-]$$

The potential of Calomel electrode depends on the concentration of Chloride (KCl) solution. If the KCl is saturated then we call it saturated calomel electrode (SCE), If 1N KCl is used then it is Normal calomel electrode (NCE) and if 0.1 N KCl is used then it is deci-normal calomel electrode. Electrode potential decreases in the order 0.1N (0.334 V) > 1 N (0.280 V) > saturated (0.241 V) KCl. Calomel electrode is convenient to handle and it can be constructed easily. It cannot be used above 50^0C due to instability of Hg_2Cl_2 and mercury is poisonous.

Batteries

2.6. Dry Cell or Leclanché Cell

It is called a dry battery as it does not have any liquid electrolyte init. The zinc vessel serves as the anode. The cathode is a graphite rod in the center of the cell. It is surrounded by the electrolyte which consists of a paste of NH_4Cl, $ZnCl_2$, and MnO_2. Traces of acetylene black, graphite powder are also added starch is added to make the mixture like a thick paste.

The cell represented as Zn/Zn^{2+}, NH_4/MnO_2, C

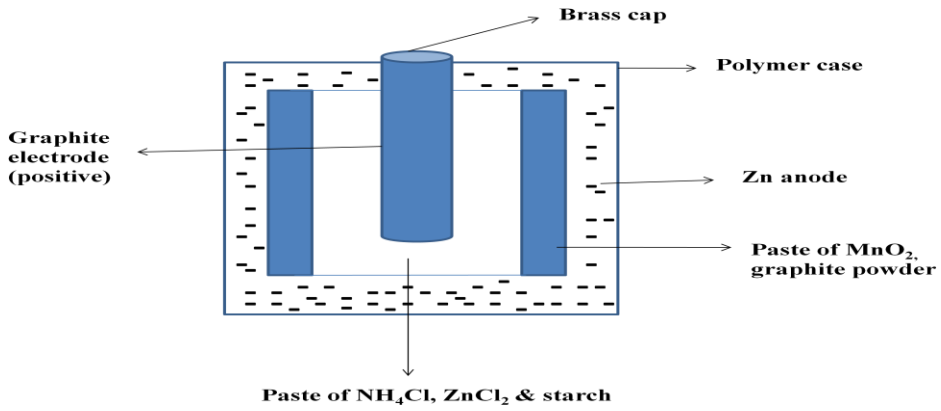
The cell reactions are involved are

At anode	Zn (S)	→	Zn^{2+} (aq)+ $2e^-$
At cathode	2 NH_4Cl(aq)+ $2MnO_2$ (S)+$2e^-$	→	$2NH_3$(aq)+Mn_2O_3(S)+H_2O+Cl^- (aq)
Net cell reaction	Zn (S)+ 2 NH_4Cl (aq)+ $2MnO_2$(S)	→	$Zn(NH_3)_2Cl_2$(s)+Mn_2O_3(S)+H_2O(l)

The NH_3 liberated reacts with Zn^{2+} to form a complex $[Zn(NH_3)_2]Cl_2$. The cell is a primary cell and gives a voltage of 1.5 V

Gradually, the products are formed and the cell voltage drops.

Applications: these cells are used in transistors, tape recorders, toys andportable electronic gadgets.



2.7. Lithium-MnO_2 Batteries or Lithium Batteries

- These are primary batteries that cannot be recharged.
- In these batteries lithium metal or lithium compounds are used as anode.
- The cathode is made of specially treated MnO_2 crystal obtained by special heat treatment.
- Since, lithium is highly reactive with water and non-aqueous solvents, these solvents cannot be used as electrolyte.
- Instead, lithium salt solution in propylene carbonate and dimethoxy salts is used the electrolyte solvents like thionyl chloride containing lithium compounds such as LiCl, LiBr, $LiAlCl_4$, $LiSO_3F_3$, $LiClO_4$ can also be used.
- The cell can be represented as

Li, non-aqueous solvents containing lithium salts/MnO_2

The cell reactions are

At anode $\quad Li \rightarrow Li^+ + e^-$

At cathode $\quad MnO_2 + Li^+ + e^- \rightarrow MnO_2(Li^+)$ Net cell reaction $\quad Li + MnO_2\ (S) \rightarrow MnO\ (Li^+)^{2+}$

Lithium batteries find use in clocks, calculators, toys, digital camera, watches, medical equipment like artificial pacemakers, thermometers and remote car locks etc.

(-) Li anode	**Li^+ Polymer electrolyte**	**(+) MnO_2 cathode**

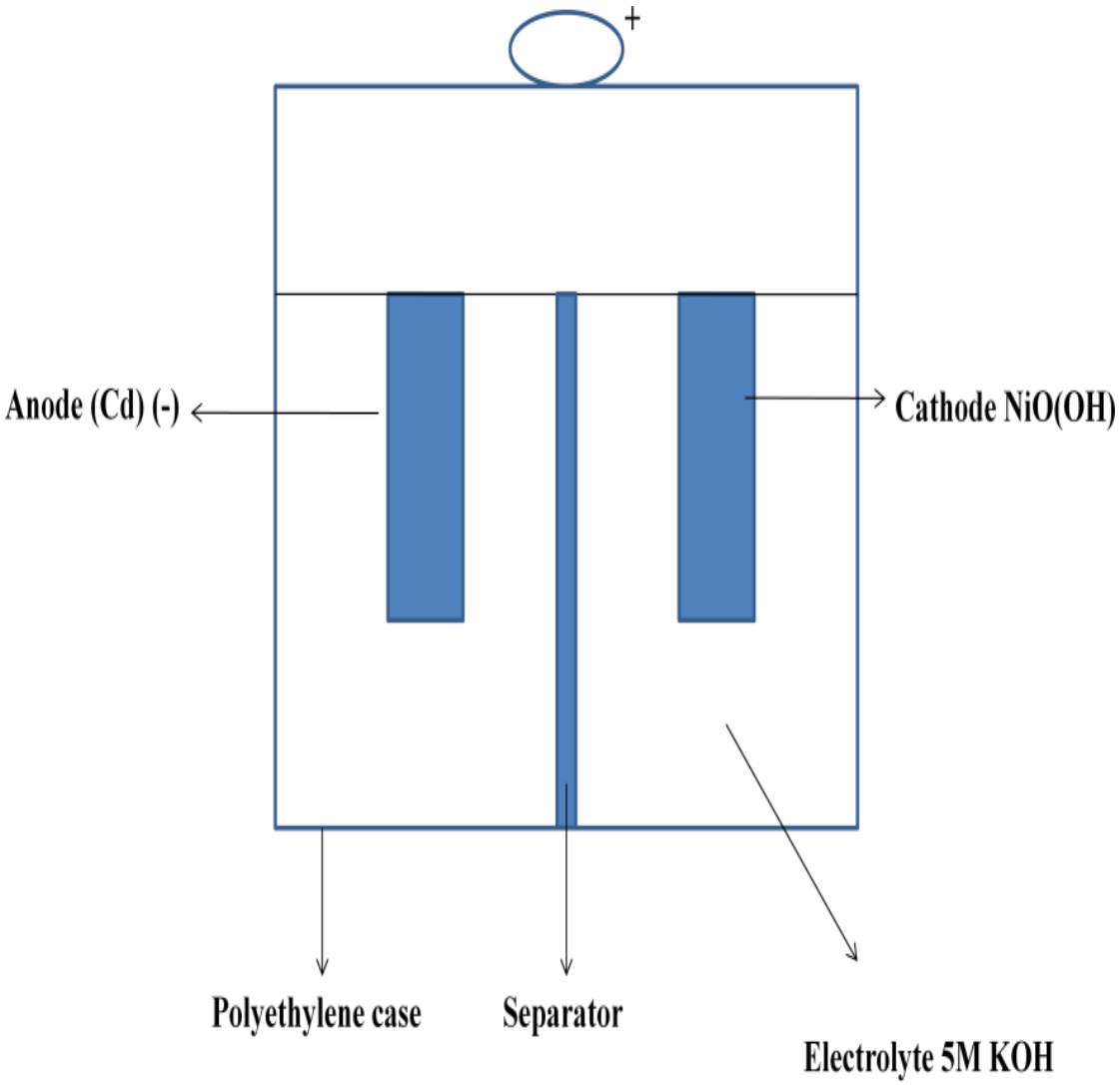
2.8. Ni-Cd Battery

- It consists of a Cd anode. The cathode is composed of a paste of NiO(OH). The electrolyte used is 5M KOH solution.
- The cell reactions are

At anode	$Cd(S) + 2OH^-(aq)$	$\rightarrow$	$Cd(OH)_2(S) + 2e^-$
At cathode	$2\,NiO(OH)(S) + 2H_2O(l) + 2e^-$	$\rightarrow$	$2Ni(OH)_2(S) + 2OH^-$
Net cell reaction	$Cd(S) + 2\,NiO(OH)(S)$	$\rightarrow$	$Cd(OH)_2(S) + 2Ni(OH)_2(S)$

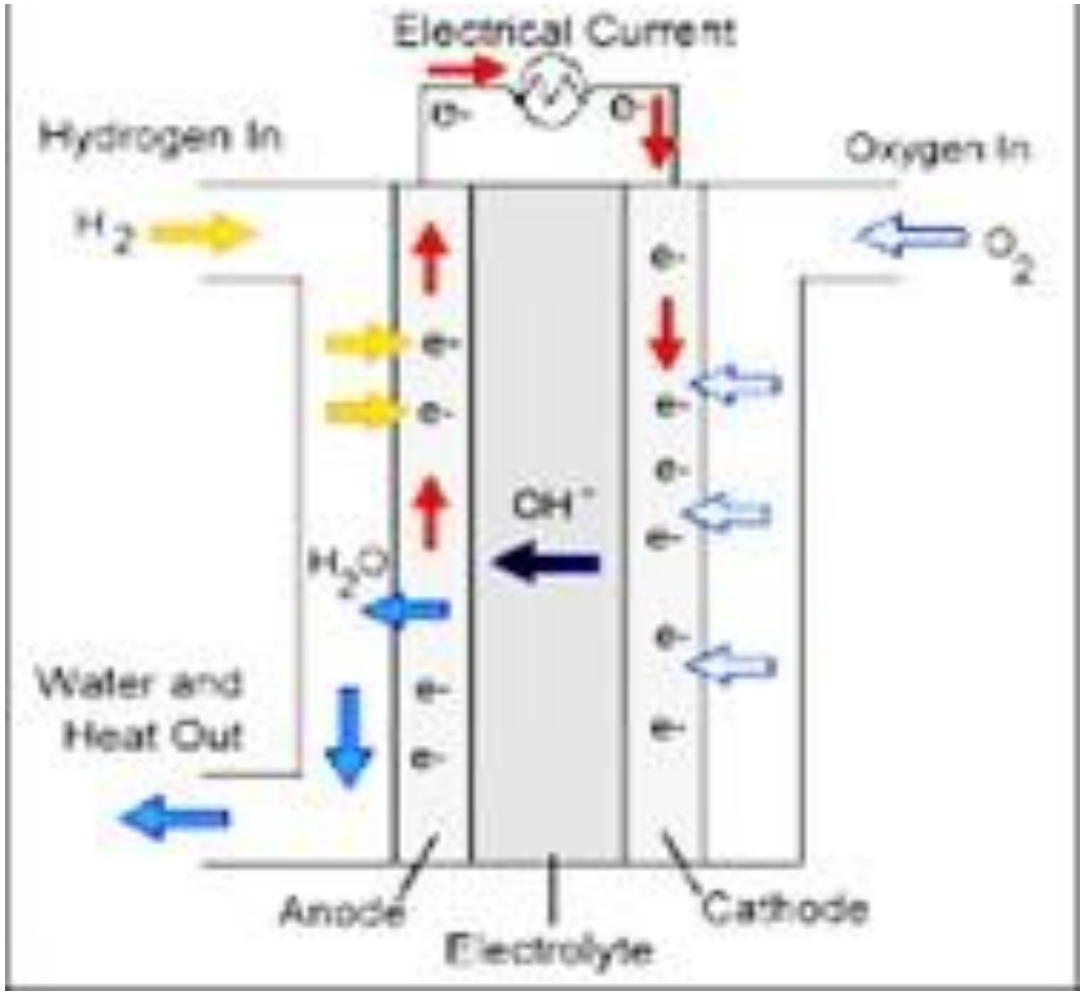
- Its voltage is ~1.35 V. however, its voltage is fairly constant.
- The reaction products adhere to the electrode surface and the reactioncan be easily reversed.
- This type of battery does not undergo deterioration as no gases areproduced.
- It is light, portable and rechargeable.

Applications: It is used in calculators, electronic flash units and cordlessshavers, transistors etc.



Fuel cells

2.9. H_2-O_2 fuel cell



Hydrogen - oxygen fuel cells consisting of two porous graphite electrodes, which is impregnated with a catalyst such as pt/pd/Ag. 25% KOH is used as an electrolyte. Hydrogen and oxygen are continuously supplied to the anode and cathode respectively. The hydrogen is oxidized and the electrons lost in this process migrate to the cathode through the external circuit. Oxygen at cathode receives the electrons and gets reducedto OH- ions. The cell reaction, hydrogen and hydroxyl ions combine to form water which is removed from the cell.

Cell reactions

At anode	:	$H_2 + OH^-$	→	$2H2O + 2e^-$
At cathode	:	$½ O_2 + H_2O$	→	$2OH^-$
Overall cell reaction	:	$H_2 + ½ O_2$	→	H_2O

Advantages

- The energy conversion is very high (75-82%)
- The product H2O is drinking water for astronauts.
- Noise and thermal pollution are low
- Maintenance cost is low for the fuels

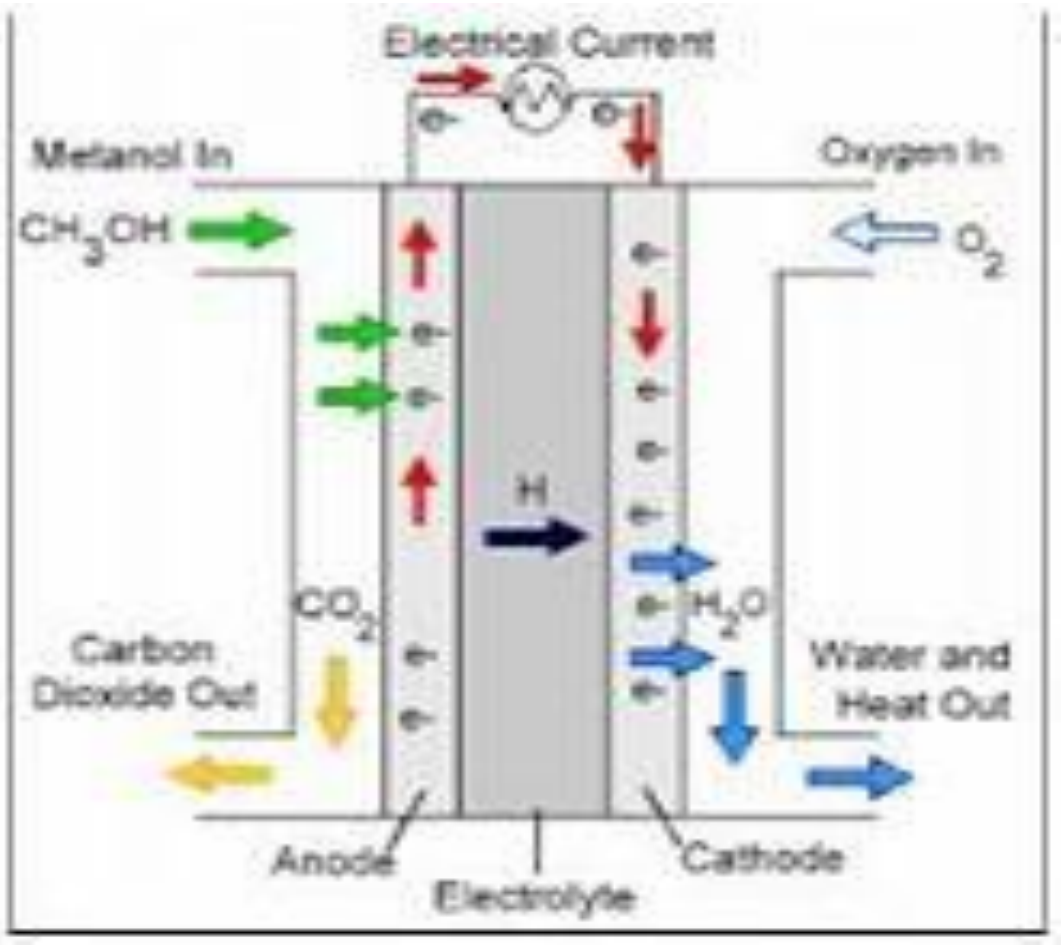
- The regenerative H2-O2 fuel cell is an energy storage system for spaceapplications.

Limitations

- The life time of fuel cell is not accurately known
- Initial cost is high
- The distribution of hydrogen is not proper.

Uses: The H_2-O_2 cells are used in Space vehicles, military and mobilepower systems.

2.10. Methanol – O_2 Fuel Cell



It consists of two electrodes made up of platinum. In between the electrodes KOHis placed as an electrolyte. Methanol and O_2 is supplied at the anode and oxygen are supplied at the cathode. The methanol is oxidized to CO_2 & H_2O with the liberation of electrical energy. The cell delivers an emf of 1.20v

Cell reactions

At anode $CH_3OH + H_2O \rightarrow CO_2 + 6\,H^+$

At cathode $O_2 + 6\,H^+ + 6\,e^- \rightarrow 3H_2O$

NCR $CH_3OH + 3/2\,O_2 \rightarrow CO_2 + 2H_2O$

Corrosion

2.11. Corrosion Definition: The process of decay of metal by environmental attack is called corrosion.

Effects or disadvantages of the corrosion

1. The valuable metallic properties conductivity, ductility. Strength and malleability etc., are lost.
2. The process of corrosion is very harmful and causing for the wastage ofmetal. It is estimated that the amount of iron wasted by corrosion each year is about ¼ of its annual world protection.
3. Life span of the metallic parts of the machines is reduced.

2.12. Theories of corrosion (chemical and electrochemical)

2.12.1 Dry Corrosion or Chemical Corrosion

This type of corrosion occurs mainly through the direct chemical action of environment or atmospheric gases such as oxygen, halogen,hydrogensulphide, Sulphur dioxide, etc. with metal surfaces in immediate proximity.

There are three main types of dry corrosion.

a. Oxidation corrosion: Direct attack of oxygen on metals in the absence of moisture at high or low temperatures is called oxidation corrosion. Except Ag, Au and Pt all metals are oxidized even though alkali (Li, Na, K, Rb etc.) and alkaline earth metals (Be, Ca, Sr etc.) are very rapidlyeven at lower temperatures.

$2M \rightarrow 2M^{+} + 2e^{-}$ (Oxidation by loss of electrons)

$O_2 + 2e^{-} \rightarrow 2O^{2-}$ (Reduction by gain of electrons)

Total reaction $2M + O_2 \rightarrow 2M^{+} + 2O^{2-} \rightarrow 2MO$

Mechanism: The oxidation of the metal occurs at the surface first, resulting in the formation metal oxide scale, which restricts further oxidation. Further oxidation can continue if the metal diffuses out of the scale or the oxygen must diffuse in through the scale of the underlyingmetal.

Nature of the metal oxide formed plays an important role in oxidationcorrosion.

1. A stable metal oxide layer tightly adheres to the metal surface is impermeable to the attacking oxygen.
   E.g.: the oxide film on Al, Sn, Pb, Cu, Cr, W etc.
2. An unstable oxide formed on the metal surface decomposes back into the metal and oxygen. In such cases oxidation corrosion isnot possible.
   ego: Ag, Au & Pt do not undergo oxidation process.
3. A volatile oxide layer formed during corrosion evaporates as soonas it is formed and leaving the metal for further attack. This leads to rapid and continuous corrosion leading to excessive corrosion to occur.
   e.g.: Molybdenum oxide (MoO_3) is volatile.
4. A porous layer of metal oxide contains pores, cracks etc. provideaccess to oxygen to reach the underlying the surface of the metal. **e.g.:** Alkali metals and alkaline earth metals (Li, K, Na, Mg) suchtype of oxide film.

b. Corrosion by other gases: The gases like SO_2, CO_2, Cl_2, H_2S and F_2 etc., also cause chemical corrosion and the extent of the corrosion mainly depends on the reactivity to the gas on the metal surface. The corrosion product is metal compound. The nature of the metal compound determines the corrosion.

c. Liquid metal corrosion: The chemical action of a flowing liquid metal at high temperature, on a solid metal or alloy produces liquid metal corrosion. The reason for this corrosion is the dissolution of the solid metal by liquid metal or internal penetration of the liquid metal into the solid phase, weakening the solid metal.

2.12.2. Mechanism of Wet Corrosion

Wet or electrochemical corrosion: Wet corrosion or electrochemical corrosion takes place under wet or moist conditions through the formation of short-circuited electrochemical cells. Wet corrosion is more common than dry corrosion.

a. Evolution of hydrogen: this type of corrosion casually occurs in acidic environments. The rusting of iron takes place in acidic environment in the following way.

At anode the dissolution (oxidation) of iron to ferrous ion with liberation of electrons takes place.

$$Fe \rightarrow Fe^{2+} + 2e^- \text{ (oxidation)}$$

The $2e^-$ flow from anodic area to cathodic area through the metal where H^+ ions (of acidic solution) are available and eliminated as hydrogen gas.

$$2H^+ + 2e^- \rightarrow H2 \uparrow \text{ (reduction)}$$

Thus Fe^{2+} ions are liberated at anode and H_2 gas is liberated at the cathode. The overall reaction in acidic medium is given by

$$Fe + 2H^+ \rightarrow Fe^{2+} + H_2$$

In hydrogen evolution type of corrosion, the anodes are usually large and the cathodes are small areas.

In neutral medium at the cathode formation of OH^- takes place along with the liberation of hydrogen.

$$2H_2O + 2e^- \rightarrow H_2 \uparrow 2OH^-$$

The Fe^{2+} and OH^- diffuse towards each other through the conducting medium forming the corrosion product rust in between cathodic and anodic areas.

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2 \downarrow \text{ (ferrous hydroxide)}$$

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \rightarrow 2Fe_2O_3.3H_2O$$

(Ferric hydroxide) (rust)

b. Absorption of oxygen: In presence of neutral, aqueous solution of electrolytes like (NaCl solution), in the presence of atmospheric oxygen and some cracks developed in iron oxide film causes this type of electrochemical corrosion. The surface of iron is always coated with a thin film iron oxide. If a crack is developed in this iron oxide film, the anodic areas are created on the surface, while the well-coated metal parts act as cathode. It follows that anodic areas are small and the cathodic areas are large.

The following reactions take place taking rusting of iron

At the anodic areas, the metal oxidizes as ferrous ions liberating two electrons which pass through metal to the cathodic area.

$$Fe \rightarrow Fe^{2+} + 2e^-$$

At the cathode the electrons were accepted by water and oxygen to generate OH^- ions by reduction.

$$O_2 + H_2O + e^- \rightarrow 4OH^-$$

The combination of Fe^{2+} and OH^- originated at anode and cathode respectively occurs at cathode, because the mobility of Fe^{2+} ions, is greater than OH- ions. Corrosion occurs at anode and the rust occurs in between cathodic and anodic areas.

The presence of oxygen accelerates the corrosion at anodic area and rust formation at cathodic area.

The Fe2+ ion at anode and which OH^- ions at cathode diffuse towards cathodic and anodic areas respectively and they form rust whenthey meet.

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2 \downarrow \text{(ferrous hydroxide)}$$

If enough oxygen is present ferrous hydroxide is easily oxidised to ferric hydroxide and then to hydrated ferric oxide which is nothing but rust.

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \rightarrow 2Fe_2O_3.3H_2O$$

(Ferric hydroxide) (rust)

2.13. Different types of Wet/ Electrochemical Corrosion

1. **Galvanic corrosion:** When two dissimilar metals are connected and jointly exposed to the corrosive atmosphere, the metal possessing higher oxidation potential or higher in the electrochemical series become anodic and undergoes corrosion.

   e.g.: When zinc and copper are connected and exposed to corroding environment or higher position in the electrochemical series, zinc under goes oxidation and corroded whereas copper undergoes reduction and protected.

2. **Differential aeration corrosion:** The different ways of creating aeration on the metal surface is listed below

   Partial immersion of metal in a solution
   The Zn rod immersed deep in NaCl --- anodic partThe Zn rod above NaCl
   --- cathodic part
   The circuit is completed by the migration of ions through the electrolyte. As a result, zinc rod starts corroding in the anodic area immersed in sodium chloride solution. This is also called waterline corrosion. Examples: Ship hulls, storage tanks.

3. **Stress corrosion:** Stress corrosion or stress cracking is the combined effect of static tensile stresses and the corrosive environment on a metal. Presence of tensile stresses is a main reason for stress corrosion. Majorly seen in fabricated articles of alloy like high zincbrasses and nickel brasses.

2.14. The Factors Influencing Corrosion

- **Nature of metal:** (1) Position in galvanic series, (2) Over voltage, (3) Relative areas of cathodic and anodic areas, (3) Purity of metal, (4) Physical state of metal, (5) Nature of surface film, (6) Passive character of metal, (8) Volatility of corrosion products, (9) Solubility of corrosionproducts.

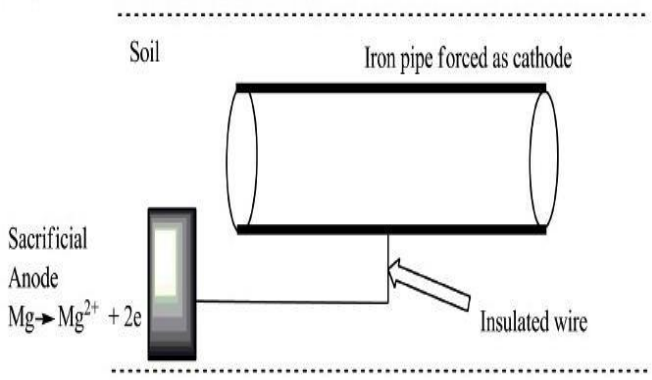
- **Nature of corroding environment:** (1) Temperature, (2) Humidity of air, (3) Presence of impurities

in atmosphere, (4) Nature of ions present,
(5) Conductance of corroding medium, (6) Amount of oxygen in atmosphere, (7) Velocity of ions which flow in the medium, (8) pH value of the medium.

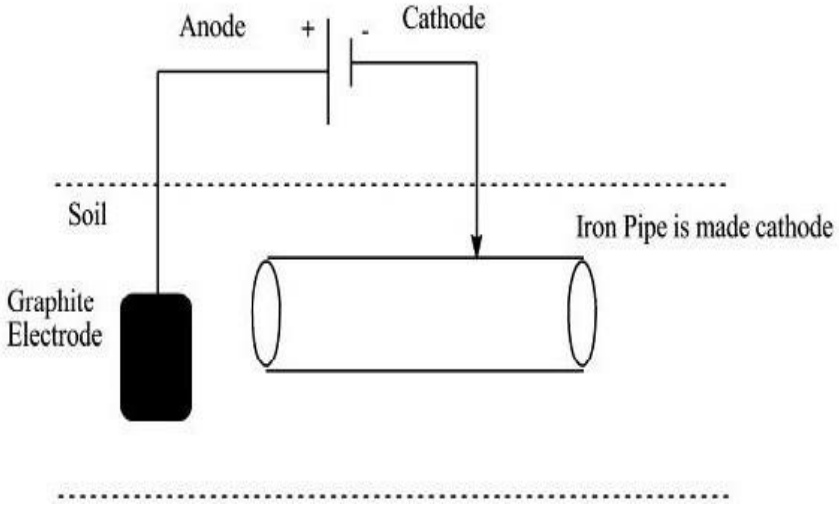
2.15. Cathodic Protection

Force the metal to be protected to behave as a cathode, thencorrosion will not take place.

a. **Sacrificial anodic protection or galvanic protection**: The metal structure to be protected is made cathode by connecting it to a more anodic material. As a result, corrosion takes place at anode and the metal structure is saved from corrosion. Metals like Al, Zn, Mg and their alloys are used as sacrificial anodes.



b. **Impressed current cathodic protection:** The metal structure to be protected is made cathode by connecting it to a negative terminal of a DC source. The positive terminal is connected to an insoluble anodelike graphite.



2.16. Protective Coating

It is a way to prohibit corrosion. It is continuous barrier between the coated surface and environment. The coating must be chemically inert to the environment and it should prevent diffusion of chemicals in the environment to the base metal. Sometimes coating will also increase mechanical and physical properties.

Two types of protective coating

i) Inorganic (Galvanizing, anodizing, chromate coating and tinning)
ii) inorganic (Paint, varnish etc.)

Inorganic Coating: It can be either metallic or non-metallic.

Metallic Coating: A metallic barrier on the surface of metal. It increases corrosion resistance, mechanical strength and change other physical properties. It also used for decoration.

2.16.1. Galvanization

Hot Dip Galvanizing: Coating Zinc on Iron or Steel (anodic coating) byhot dip process. It is done to prevent iron from rusting.

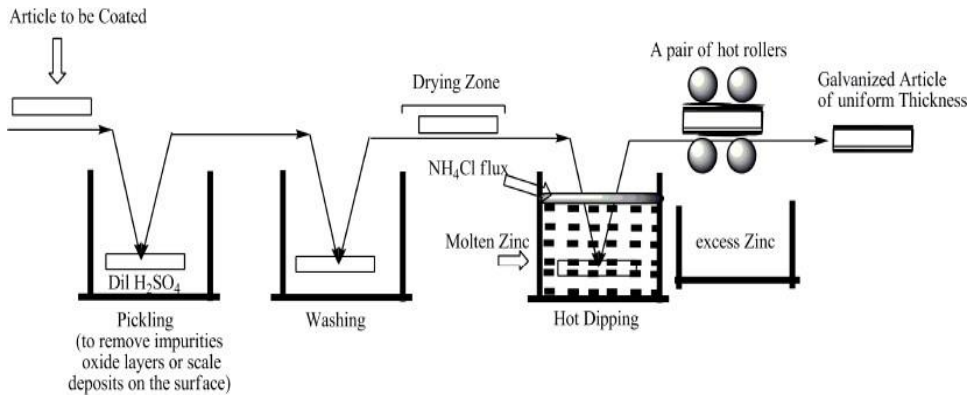
Process

Step 1: The Iron or steel is first cleansed by pickling with dil. H_2SO_4 solution. This treatment removes any impurities, scales or oxide layer formed in the base metal Iron.

Step 2: The article is then washed and dried.

Step 3: It is then dipped in a bath of molten Zinc. The surface of bath is kept covered with a flux of NH_4Cl to prevent oxidation.

Step 4: The dipped article is then passed through a pair of hot rollers to remove excess zinc and to get a coating of uniform thickness.



Roofing sheets, pipes, wires, nails, bolts, nuts etc. are galvanized to prevent rusting. Zinc gets dissolved in dil. Acids to form highly poisonous compounds. Hence galvanized utensils cannot be used for storing and preparation of food stuffs.

2.16.2. Tinning

Hot Dip Tinning

Coating Tin on Iron or Steel (cathodic coating) by hot dip process is known as tinning. It is done to prevent iron from rusting.

Step 1: The Iron or steel is first cleansed by pickling with dil. H_2SO_4 solution. This treatment removes any impurities, scales or oxide layer formed in the base metal Iron.

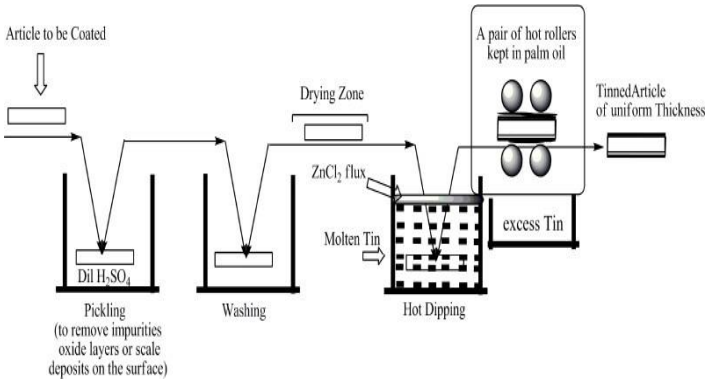
Step 2: The article is then washed and dried.

Step 3: It is then dipped in a bath of molten tin. The surface of bath is keptcovered with a flux of $ZnCl_2$ to prevent oxidation.

Step 4: The dipped article is then passed through a pair of hot rollers kept in palm oil to remove excess tin and to get a coating of uniform thickness. The palm oil prevents contact with oxygen.

Tinning is widely used for coating steel, copper, brass etc. Tin is non-toxic therefore tinned items

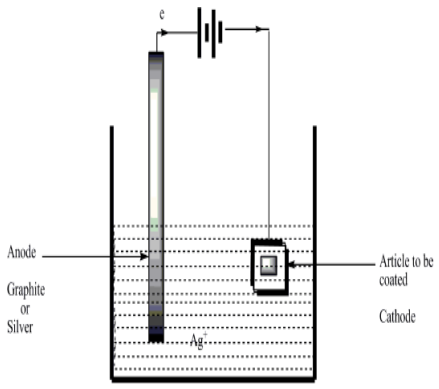
can be used to store food items, used to make cooking utensils, refrigeration equipment etc.



2.17. Electroplating or Electro-Deposition Process

Electroplating is a method of application of metallic coating. Coating is done by electrochemical process.

Theory of electroplating: The object to be plated is made cathode of electrolytic cell. The anode is either coating metal or an inert electrode likegraphite. The electrolyte is a soluble salt of coating metal. Example is copper plating.



The article to be electroplated is subjected to acid pickling with dil.HCl or dil.H_2SO_4 to remove any scales, oxides etc. Then it is subjectedto remove any scales, oxides etc. The cleaned article is made cathode of the electrolytic cell and the anode can be made of the pure coating metal or graphite. The electrodes were dipped in the salt solution of the coating metal, which acts as an electrolyte. When direct current is passed, the coating metal ions migrate to the cathode and deposit on the base metal article in the form of a thin layer. For brighter and smooth deposits, conditions like low temperatures, medium current density, and low metal ion concentration are used.

For example, for electroplating of copper on iron article, the following are maintained

1	Electrolytic bath solution	:	Copper sulphate
2	Temperature maintained	:	40-70^{0}C (low temperature for brighter and smooth surface)
3	Current density	:	20-30 mA/cm^2 (medium current density for high efficiency)

4	Anode	:	Copper pellets (99 % pure) taken in a titanium mesh basket or Graphite
5	cathode	:	Base material article

Application of electroplating

- Base metals like Al, Cu, Fe are electroplated with gold or silver toincrease their commercial and decorative value.
- In internal combustion engines electroplating with chromium increasesthe wear resistance.

2.18. Paints

Organic coatings are inert-barriers, applied as metallic surfaces andother constructional materials for both protection and decoration.

Factors determine the protective value of organic coatings:

1. Protects from corrosion environment.
2. Good surface adhesion.
3. Impermeability to salts, gases and water.

Paint: Paint is a mechanical dispersion mixture one or more pigments indrying oil.

Constituents of paints

1. **Pigment:** Pigment is a solid substituent and an essential constituent of paint. A good pigment must be opaque, chemically inert, nontoxic, freely mixable and cheap.

   The functions of pigment are: give strength, provide colour, give pleasant appearance, improves impermeability of paint from moisture and increases weather resistance.
   e.g.: White lead, zinc oxide (white), ferric oxide and chrome red (red colour), chromium oxide (green) and carbon black (black)

2. **Drying oil:** Drying oil is film forming constituent of paint. The functions of drying oils are: The main function of drying provides toughness, adhesion, durability and water proof.
   e.g.: dehydrated castor oil, soyabean oil and sunflower oil

3. **Thinners:** Function of thinners are reducing viscosity of paint, helps in dissolving additives, increase elasticity of the paint film and helps in drying of the paint.
   e.g.: Turpentine, benzene, naphtha, xylol and kerosene etc.

4. **Driers:** Driers are oxygen catalysts and accelerate the drying quality of the paint film.
   e.g.: Resinates, tungstates and naphtholates of Co, Mn, Pb and Zn.

5. **Fillers:** Fillers help to reduce the cost, increase durability of paint, reduce the cracking of dry paint film and act as carrier for pigment colour.
   e.g.: Barytes ($BaSO_4$), talc, asbestos, ground silica, gypsum and China clay

6. **Plasticizers:** The functions are to provide elasticity to the fil andminimize its cracking.
   e.g.: Tricresyl phosphate, triphenyl phosphate and tributyl phthalate.

7. **Anti skinning agents:** To prevent gelling and skinning of the paint film.
   e.g.: polyhydroxy phenols.

FAQ

1. What is electrochemical series? Mention its applications and uses.
2. Write a note on standard hydrogen electrode.
3. Explain standard calomel electrode.
4. Describe dry cell.
5. Write about Li – MnO_2 cell.
6. Write the function and working of Ni – Cd cell.
7. Explain working of $H_2 – O_2$ fuel cell.
8. How methanol – O_2 fuel cell working and explain with neat sketch.
9. Explain dry corrosion or chemical corrosion.
10. Write the mechanism of wet corrosion or electrochemical corrosion
11. Describe types of wet corrosion.
12. Write about galvanic cell corrosion.
13. Explain stress corrosion.
14. List out the factors influence on rate of corrosion.
15. Write a note on cathodic protection.
16. How does the galvanization process work?
17. Illustrate tinning process.
18. Explain electroplating.
19. Define paints. What are the constituents in paints and mention their functions.

Unit 3

Chemistry of materials

3.1. Introduction

The reason for giving a special introduction to this chapter is that this is a unique chapter where a few topics have been merged here without having any interconnection between the topics. Each and every topic is very unique. This chapter will go over the preparations, properties, and applications of carbon nano tubes and fullerenes. Following that, lubricants and their properties, as well as the mechanisms of lubricant function, were discussed, followed by refractories, and the discussion concluded with cement and the chemistry involved in its manufacturing and operation.

3.2. Carbon Nanotubes

3.2.1. *Introduction*

Sumio Iijima made the discovery of carbon nanotubes, which are long, thin carbon cylinders, in 1991. These are massive macromolecules that stand out due to their distinctive size, shape, and exceptional physical characteristics. They can be compared to a sheet of graphite, a hexagonal carbon lattice, that has been rolled into a cylindrical shape. In addition to having a single cylindrical wall (SWNTs), nanotubes can also have multiple walls (MWNTs), with one cylinder inside another.

3.2.2. *Key properties of Carbon Nanotubes: Carbon Nanotubes are an example of true nanotechnology*

- They are less than 100 nanometers in diameter and can be as thin as 1 or 2 nm. They are molecules that can be manipulated chemically and physically in very useful ways.
- They open an incredible range of applications in materials science, electronics, chemical processing, energy management, and many other fields. Some properties include
- Extraordinary electrical conductivity, heat conductivity, and mechanical properties.
- They are probably the best electron field-emitter known, largely due to their high length-to-diameter ratios
- As pure carbon polymers, they can be manipulated using the well-known and the tremendously rich chemistry of that element.

Some of the above properties provide opportunity to modify their structure, and to optimize their solubility and dispersion. These extraordinary characteristics give CNTs potential in numerous applications.

3.2.3. *Key application areas*:

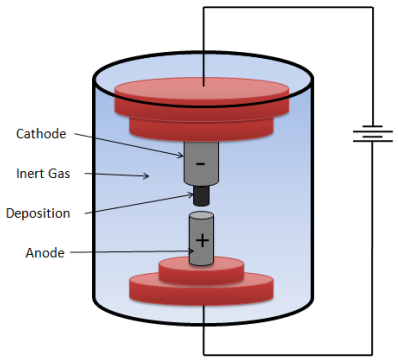
- Field Emitters/Emission:
- Conductive or reinforced plastics
- Molecular electronics: CNT based nonvolatile RAM • CNT based transistors

- Energy Storage
- CNT based fibers, fabrics and ceramics
- Biomedical applications etc. ...

3.2.4. Synthesis of CNT:

Arc Discharge Method:

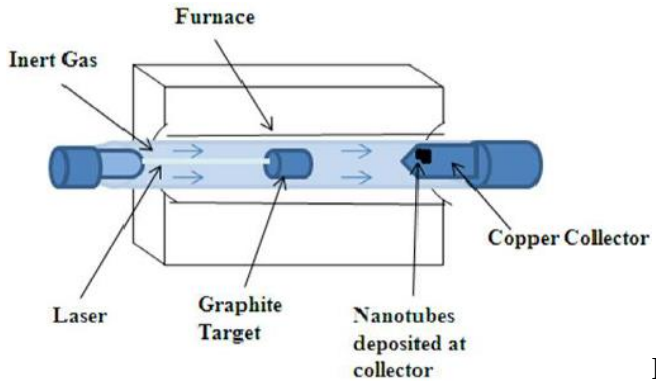
A chamber containing a graphite cathode and anode contains evaporated carbon molecules in a buffer gas such as helium. A small number of metal catalyst particles (such as cobalt, nickel, and/or iron) are also present in the chamber. The chamber is pressurized, heated, and subjected to DC current while being pressurized and heated to 4000K. During this process, roughly half of the evaporated carbon forms a "cylindrical hard deposit" on the cathode tip. The remaining carbon condenses into "cathode soot" on the cathode and "chamber soot" on the chamber walls. Carbon nanotubes with one or more walls can be produced from cathode and chamber soot. The cylinder-shaped hard deposit produces nothing particularly noteworthy. the chamber's pressure, the buffer gas selection, and the metallic catalyst that has been added. It appears that the metallic catalyst particle surfaces are where the nanotubes originate. The shape and number of walls are determined by these decisions. This technique has the advantage of producing a lot of nanotubes. The primary drawback, however, is the lack of sufficient control over the nanotubes' alignment, or chirality, which is essential to their characterization and function. Additionally, the products must be purified later because the reaction uses a metallic catalyst. Many techniques have been employed, including oxidation, centrifugation, filtration, and acid treatment.



Arc discharge method

Laser Ablation Method:

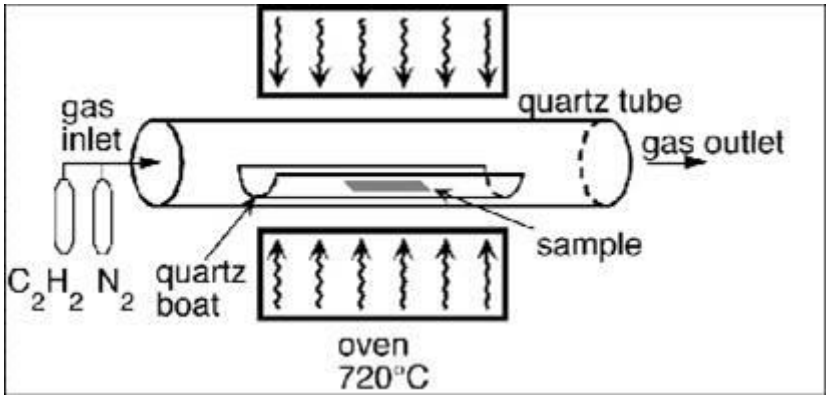
A quartz tube containing a block of graphite is heated in a furnace. A flow of argon gas is maintained throughout the reaction. A laser is used to vaporize the graphite within the quartz. The carbon vaporizes, is carried away by the argon, and condenses downstream on the cooler walls of the quartz. This condensation is SWNT and metallic particles. Thereafter, purification methods are applied to this mixture.



Laser ablation

Chemical Vapor Deposition (CVD):

The CVD method makes it more practical to incorporate CNTs into currently used processes for synthesizing electronics because it enables CNTs to grow on a variety of materials. A hydrocarbon on a substrate is chemically broken down in this process. It has already been demonstrated that one of the primary ways to grow carbon nanotubes is by exciting carbon atoms that are in contact with metallic catalyst particles in earlier techniques, such as the arc discharge method.



By embedding these metallic particles (iron in the case of the seminal paper) in precisely positioned holes in a substrate, the CVD method expands on this concept (silicon, in this case). Iron nanoparticles are essentially implanted at the bottom of tubes that are drilled into silicon. The hydrocarbon is then heated and broken down onto the substrate. The metal particles in the holes come into contact with the carbon, which then begins to form nanotubes that are "templated" from the tunnel's shape. It turns out that the carbon nanotubes grow in the tunnel's angle very long and very well aligned.

This method has the advantages of a very high yield, consistent nanotube alignment (important for producing specific types of nanotubes, such as metallic or semiconductor ones), and theoretically arbitrary growth area size. The main drawback is that large sized areas (several millimeters) have a propensity to crack, shrink, and otherwise warp, despite the fact that the size of the growth area is essentially arbitrary. To avoid this, the substrates must be dried very thoroughly.

3.3. Fullerenes

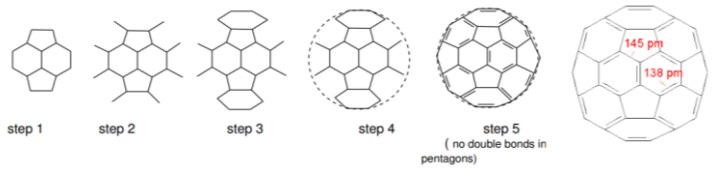
Fullerenes are a family of molecules entirely composed carbon in the form of hollow sphere, ellipsoid, tube or plane. They are of following types.

a) Spherical fullerenes: They look like a soccer ball and called as bucky balls

b) cylindrical fullerenes: These are called as carbon nanotubes or bucky tubes.

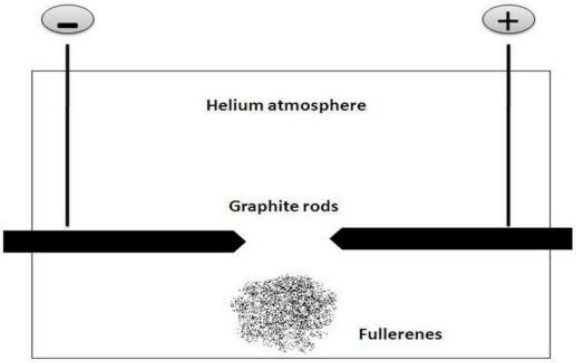
c) Planar fullerenes: Graphene is an example of planar fullerene sheet.

How to Draw a fullerene:



Preparation by electric arc heating of Graphite:

Kratschmer's methodology for preparation of fullerenes is the one widely used currently which has been modified by various workers to increase yields. It was Kroto again who separated C_{60} and C_{70} for the first time in pure forms. Graphite electrodes are evaporated in an atmosphere of ~ 100 torr of Helium (Kratschmer) or 50 -100 torr of Argon (Kroto) in a glass vessel (modified RB flask, Power from a transformer). The soot formed is scrapped and dispersed in benzene whereupon a wine-red solution is obtained. This is filtered from the insoluble solids and concentrated. This mixture of C60 and C70 is then run on an alumina column using hexane as eluant. The magenta colored C60 elutes out first followed by the port wine colored C70. In a typical case the ratio of C60 to C70 will be 5:1. Solid pure C60 will be mustard colored [UV 596, 604, 625nm] and C70 will be red [600, 617, 644 nm]. (Laser vaporization of graphite and graphite doped with other elements and compounds such as lanthanide metals, boron nitride has also been used for synthesis of fullerenes).



Structure: Fullerenes are a class of closed – cage carbon molecules. C_n which generally contain 12 pentagons and variable number of hexagons.

C_{60} is fullerenes are stable with SP^2 hybridised carbon atoms. Each carbon is trigonally linked to its three neighboring carbons by two single bonds and one double bond.

C_{60} is a buckyball with 60 carbon atoms arranges in interlocking hexagons and pentagons, like the patches on a soccer ball. It is a spherical cage structure. (C_{60} – 12 pentagons and 20 hexagons)

C_{70} is elongated like a rugby ball. The smallest number of fullerene family is C_{20} and highest is C_{600}.

Properties:

1. They are very tough.

2. They have high tensile strength.

3. Highest packing density.

4. They are thermally stable up to 600^0C.

Engineering applications:

1. They can be used as organic photovoltaics and alkali metal fullerenes are superconductors.

2. Their spherical structures makes them suitable as lubricants.

3. The water-soluble derivatives of fullerenes inhibit HIV – 1, which leads to AIDS>

4. They exhibit amazing conducting, magnetic optical and mechanical properties.

5. Fullerenes are related compounds have the potential as catalysts.

6. They are used in cosmetics for skin protection.

3.4. Refractories

Refractories are inorganic materials that can withstand high temperatures without deforming and can withstand the abrasive and corrosive action of molten metals, slags, and glasses.

Refractories are divided into the following groups based on their chemical characteristics, constituent materials, and fusion temperature:

Type of refractory	**Property**	**Examples**
Normal refractory	Fusion temperature = 1580 – 1780^0C	Fire clay
High refractory	Fusion temperature = 1780 – 2000^0C	Chromite
Super refractory	Fusion temperature = > 2000^0C	Zircon
Acidic refractory	Consist of acidic materials. Resistant to acidic slags and are readily attacked by basic slags	Alumina, Silica and fire clay ($Al_2O_3.2SiO_2.2H_2O$)
Basic refractory	Consist of basic materials. Resistant to acidic slags and are readily attacked by acidic slags	Magnetite and dolomite bricks
Neutral refractory	Made from weakly acidic or basic materials. These are showing resistance to the action of acidic and basic materials and also show good	Graphite, Zirconia and SiC (carborundum)

	chemical stability.	
Oxide refractory	Single oxide refractory	Alumina, Magnesia and Zirconia refractory
	Mixed oxide refractory	Spinel and mullite
	Non – Oxide refractory	Borides, Carbides and Silicates

Characteristics of refractory materials:

1. They should possess high temperature resistance under working conditions.
2. They should possess good abrasion resistance by dusty gases and molten metals.
3. They should possess low permeability or ability to contain heat.
4. They should possess high mechanical strength and crack resistance to withstand overlying load.
5. They should possess thermal strength to withstand thermal shock due to rapid and repeated temperature fluctuations.
6. They should possess high resistance to change in physical, chemical and mechanical properties.

Reasons leading to failure of refractory materials:

1. Using a refractory of less refractoriness than that of the operating materials.
2. Using refractories which cannot withstand the load of raw materials and products.
3. Rapid changes in temperature of the furnace.
4. Using bricks of higher thermal expansion.
5. Using refractory bricks which are not properly fired.
6. Using bricks when undergo considerable volume changes during their use at high temperatures.
7. Using acidic/basic refractory in a furnace in which basic/acidic reactants and / or products are benign processed.
8. Using heavy – weight refractory bricks.

Applications of refractories:

1. Refractories are extensively used for construction of furnaces, kilns, chimney stacks, checkers, and boiler settings.
2. The lining of Portland cement rotary kilns soaking pits and furnace hearths are made from refractory materials.
3. The construction of crown and arches of furnaces, combustion zones and melting furnaces are made from refractory materials.
4. The coke oven walls, Cowper stove domes, roots of electric furnaces and glass retort settings are made from refractory materials.
5. The pyrometer sheaths and crucibles, refining furnaces, and hot zones of cement rotary kilns are made from refractories.

3.4.1. Properties of refractories:

1. Refractoriness: Refractoriness is the ability of a material to withstand high temperature without appreciable deformation or softening. It is generally measured by the softening or fusion temperature of the material. A refractory material should have a softening temperature higher than the operating temperature of the furnace.

2. Strength/ refractoriness under load: Refractories used inside the furnace have to withstand high pressure and the load of the charge inside the furnace. For example, fireclay refractories soften over a range of temperature but under conditions of heavy load, they soften far below their true fusion temperatures as determined by the Seger cone test.

3. Dimensional stability – Thermal expansion and contraction:

It is the ability of a refractory to resist any permanent change in volume when it is exposed to high temperatures for a prolonged period of time.

There are two types of dimensional changes.

Permanent expansion: Permanent expansion occurs when one crystalline form of refractory is transformed into another crystalline form of low density.

Quartz → tridymite → cristobalite

Permanent contraction: When a refractory is subjected to high temperature for a long duration, permanent contraction occurs due to following reasons. Melting of low fusible constituents forming liquid which fills the pores of the refractory body causing shrinkage and vitrification: for example, fireclay bricks. Transformation of one crystalline form into another denser form: for example, Magnesite brick.

Amorphous MgO → Periclase

4. Porosity: The proportion of pore volume to bulk volume determines how porous a refractory material is. Every refractory material has pores. High porosity refractory materials are not advised for furnace lining, though, as molten charge, gases, and slag can seep through the pores and shorten the refractory's lifespan. High porosity refractory is weaker, less resistant to abrasion, and more susceptible to slag corrosion. In general, a good refractory has little porosity.

5. Thermal spalling: It is when a refractory material fractures, cracks, peels off, or breaks as a result of temperature changes. Rapid temperature changes that result in uneven expansion and contraction that cause internal stress or strains are what cause spalling. By avoiding abrupt temperature changes, using bricks with low thermal coefficients of expansion, good thermal conductivity, and relatively high porosity within acceptable bounds, as well as by modifying the furnace design to prevent stress from building up during heating, thermal spalling can be reduced.

6. Permeability: It is a measure of the rate of diffusion of molten solids, liquids and gases through the interconnected pores of a refractory. Permeability increases with the rise in temperature since viscosity of the molten metal decreases.

7. Chemical inertness: The refractory material should not react with reactants, slag, furnace gases, fuel, ashes and products used inside the furnace. Hence, if the furnace process acidic material, then a basic refractory should not be employed and vice versa.

8. Thermal conductivity: In industrial operations, refractories of both low and high thermal conductivities are required, depending upon the type of furnace. Generally, furnaces are lined with refractories of low thermal conductivity to reduce heat losses by radiation; otherwise, it would be difficult to maintain high temperature inside the furnace. However, in some cases, refractories of high thermal conductivity are required, for example, coke oven walls, muffle furnace and retorts which are heated from outside.

9. Resistance to abrasion and erosion: Abrasion of a refractory material leads to erosion. Slags, molten gases, molten metals, etc., can also cause the corrosion and erosion of refractory lining due to the chemical reaction between the slag and the refractory lining. Thus, a refractory material should have erosion and corrosion resistance.

10. Electrical conductivity: Generally, all refractories, except graphite, are poor conductors of electricity. The refractories used for the lining of an electric furnace should have low thermal conductivity. While selecting a refractory, it should also be remembered that electrical conductivities of these materials increase with the rise in temperature.

11. Heat capacity: Heat capacity of a furnace depends upon thermal conductivity, specific heat and specific gravity of the refractory. Light weight refractory bricks have low heat capacity and are suitable for intermittently operated furnaces because the working temperature of the furnace can be achieved in lesser time with lesser consumption of fuel.

3.5. Lubricants

Lubricant may be defined as a substance which reduces the friction when introduced between two surfaces and the phenomenon is known as lubrication.

3.5.1*Classification of lubricants:*

Lubricating oils (or liquid lubricants): Animal oils (Tallow oil, whale oil); Vegetable oils (Castor oil, Palm oil); Petroleum oils (petroleum fractions); Synthetic lubricants (polyglycol, silicones)

Greases (Semi – solid lubricants): Emulsifying oil and fat with thickening agents like sodium, calcium and aluminum

Solid lubricants: Graphite, Molybdenum disulphide (MoS_2) and Boron nitride (BN_x)

3.5.2. *Properties of lubricants:*

1. Viscosity: A fluid's viscosity is a characteristic that determines how much flow resistance it has. It serves as a gauge for lubricating oil flow capacity. The ability to flow will be greater the lower the viscosity. The lubricating oil's viscosity decreases as temperature rises, and as pressure rises, lubricating oil viscosity rises. In a summary, good lubricating oil is defined as having a viscosity that does not change with temperature.

2. Flash point: The lowest temperature at which vapors of a volatile substance will momentarily ignite when an ignition source is brought close to it is known as the flash point of the substance. The flash point of the lubricating oil should be somewhat higher than its working temperature.
3. Fire point: The lowest temperature at which the vapor of a fuel will continue to burn when brought close to an ignition source for at least five seconds is known as the fire point of that fuel. Fire point is roughly $10^{o}C$ greater than flash point.
4. Cloud point: The cloud point is the temperature at which lubricating oil starts to appear cloudy.
5. Pour point: Pour point is the lowest temperature at which lubricant oil turns semi-solid and stops flowing. It shows whether lubricants used in cold conditions are suitable. Low pour points are characteristics of good lubricants.
6. Viscosity: A fluid's viscosity is a characteristic that governs how much flow resistance it has. It serves as a gauge for lubricating oil flow capacity. The ability to flow will be greater the lower the viscosity. If the temperature rises, the lubricating oil's viscosity decreases while its viscosity rises under pressure. In a nutshell, good lubricating oil is defined as having a viscosity that does not change with temperature.
7. Viscosity Index: Viscosity index is the measure of how a liquid's viscosity changes as a function of temperature. High viscosity indexes show a relatively small change in viscosity with temperature, whereas low viscosity indexes show a relatively large change in viscosity with temperature.
8. Oiliness: This characteristic makes it possible for oil to cover and stick to the bearing's surface. It plays a crucial role in boundary lubrication. The ability of a lubricant to stick to the surface of machine parts while being subjected to high pressure and load is one of its key properties.
9. Carbon residue: Lubricating oils have a high combined percentage of carbon. They decompose when heated, leaving behind a certain amount of carbon. The least amount of carbon should be deposited when using a good lubricant. The Conradson method is typically used to estimate carbon residue.
10. Volatility: It shows the evaporation behavior of lubricant at high temperature. A good lubricant should have low volatility.
11. Neutralization number: a sign of an oil's basicity (alkalinity) or acidity. The amount of hydrochloric acid or potassium hydroxide required to neutralize one gramme of oil of any acid or base is known as the neutralization number. Potassium hydroxide is used as a neutralizing agent in the acid number (AN) test and hydrochloric acid is used in the base number (BN) test.
12. Saponification number: The number of milligrams of KOH needed to saponify 1 g of oil is known as the saponification number. Vegetable and animal oils saponify, whereas mineral oils do not. As a result, this test enables us to determine whether the oil in question is mineral, a compounded oil made up of mineral and vegetable oils, or animal and vegetable oil.
13. Aniline point: The lowest temperature at which an equilibrium solution exists is known as the aniline point of oil. for a sample of oil and aniline of equal volume. An aniline point indicates the potential

for oil to degrade when it comes into contact with rubber pickings, sealings, etc. Natural rubber and some varieties of synthetic rubber are susceptible to dissolution by aromatic hydrocarbons. As a result, it is preferable for lubricants to have little aromatic content. Higher paraffinic hydrocarbon content and, consequently, lower aromatic hydrocarbon content are indicators of higher aniline-point.

3.5.3. *Conradson method for determination of carbon residue:*

An accurately measured amount of oil is added to a silica crucible (about 65-85 ml capacity). A small tube-shaped opening in the lid of the Skidmore crucible allows volatile material to escape. The mixture is then placed in a wrought iron crucible that is about 8 cm in diameter and 6 mm high and has an iron hood in the shape of a chimney on top (of about 10 cm diameter). For 10 minutes, the wrought iron crucible is slowly heated until flame appears. Continued slow heating lasts for a further five minutes. Finally, intense heating is applied for about 15 minutes, or until all volatile matter's vapors have been completely burned. After allowing the apparatus to cool, the weight of the remaining residue is calculated. The outcome is displayed as a percentage of the starting weight.

3.5.4. *Determination of aniline point:*

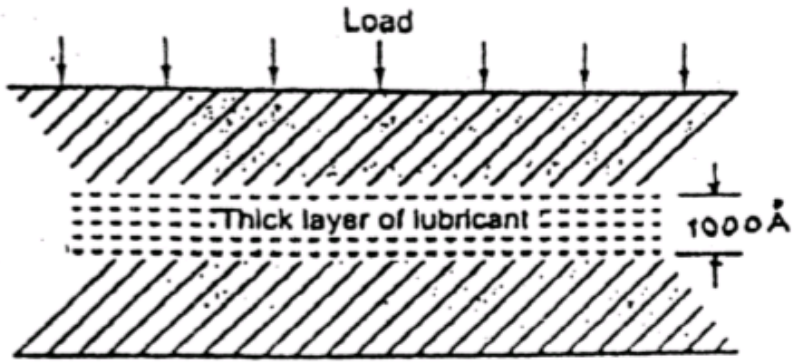
In a test tube, mechanically equal volumes of the oil sample and aniline are combined to determine the aniline point. Heat is applied to the mixture until a homogeneous solution is achieved. The tube is then given time to cool down at a controlled rate. At the aniline point, the temperature at which the two phases (oil and aniline) separate is noted.

3.5.5. *Mechanism of Lubrication:*

The phenomenon of lubrication can be explained with the help of the following mechanism; (a) **Thick-Film lubrication** (Fluid-Film or hydrodynamic lubrication) (b) **Thin Film lubrication** (Boundary lubrication) and (c) **Extreme Pressure lubrication**

(a) Thick-Film lubrication:

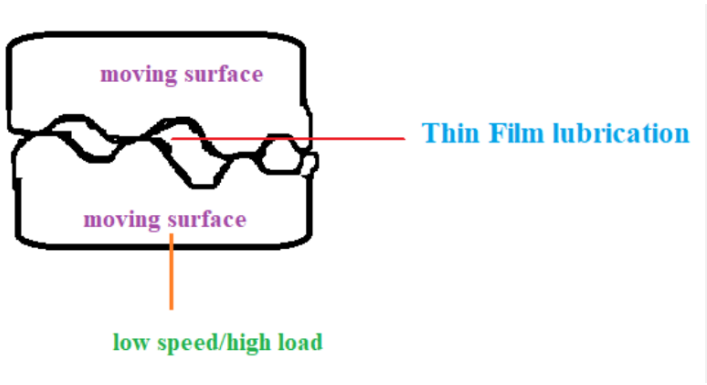
This prevents direct surface to surface contact and the welding of junctions by keeping moving/sliding surfaces separated from one another by a thick fluid film that is at least 1000 A° thick. In order to prevent direct contact between the material surfaces, the lubricant film forms a thick layer between moving or sliding surfaces and fills in any irregularities on those surfaces. Friction is consequently decreased. In order to reduce internal resistance between the lubricant particles under working conditions, the lubricant should have the least amount of viscosity possible while still holding in place and separating the surfaces.



For thick-film lubrication, hydrocarbon oils—mineral oils that have lower molecular weight hydrocarbons with 12 to 50 carbon atoms—are regarded as suitable lubricants. Select long chain polymers are blended with regular hydrocarbon lubricants to maintain the viscosity of the oil throughout the year.

(b) Thin Film lubrication:

Where a continuous film of lubricant cannot last, this type of lubrication is preferred. In these circumstances, a substance that can be adsorbate onto both metallic surfaces by physical or chemical forces is used to lubricate the clearance space between the moving or sliding surfaces. The metal surfaces are kept apart from one another by the adsorbed film, at least up to the height of the peaks on the surface.



(c) Extreme Pressure lubrication:

A high local temperature is reached when the moving/sliding surfaces are subjected to extremely high pressure and speed. Under these circumstances, liquid lubricants are unable to adhere and may even break down and vaporize. Mineral oils are given special additives to meet these high-pressure conditions. Extreme pressure additives are what these are. On metal surfaces, these additives create stronger films that can withstand extremely high loads and high temperatures. Organic compounds with active radicals or groups, such as chlorine (found in chlorinated esters), Sulphur (found in sulphurized oils), or phosphorus, are important additives (as in tricresyl phosphate). At the current high temperatures, these compounds react with metallic surfaces to produce metallic chlorides, sulphides, or phosphides.

3.5.6. *Characteristics of good lubricants:*

1. The lubricants should keep moving parts apart.
2. The foremost function of a lubricant is to reduce friction.

3. It should transfer heat and act as a coolant.

4. It should reduce the wear and tear as well as surface deformation, caused due to rubbing action of a two sliding surfaces.

5. It prevents rust and corrosion and there by reduces the maintenance and running cost of the machines.

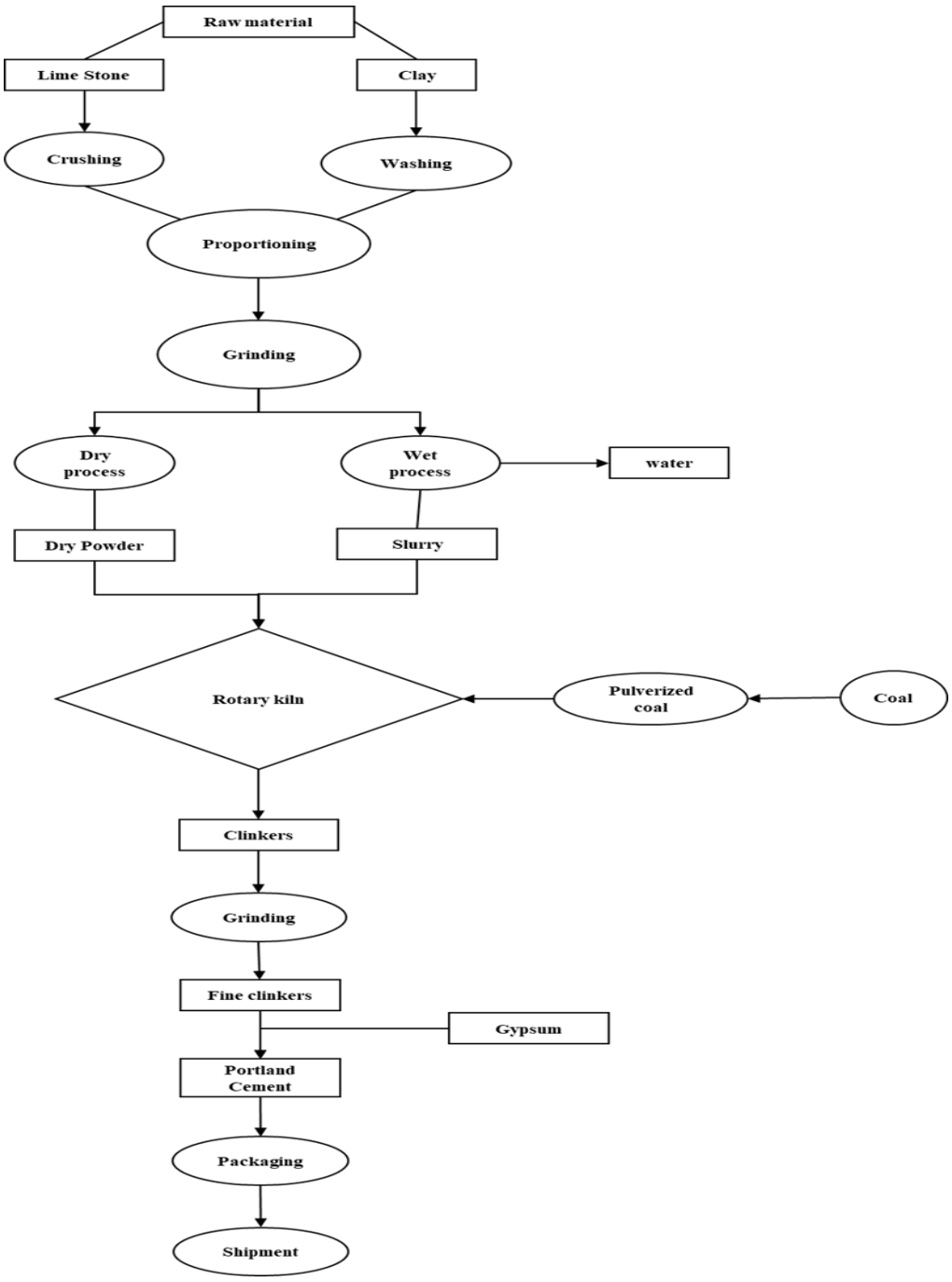
6. It carries away contaminants and debris which would otherwise damage the surfaces of the machinery.

7. It acts as a seal.

8. It should also reduce the loss of energy in the form of heat.

3.6. Cement

Manufacturing process of cement a flow chart



3.6.1. Composition of Portland cement:

Calcium oxide or lime (CaO) = 60 - 70 %,

Silica (SiO_2) = 20 - 24 %

Alumina (Al_2O_3) = 5 -7 %

Magnesia (MgO) = 2 - 3 %

Ferric oxide (Fe_2O_3) = 1 - 2.5%

Sulphur trioxide (SO_3) = 1 - 1.5 %

Sodium oxide (Na_2O) = 1%

Potassium oxide (K_2O) = 1%

3.6.2. Manufacture of cement

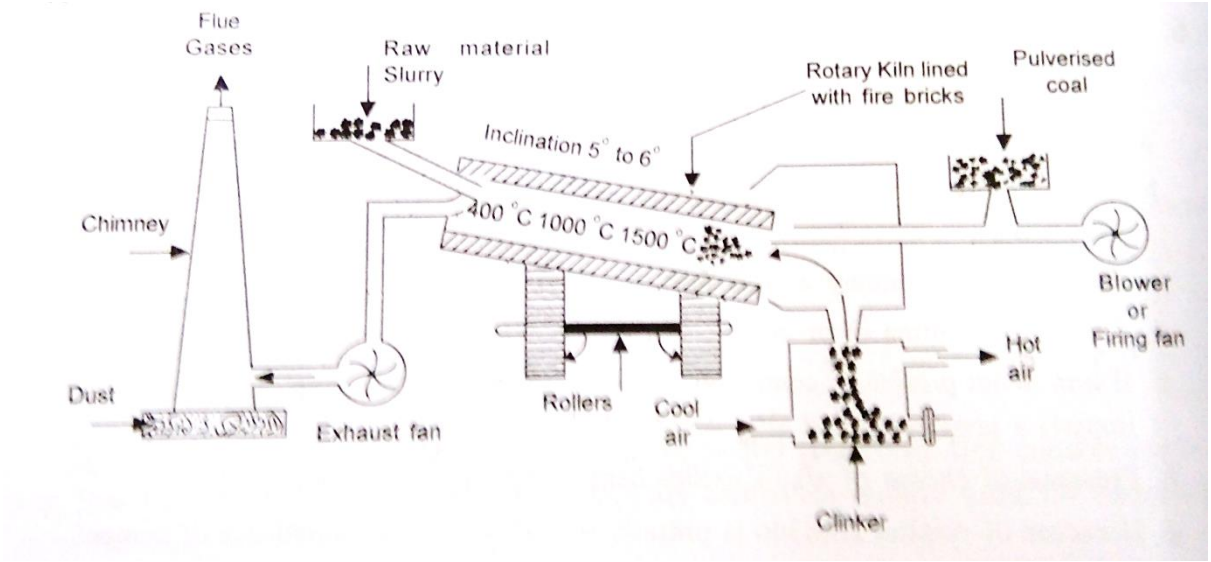
It involves the following steps

1. Mixing of raw material 2. Burning, 3. Grinding 4. Storage and packaging

1. Mixing of raw material

Mixing can be done by any one of the following two processes (a) Dry process (b) Wet process

a) Dry process: Lime stone or chalk and clay are crushed into gyratory crusher to get 2-5 cm size pieces. Crushed material is ground to get fine particle into ball mill or tube mill. Each material after screening stored in a separate hopper. The powder is mixed in require proportions to get dry raw mix which is stored in silos (storage tank) and kept ready to be fed into the rotary kiln. Raw materials are mixed in required proportions so that average composition of the final product is maintained properly.



b) Wet process: Raw materials are crushed, powdered and stored in silos. The clay is washed with water in wash mills to remove adhering organic matter. The washed clay is stored separately. Powdered lime stone and wet clay are allowed to flow in channel and transfer to grinding mills where they are intimately mixed and paste is formed known as slurry. Grinding may be done either in ball mill or tube mill or both. Then slurry is led to correcting basin where chemical composition may be adjusted. The slurry contains 38-40% water stored in storage tank and kept ready for feeding to a rotary kiln. The remaining two operations burning and grinding are same for both the process.

2. Burning

Burning is carried out in rotary kiln which rotating at 1-2 rpm at its longitudinal axis. Rotary kiln is steel tubes having diameter in between 2.5-3.0meter and length varies from 90-120meter. The inner side of kiln is lined with refractory bricks. The kiln is rested on roller bearing and supported columns of masonry or concrete in slightly inclined position at gradient of 1 in 25 to 1 in 30. The raw mix or corrected slurry is injected into the kiln from its upper end. Burning fuel like powdered coal or oil or hot gases are forced through the lower end of the kiln so long hot flame is produced. Due to inclined position and slow rotation of the kiln, the material charged from upper end is moving towards lower end (hottest zone) at a speed of 15meter/hour.

As gradually descends the temperature is rises. In the upper part, water or moisture in the material is evaporated at 400^0C temperature, so it is known as drying zone. In this zone most of the water in the slurry get evaporated because of the hot gases. The clay is broken as Al_2O_3, SiO_2 and Fe_2O_3 i.e.,

Al_2O_3. 2SiO2.Fe_2O_3.2H_2O → Al_2O_3 + 2SiO_2 + Fe_2O_3 + 2H_2O

In the central part (calcination zone), temperature is around 1000^0C, where decomposition of lime stone takes place. After escapes of CO_2, the remaining material in the forms small lumps called nodules.

$CaCO_3$ → CaO + CO_2

The lower part (clinkering zone) has temperature in between 1400-1700^0C where lime and clay are reacts to yielding calcium aluminates and calcium silicates. These aluminates and silicates of calcium fuse to gather to form small and hard stones are known as clinkers. The size of the clinker is varying from 5-10mm.

2CaO + SiO_2 →2 CaO. SiO_2(dicalcium silicate (C_2S))

3CaO + SiO2 → 3 CaO. SiO_2. (Tricalcium silicate (C_3S))

3CaO + Al_2O_3 →3 CaO. Al_2O_3 (dicalcium aluminate (C_2A))

4CaO + Al_2O_3 + Fe2O3 → 4CaO. Al_2O_3.Fe_2O_3 (tetra calcium alumina ferrite(C_4AF))

As clinkers are coming from burning zone, they are very hot. The clinkers are cooled down by air admitting counter current direction at the base of rotary kiln. Resulting hot air is used for burning powdered coal or oil and cooled clinkers are collected in small trolleys or in small rotary kiln.

3. Grinding

Cooled clinkers are ground to fine powder in ball mill or tube mill. 2-3% powdered gypsum is added as retarding agent during final grinding. So that, resulting cement does not settle quickly, when comes in contact with water. After initial set, cement - water paste becomes stiff, but gypsum retards the dissolution of tri-calcium aluminates by forming tricalcium sulfoaluminate which is insoluble and prevents too early further reactions of setting and hardening.

4. Storage and packaging

The ground cement is stored in silos, from which it is marketed either in container load or 50kg bags.

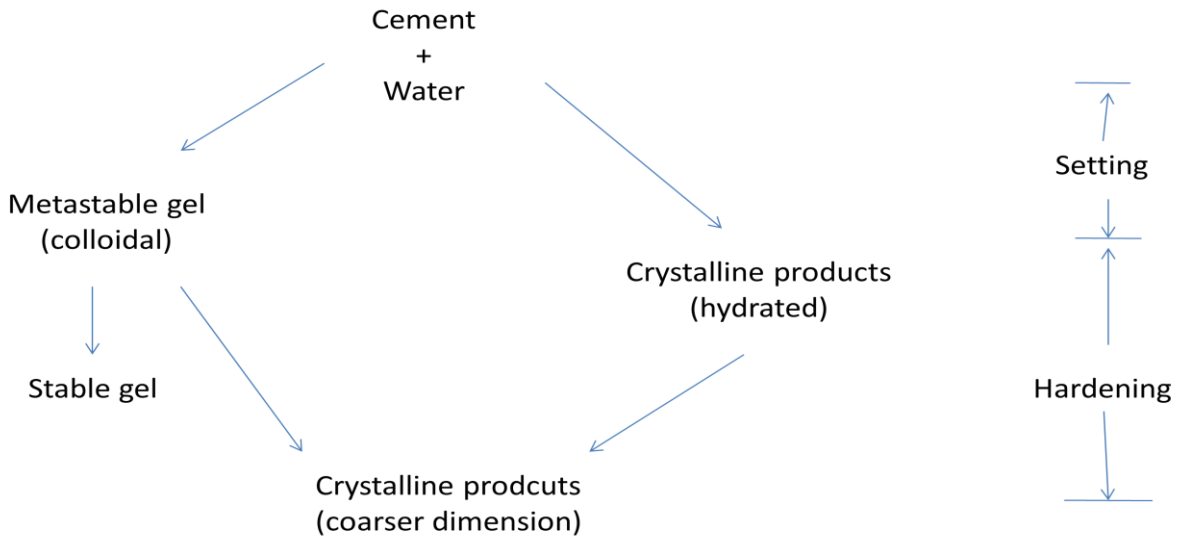
3.6.3. Setting and hardening of cement:

Portland cement on mixing with water, changes to a plastic mass called cement paste. Which slowly loses its plasticity and becomes a stiff and ultimately a rocky mass is obtained. This process is known as setting.

After hydration, anhydrate compounds become hydrated, which have less solubility. Hence, they are precipitated as insoluble gels or crystals.

These have the ability to surround sand, crushed stones, other inert materials and bind them very strongly.

Flow chart of setting and hardening of cement:



Hardening of cement can be explained on the basis of two theories:

a) Crystalline theory (given by Le-Chatelier): According to this theory, constitutional compounds after hydration form crystalline products. These crystalline products undergo interlocking which is responsible for hardening of cement.
b) Colloidal theory (given by Michaelis): According to this theory, during hydration silicate gels are formed which undergo hardening and are responsible for the hardening of cement.
c) Setting and hardening of cement is due to the formation of interlocking crystals reinforced by the rigid gels Portland cements exhibit initial set in about 3 hours and final set in the about 7 hours.

Setting can be tested by a standard needle. Need le does not penetrate at all, the cement is said to have reached final setting stage.

Reactions involved in setting and hardening of cement:

The basic chemicals compounds in the Portland cement are:

Name	**Chemical formula**	**Abbreviation**
Tricalcium silicate	$3CaO.SiO_2$	C_3S
Dicalcium silicate	$2CaO.SiO_2$	C_2S
Tricalcium silicate	$3CaO.Al_2O_3$	C_3A
Tetra calcium alumino ferrite	$4CaO.Al_2O_3.Fe_2O_3$	C_4AF

When cement is mixed with water, the paste becomes quite rigid within a short time which is known as initial set of flash set. This is due to C_3A which hydrates rapidly as follows:

$3CaO.Al_2O_3 + 6H_2O \rightarrow 3CaO.Al_2O_3.6H_2O$

(Crystals)

These crystals prevent the hydration reactions of other constitutional compounds forming barrier over them. In order to retard this flash set, gypsum is added during the pulverization of cement clinkers. Gypsum retards the dissolution of C_3A by interacting with it forming insoluble complex of sulfoaluminate which does not have quick hydrating property.

$3CaO.Al_2O_3.Fe_2O_3 + X\ H_2O + Y\ CaSO_4.2H_2O \rightarrow 3\ CaO.Al_2O_3.Y\ CaSO_4.ZH_2O + CaO.Fe_2O_3 + H_2O$

The tertacalcium alumino ferrite (C_4AF) then reacts with water forming both gels and crystalline compounds as follows:

$4CaO.Al_2FeO_3 + 7\ H_2O \rightarrow 3CaO.Al_2O_3.6H_2O + CaO.Fe_2O_3.H_2O$

Crystals gels

These gels shrink with passage of time and leave some capillaries for the water to come in contact with C_3S and C_2S to undergo further hydration and hydrolysis reactions enabling the development of greater strength over a length of time.

Final setting and hardening of cement paste is due to the formation of tobemonite gel plus crystallization of calcium hydroxide and hydrated tricalcium aluminate.

$2CaO.SiO_2 + X\ H2O \rightarrow 2CaO.SiO_2.\ X\ H2O$

Gels

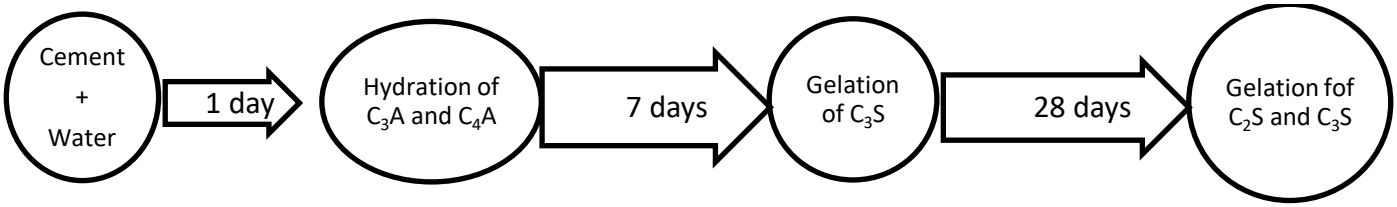
$2(3CaO.SiO2) + 6\ H2O \rightarrow 3Cao.SiO2.3H2O + 3Ca(OH)_2$

Tobermonite gel

During the setting and hardening of Portland cement, some amount of heat is liberated due to hydration and hydrolysis reactions. It is essential to dissipate the heat generated during hydration as quickly as possible, to prevent shrinkage cracks on setting and hardening.

Steps during setting and hardening of cement:

When water is added to cement, at first hydration of tricalcium aluminate (C_3A) and tetra calcium aliminoferrrite (C_4AF) takes place.



Hydration of tricalcium silicate (C_3S) begins within 24 hours and gets completed in 7 days. The gel of aluminate begins to crystallize and at the same time, dicalcium silicate (C_2S) begins to crystallize and at the same time, dicalcium silicate (C_2S) begins to hydrate in 7 to 28 days.

The initial setting of cement is due to hydration of aluminate. Hydration of tricalcium silicate and further hydration of aluminate leads to the development of early strength 1 to 7 days. Further hydration of dicalcium silicate and continued hydration of tricalcium silicate between 7 to 28 days, is responsible for the increase of strength of cement.

3.6.4. Deterioration of cement:

Deterioration of cement concrete takes place due to attack of salty water and other acidic conditions. The rate of attack will be minimized by the impermeable surface. On long exposure to salty water and acidic environment, the structure can be weakened. The rate of attack increases with decreasing pH. The acidic environment is mainly due to chemical action of dissolved CO_2 and other organic and inorganic acids. Due to action of CO2 leaching out of free lime from cement concrete takes place.

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$

$Ca(OH)_2 + Ca(HCO_3)_2 \rightarrow 2CaCO_3 + 2H_2O$

Initially insoluble $CaCO_3$ is formed, which on further reaction with CO_2 and H_2O. The cycle is repeated causing deterioration of cement concrete. Hydrolysis of silicates and aluminates also cause deterioration of cement.

Ex: concrete pipes carrying sewage.

Prevention of deterioration:

By making use of a surface coating with epoxy resin paint deterioration is minimized by the fact that coating makes the concrete surface impermeable to acidic environment. To prevent the hydrolysis of silicates coating makes the concrete surface impermeable to acidic environment. To prevent the hydrolysis of silicates coating the surface with SiF_4 which forms CaF_2 which is insoluble resistant to hydrolysis is carried out.

3.6.5. Parameters to characterize the clinker formation

When describing the properties of clinker, chemical parameters based on the oxide composition are extremely helpful. The following variables—which correspond to weight percentages in chemical formulas—are frequently used:

Lime Saturation Factor (LSF):

The ratio of CaO to the other three major oxides is known as the lime saturation factor. It is calculated as follows when used with clinker:

$LSF = CaO / (2.8SiO_2 + 1.2Al_2O_3 + 0.65Fe_2O_3)$

This is frequently described as a percentage; in which case it is multiplied by 100.

- The LSF regulates the clinker's alite to belite ratio. In comparison to clinkers with low LSF, those with higher LSF will have a higher ratio of alite to belite.
- Modern clinkers typically have LSF values of 0.92-0.98, or 92%-98%. Values greater than 1.0 suggest that the clinker is likely to contain free lime.

- This is due to the fact that, in theory, all of the free lime should have combined with belite to form alite at LSF=1.0. If the LSF is greater than 1.0, the excess free lime will remain free lime because it has nothing to combine with.
- In reality, there are always areas within the clinker where the LSF is locally a little above or a little below the target for the clinker as a whole because raw material mixing is never completely accurate.
- This indicates that even in cases where the LSF is significantly below 1.0, there is almost always some residual free lime. Additionally, it means that an LSF just above 1.0 is required to convert almost all of the belite to alite.
- If (0.7 x SO_3) is subtracted from the CaO content, Portland cement containing clinker and gypsum can also use the LSF calculation.

Silica Ratio (SR):

The Silica Ratio (also known as the Silica Modulus) is defined as:

SR = SiO_2 / (Al_2O_3 + Fe_2O_3)

A high silica ratio indicates that the clinker contains more calcium silicates and less aluminate and ferrite. SR typically ranges from 2.0 to 3.0.

Alumina Ratio (AR):

The alumina ratio is defined as: **AR = Al_2O_3 / Fe_2O_3**

- This establishes the aluminate and ferrite phases' potential relative ratios in the clinker.
- There will be proportionally more aluminate and less ferrite in the clinker as clinker AR (also referred to as A/F) increases.
- The AR in common Portland cement clinker typically ranges from 1 to 4.
- The most frequently used parameters are the three listed above. A fourth parameter, the "Lime Combination Factor" (LCF), is the same as the LSF parameter but subtracts the total amount of clinker-free lime.
- Therefore, the maximum amount of silica is present as C_3S when LCF = 1.0.

FAQ

1. What are the properties of Carbon Nanotubes?
2. Describe synthesis of carbon nanotubes by arc discharge method.
3. Explain laser ablation method for synthesis of carbon nanotubes.
4. Describe how chemical vapor deposition method synthesis carbon nanotubes.
5. Write a note on fullerenes.
6. List out the properties and applications of fullerenes.
7. Write a preparation method for synthesis of fullerenes.
8. What is refractory? Classify them.
9. Mention applications of refractories.
10. Identify the reasons for loss of refractoriness.
11. Describe the properties of refractories.
12. Define lubricant and describe properties of lubricants,
13. How to determine carbon residue by Conradson method?
14. How to determine aniline point?
15. Write the mechanisms of lubricants.
16. List out the characteristics of good lubricant.
17. What is the composition of Portland cement.
18. Describe the process of cement manufacturing,
19. Explain hardening and setting of cement.
20. Describe deterioration of cement.

Fuels

Introduction

A fuel is a substance that, when burned with oxygen, produces a large amount of heat. A fuel is primarily made up of carbon and hydrogen. Chemical energy is the heat energy produced by the combustion of fuel.

Fuels classification, Natural fuels and artificial (or prepared) fuels are classified based on their occurrence. Natural fuels occur naturally and are also known as primary fuels, whereas artificial fuels are created by further processing of primary fuels and are known as secondary fuels.

They are classified as solid fuels, liquid fuels, or gaseous fuels based on their state.

4.1 Fuel Definition

Fuel is a combustible substance that, when burned, generates a large amount of heat that can be used for a variety of domestic and industrial purposes. Wood, charcoal, kerosene, diesel, producer gas, and other common fuels contain carbon as the primary constituent.

4.2 Gross Calorific Value and Net Calorific Value

The amount of heat produced by the combustion of a unit quantity of fuel is known as its gross calorific value (GCV). GCV or HCV is the amount of heat liberated by combusting a unit mass of fuel in oxygen, with the original material and final product of combustion being at a reference temperature of 25°C and the water obtained in the liquid state.

The amount of heat evolved when a unit quantity of fuel is burned in oxygen, with the original material and final products of combustion at a reference temperature of 25°C and the water obtained from the fuel in the vapor state. As a result, the net calorific value is always less than the gross calorific value by the amount corresponding to the heat of condensation of water vapors, which is 587.0 kcal/kg.

NCV = HCV — Latent heat of water vapor formed

= HCV – (Mass of hydrogen x 9 x Latent heat of steam)

Dulong' Formula

If ultimate analysis is available and the calorific value of elementary combustibles is known, Dulong's formula is used to calculate the theoretical calorific value of fuel.

Fuel calorific value = 33800 C+144000 (H_2-O_2/8) + 9270 S kJ/kg

Where C, H_2, O_2, and S denote the mass of carbon, hydrogen, oxygen, and Sulphur in kJ/kg

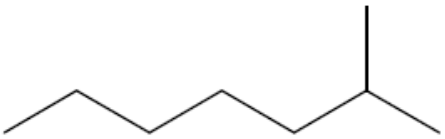
Analysis of solid fuels: We all know that in today's world, coal is used as a solid fuel and that it is commonly found in nature, though not in its purest form. There is also no consistent composition. Coal does not have a specific chemical formula. So, the following two methods are used to determine the composition of coal.

4. 3 Octane Number and Cetane Number

Octane number: It is defined as the percentage of isooctane in isooctane and n-heptane that produces the same knocking as gasoline when burned in a standard test engine under standard conditions.



N-HEPTANE

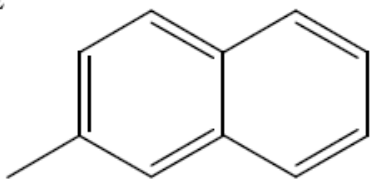


ISOOCTANE

Cetane number: It is defined as the percentage of cetane in a mixture of cetane and 2-methyl naphthalene that has the same ignition delay as diesel oil when burned in a standard test engine under standard conditions.



CETANE



2-METHYL NAPHTHALENE

4.4 Advantages and disadvantages of a gaseous fuel.

Advantages:	**Disadvantages:**
• Can be distributed over a large area via pipeline. • Quiet combustion with no smoke or ash. • The temperature of the heating can be adjusted by adjusting the gas flow to the burner. • Higher calorific values.	• Explosion danger. • Fire danger on a larger scale.

4.5 Proximate Analysis of Coal

i. **Determination of Moisture content**: 1 g of finely powdered air-dried sample is placed in a crucible and heated for 1 hour in an electrically heated hot air oven set to 105°C—110°C. The crucible is taken outside after heating, cooled in a desiccator, and weighed.

$$\text{Percentage of moisture content} = \frac{\text{loss in weight of coal}}{\text{weight of coal taken}} \times 100$$

ii. **Determination of Volatile matter***:* The dried sample left in the crucible with the lid is heated in a muffle furnace at 950 25°C for 7 minutes before being cooled in a desiccator and weighed.

$$\text{Percentage of volatile content} = \frac{\text{loss in weight of coal}}{\text{weight of coal taken}} \times 100$$

iii. **Determination of Ash content***:* After the two preceding experiments in the crucible, the residual sample is heated in the furnace at 750°C for 30 minutes without the lid. The product is then cooled in a desiccator and weighed.

$$\text{Percentage of ash content} = \frac{\text{weight of ash left}}{\text{weight of coal taken}} \times 100$$

iv. **Determination of Fixed carbon***:* The fixed carbon content can be determined indirectly by subtracting the percentage of moisture, volatile and ash content from 100.

Percentage of fixed carbon =100 — % of (moisture + volatile matter + ash)

Significance:

1. A higher percentage of moisture content reduces coal's calorific value. As a result, the lower the moisture content, the higher the quality of coal.
2. A higher percentage of volatile matter reduces coal's calorific value. Low volatile matter also reduces coal's coking property.
3. Because ash is non-combustible, it reduces the calorific value of coal. Ash deposition also causes problems in the furnace walls, and ash disposal is an issue. The higher the percentage of fixed carbon, the higher the calorific value and the higher the coal quality.

4.6. Ultimate analysis of coal.

The determination of the weight percentages of carbon, hydrogen, nitrogen, oxygen, and Sulphur is referred to as ultimate analysis.

i. **Carbon and hydrogen***:* A known amount of coal is taken and burned in a current of O2 in a combustion apparatus, producing CO_2 and H_2O. CO_2 and H_2O are absorbed by tubes previously weighed with KOH and anhydrous $CaCl_2$. The increase in weight results in the following C and H content:

$$C+O_2 \rightarrow CO_2$$

and $$H_2 + 1/2\ O_2 \rightarrow H_2O$$

and $$2KOH+CO_2 \rightarrow K_2CO_3 + H_2O$$

$$CaCl_2 + 7H_2O \rightarrow CaCl_2.7H_2O$$

$$\% \text{ of C} = \frac{\text{Increase in wt. of KOH tube}}{\text{Wt.of coal taken}} \times \frac{12}{44} \times 100$$

$$\% \text{ of H} = \frac{\text{Increase in wt.of } CaCl_2 \text{ tube}}{\text{wt.of coal taken}} \times \frac{2}{18} \times 100$$

ii. **Nitrogen***:* A 1 g accurately weighed coal sample is heated in a long-necked flask with conc. H_2SO_4, K_2SO_4. The excess NaOH is then removed, and the liberated NH_3 is absorbed in a known excess of standard acid solution. The excess acid is titrated back down with standard NaOH solution. The N content is calculated using the volume of acid consumed as follows:

$$\% \text{ of N} = \frac{\text{volume of acid consumed} \times \text{Normality}}{\text{wt.of coal taken}} \times \frac{14}{1000} \times 100$$

iii. **Sulphur**: The S in coal is converted to sulphate during the calorific value determination process in a bomb calorimeter. Finally, the sulphate-containing washings are treated with a diluted HC1 and BaCl2 solution, which precipitates BaSO4, which is then filtered in a sintered glass crucible, washed with water, and heated to a constant weight.

$$\% \text{ of S} = \frac{\text{wt.of } BaSO_4}{\text{wt.of coal taken}} \times \frac{32}{233} \times 100$$

iv. **Oxygen content** = 100 - % of (C + H + S + N)

Significance:

1. Higher percentage of C and H increases the calorific value of coal and hence better is the coal.
2. Higher the percentage of O_2 lower is the calorific value and lower is the coking power. Also, O_2 when combined with Hydrogen in the coal, Hydrogen available for combustion becomes unavailable.
3. S, although contributes to calorific value, is undesirable due to its polluting properties as it forms SO_2 on combustion.

4.7. Knocking

An explosive sound produced by an internal combustion engines rapid pressure rise. Petrol's knocking characteristics can be expressed in terms of octane number.

Octane Number:

Octane number is equal to the percentage by volume of iso-octane (2,2,4- trimethyl pentane) in a mixture of n-heptane and iso-octane having the same knocking tendency compared to the sample of gasoline being tested; iso-octane has the best antiknocking properties and assigned an octane number of 100 whereas n-heptane has poor antiknocking property and assigned an octane number of zero.

The knocking properties of gasoline are influenced by the hydrocarbons present, which vary according to the series: straight chain paraffin > branched chain paraffin > olefin > cycloparaffin > aromatics.

The fuel with the same knocking tendency as the 80% iso-octane mixture has an octane number of 80. Tetraethyl lead (TEL) is the most effective antiknock agent added, along with ethylene dibromide, which prevents lead deposition by forming volatile lead halides. Tetramethyl lead (TML), tertiary butyl acetate, and diethyl telluride are some other antiknocking agents. Per liter of gasoline, 1.0—1.5 mL of TEL is added. TEL works by converting itself into a cloud of finely divided lead oxide particles, which react with any hydrocarbon peroxide molecules formed in the engine cylinder, solving the chain oxidation reaction and preventing knocking.

Cetane Number:

The knocking characteristics of diesel can be expressed in terms of cetane number. Between diesel fuel injection and ignition, there is a time lag. When the delay period is too long, too much fuel accumulates in the cylinder and burns too quickly, causing "diesel knock."

This delay is caused by the hydrocarbons found in diesel. Paraffins, olefins, naphthene, isoparaffins, and aromatics have the longest delay times. The order is reversed for gasoline anti-knock quality; n-hexadecane is assigned a cetane number of 100, while a-methyl naphthalene is assigned a cetane number of zero.

The cetane number 40 is assigned to a fuel whose quality matches the 40/60 blend of cetane and a-methyl naphthalene. The cetane number is defined as the percentage of cetane present in a mixture of cetane and 2-methyl naphthalene with the same ignition lag as the fuel under test. The cetane number of diesels can be raised by adding amyl/butyl nitrate, carbamates, Di tertiary butyl peroxide, and metal organic compounds.

4.8. Cracking Process

In an oil refinery, cracking is the process by which heavier fractions from fractional distillation are converted into useful lighter fractions by applying heat, with or without a catalyst, i.e., cracking is a process by which larger molecules break up into smaller ones. The primary use of commercial cracking in all refineries is to produce gasoline from petrol oils. Other applications include the synthesis of olefins from naphtha and gas oils. The excess of heavier petroleum fractions is also cracked to produce petrol. There are two ways to crack.

a. Thermal Cracking

Thermal cracking occurs when cracking occurs without the use of a catalyst at high temperatures ranging from 450°C to 750°C and pressures ranging from 1—70 atms. Decomposition, dehydrogenation, isomerization, and polymerization are important reactions. The paraffins decompose into lower molecular weight compounds such as paraffin and an olefin.

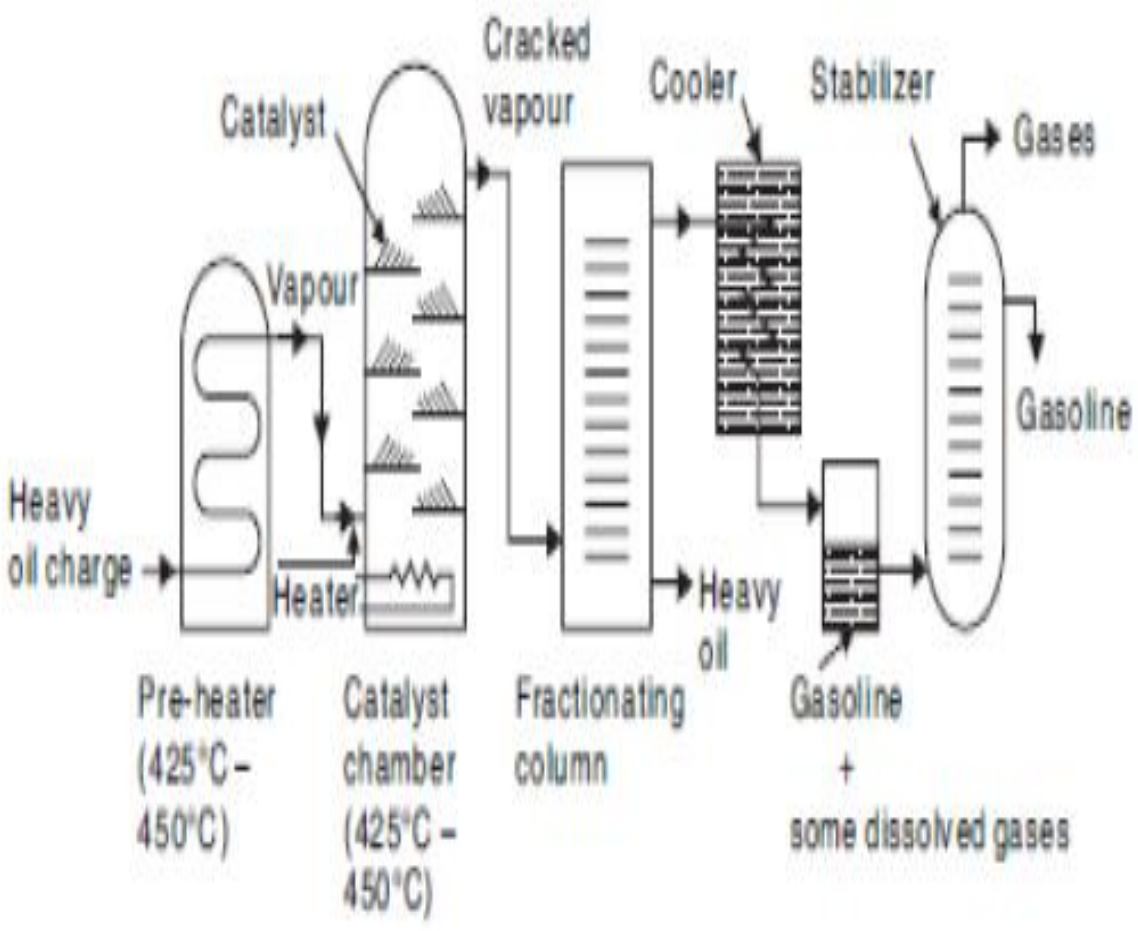
b. Catalytic cracking:

By using a suitable catalyst, such as aluminum silicate, the quality and yield of gasoline produced by cracking can be greatly improved.

There are two main types of catalytic cracking:

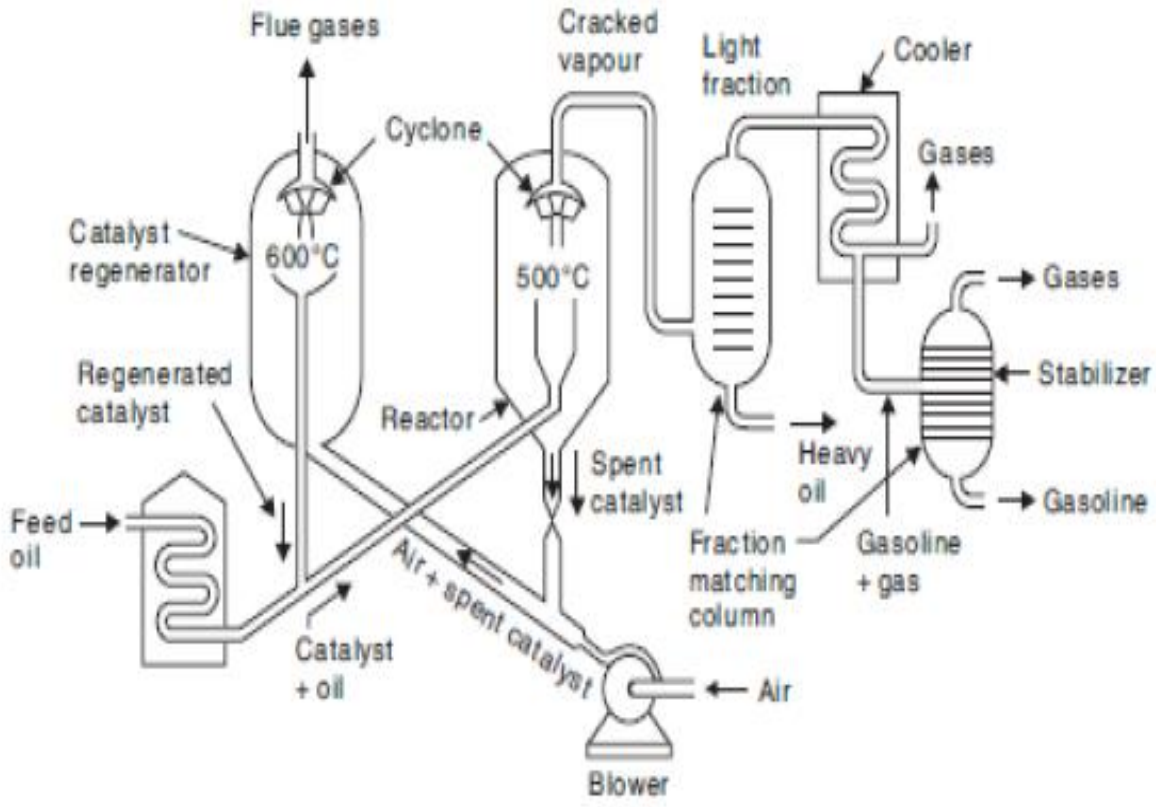
(i) *Fixed-bed catalytic cracking:*

The catalysts are installed in towers through which hot oil (500° C) flows from the top and descends. 40% of the charge is converted to petrol, and 2 — 4% of the charge is converted to carbon. Carbon accumulates on the catalyst beds. The bottom liquid is re-boiled and recycled to the fractionating column, resulting in high-octane gas oil. The petrol is purified after being stripped of dissolved gases. While the other catalyst chambers are active, the carbon deposits on the catalysts are burned by compressed air in one chamber for reactivation.



(ii) Moving bed or fluidized bed catalytic cracking:

When suspended in gas or oil vapor, the finely powdered catalyst behaves as a fluid. The preheated heavy oil, along with the fluidized catalyst, is forced through the tower. A cyclone separator at the top of the tower separates the cracked oil vapor and transports it to the fractionating column. The catalyst powder is saved and returned. A deposition deactivates the catalyst, which is reactivated by burning away the deposits with compressed air in a regenerator.



4.9. Method of Analysis of Flue Gas (Orsat Apparatus)

Analysis of flue gas will indicate the complete or incomplete combustion of a fuel; which is very essential in respect of efficient utilization of the fuel. The analysis is done in Orsat's apparatus and is based on the principle of absorption of

• CO_2 in KOH solution (500 g\ 500 ml)

• O_2 in alkaline pyrogallic acid (25 g of pyrogallic acid in 400 g KOH solution, 500 ml)

• CO in ammoniacal cuprous chloride solution (100 g Cu_2Cl_2 + 125 ml liquor ammonia Rest water to make up the volume, 500 ml).

Description of the Apparatus

It consists of a water jacketed measuring burette, connected in series with the absorption bulbs containing the above three solutions successively as depicted in the Fig.

The water jacket maintains the temperature of the gas constant. The absorption bulbs are filled with glass tubes for better absorption of the gases.

Procedure

• The apparatus is tested for its air-lightness.

• The flue gas (100 ml) is taken in the apparatus in the measuring burette by adjusting the volume by water reservoir atmospheric pressure.

• The stopper (1) is opened for CO_2 absorption and water reservoir is raised to force the gas inside the bulb. The gas is finally taken in the burette and the volume of the gas is measured by making the levels of water inside the burette and water reservoir equal. The decrease in volume gives the percentage by volume of CO_2 in the flue gas.

• The stopper (2) is opened and provided as usual to get the percentage by volume of O_2 in the flue gas.

• The stopper (3) is opened and proceeded as usual to get the percentage by volume of CO in the gas.

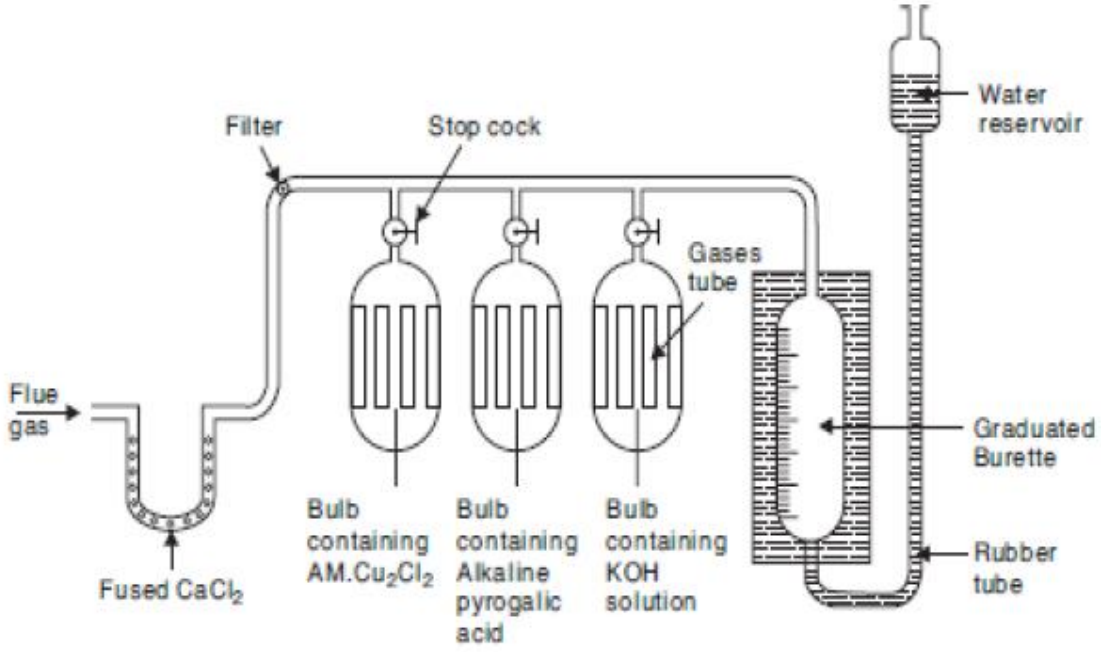
• The sequence of the bulbs should be strictly followed.

Implications of the Analysis

• If the flue gas contains greater percentage of CO, it is implied that considerable wastage of fuel is taking place due to incomplete combustion and the O2 supply is insufficient.

• The greater percentage of O2 in the flue gas indicates that O2 supply is in excess.

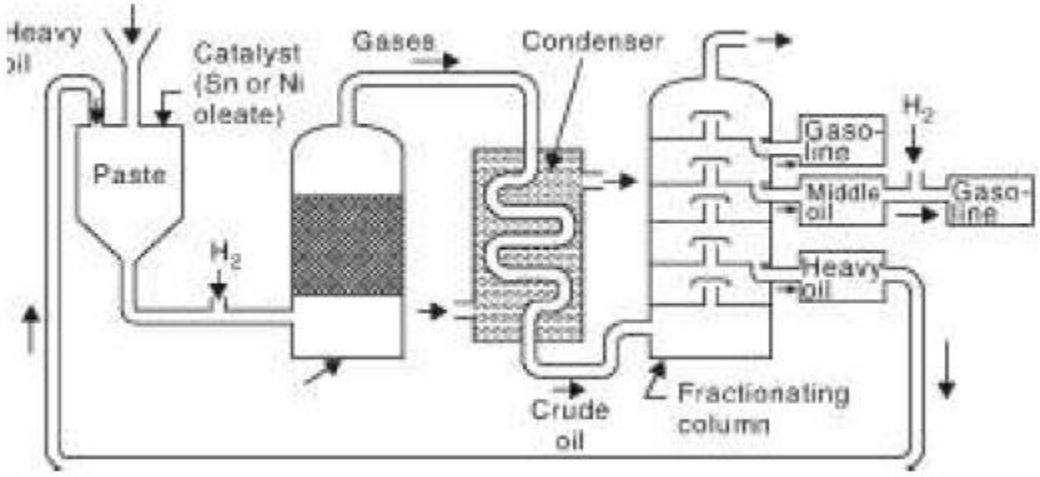
• Result of the analysis will help to control the combustion process.



4.10. Bergius Process

Oil is produced by destructive hydrogenation of coal in the presence of a catalyst, but it is not yet competitive with petroleum refining. Coal is ground and mixed with heavy recycle oil and a catalyst such as iron, tin, or molybdenum compound to form a paste. The paste is preheated and treated with H2 at 450°C — 500°C temperature and 250 — 350 atm pressure. The unreacted coal is removed and the liquid product is distilled. When hydrogen reacts with coal, it forms saturated hydrocarbons, which decompose at high temperatures to produce low-boiling hydrocarbons. The crude oil is fractionated and recycled to produce (I)

gasoline, (ii) middle oil, and (iii) heavy oil. To produce more gasoline, middle oil is hydrogenated in the vapor phase with a catalyst. The yield of petrol is 60% that of coal dust.



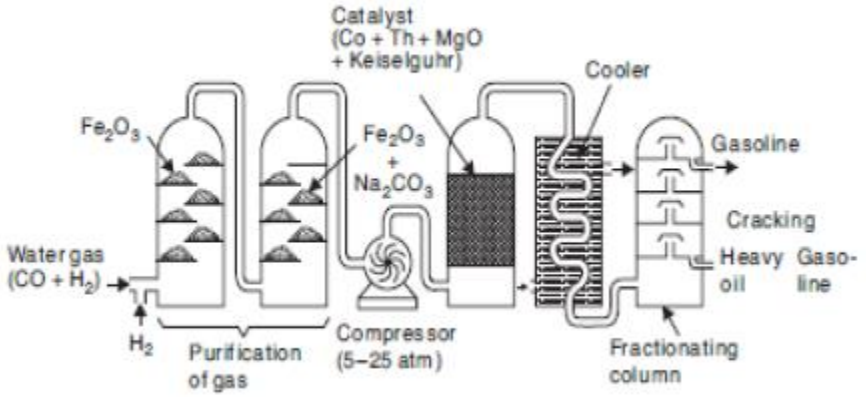
4.11. Fischer – Tropsch Process

The process is based on the catalytic hydrogenation of carbon monoxide leading to the following reaction:

$n\ CO + 2n\ H_2 \rightarrow C_nH_{2n} + n\ H2O$

$n\ CO + (2n+1)\ H_2 \rightarrow C_nH_{2n+2} + n\ H2O$

Olefin Water gas ($CO + H_2$), produced by passing steam over heated coke is mixed with hydrogen. The gas is purified by passing over Fe_2O_3 to remove H_2S, then passing over a mixture of $Fe_2O3.Na_2CO_3$ to remove organic sulfur compounds. The purified gas is compressed to 5 to 25 atm. at 200°C — 300°C passed through a converter containing catalyst which is a mixture of 100 parts cobalt, 5 parts thorium, 8 parts magnesium and 200 parts kieselguhr.



4.12. Various gaseous fuels.

a) Natural gas: Petroleum mines are generally associated with natural gas, which is obtained from petroleum wells dug in oil-bearing regions. Wet gas refers to natural gas that occurs in oil wells alongside petroleum. Drilling in a petroleum mine produces a dark green, viscous, highly volatile gas associated with lower hydrocarbons. This gas is known as "dry gas" because it is associated with crude oil.

The natural gas extracted from the well is purified to remove impurities like dust, air, grit, H_2S, CO_2, N_2 and heavier liquefiable hydrocarbons.

Natural gas composition: CH_4 = 70-90%, C_6H_6 = 5-10%, H_2 = 3%, $CO+CO_2$ = the remainder.

12000-14000 Kcal/M^3 is the calorific value.

Applications:

1. It is an excellent domestic fuel that can be transported over longer distances via pipeline.
2. Natural gas is used to synthesize a wide range of chemicals.
3. Natural gas is used as a raw material in the production of carbon black, which is used as filler in rubber, as well as hydrogen, which is used in the production of ammonia.

b) LPG (Liquid petroleum gas or Bottled gas or refinery gas)

LPG is obtained as a byproduct of heavy oil or natural gas cracking. LPG is dehydrated and desulphurized. Under pressure, LPG is a highly volatile, colourless liquid. LPG is extremely flammable and is composed of hydrocarbons that can be easily liquefied under pressure and exist as gas at atmospheric pressure. Mercaptan was added to detect LPG leakage.

n-Butane, isobutane, butylene, and propane are the constituents and 27,800 Kcal/M^3 calorific value

Advantages: Resists knocking, burns cleanly with little residue, and extends engine life.

Uses:

1. The largest use is as domestic fuel.
2. This is also used widely as an industrial fuel.
3. LPG is used as a fuel in certain class of vehicles like trucks and tractors.
4. LPG leaded with (TML) tetramethyl lead can be used as fuel for the railway diesel locomotives.
5. It is advantageously used in engines working under high compression ratio.

Disadvantages: Handling has to be done extremely carefully to avoid hazards.

c) CNG (Compressed natural gas)

CNG is produced by compressing natural gas (which contains nearly 90% CH4) to less than 1% of its original volume at STP. CNG is stored and distributed in hard spherical containers at pressures ranging from 200 to 248% Kg/cm^2. It is a non-toxic, odorless gaseous mixture. CNG is measured in Gallons Equivalent (GGE) (GGE).Composition: CH_4 – 90%, other constituents are ethane, propane and gases like N_2, CO etc.

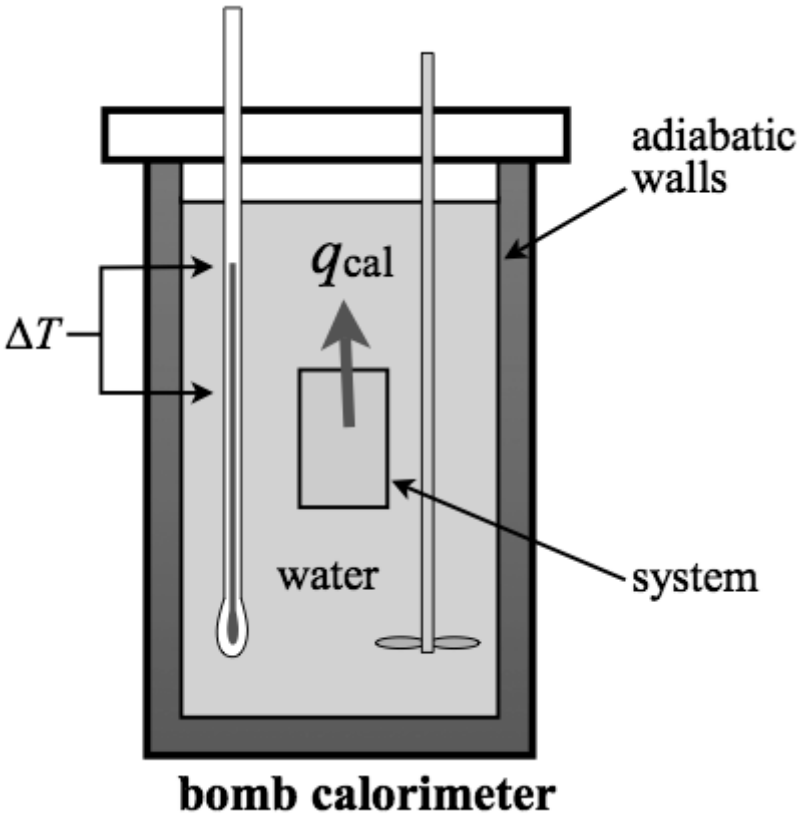
Properties: Light weight, gaseous fuel. It mixes with air easily. It has a high auto-ignition temperature (540^0C) and a narrow range of flammability. CNG does not contaminate with lubricating oils, thus increases the life of internal combustible (I/C) engines. It requires more space for storage.

Calorific value: 900 KJ/mole.

Uses:

1. Majorly used as fuel in automobiles like cars, autos, trucks, buses etc.
2. It is also used as fuel for locomotive diesel generators to generate electricity that drive the motors of trains.

4.13. Bomb Calorimeter



bomb calorimeter

A bomb calorimeter is a calorimeter with a constant volume (constant volume is isochoric). As a result, the heat measured by such a device is equivalent to the change in internal energy or ΔU. The heat can be calculated using the temperature change and the calorimeter's heat capacity ΔT and the heat capacity C_{cal} of a bomb calorimeter is typically quite large due to all of the water and hardware (stirring paddles, blades, the stainless steel "bomb" holding the reactants, the wiring, the walls of the calorimeter, etc...). This is the value C_{cal}. It is q_v used to find the system:

$$q_{cal} = C_{cal}\Delta T = -q_{v,\ system}$$

It is sometimes more convenient to divide the calorimeter's overall heat capacity into two parts: (1) the water and (2) the hardware.

$$C_{cal} = C_{hardware} + m_{water}C_{s,\ water}$$

The hardware heat capacity will be measured in J/K or kJ/K units, whereas the water heat capacity must be calculated using the water mass and specific heat of water (4.184 $Jg^{-1}K^{-1}$). Because you can fill the calorimeter with slightly different masses of water each time you use it, the water part is slightly variable. When you combine all of that into a bomb calorimeter formula, you get:

$$q_{cal} = C_{hardware}\Delta T + m_{water}C_{s,\ water}\Delta T$$

Relation between ΔU and ΔH: $\Delta H = \Delta U + \Delta n_{gases}RT$

Where $\Delta n_{gases} = \sum n_{gas\ products} - \sum n_{gas\ reactants}$

4.14 Problems

1. Calculate the net calorific value of a coal sample having the following composition:

C=80%, H=5%, O=4%, N=3%, S=3.5% and ash=5%.

Solution: Apply the Dulong's formula that is:

HCV = 1/100[8080C + 34500(H-O/8) + 22400S]

=1/100[(8080*80)+(345008*(5-(4/8)))+(22400*3.5)]

= 809.4 Cal /gm

NCV = HCV – (0.09 * H*587)

Where 587 Cal/gm is latent heat of steam

= 809.4 – (0.09*5*587)

= 7830.75 Cal /gm

2. **A coal has the following composition by weight: C=90%, O=4%, N=1%, S=0.5% and ash=5%.The NCV of the fuel was found to be 8480cal/g. Calculate the percentage of hydrogen and HCV of the fuel.**

Solution:

Apply the Dulong's formula that is:

HCV = 1/100[8080C + 34500(H-O/8) + 22400S]

HCV = 1/100[(8080*90)+(345008*(H-(4/8)))+(22400*0.5)]

HCV = [7110.7+345H] Cal/gm…… (1)

NCV = HCV – (0.09 * H*587)

Where 587 Cal/gm is latent heat of steam

8480 = HCV – (0.09*H*587)

HCV = 8480 + 52.83 H Cal/ gm……. (2)

On solving eq (1) & eq (2)

H = 4.686 %

Substitute H value in eq (1)

HCV = 8727.37 Cal/gm

3. C=70%, O=10%, N=1%, S=5% and ash=4%.The NCV of the fuel was found to be 9210 Cal/g. percentage of hydrogen and HCV of the fuel.

Solution:

Apply the Dulong's formula that is:

HCV = 1/100[8080C + 34500(H-O/8) + 22400S]

HCV = 1/100[(8080*70)+(345008*(H-(10/8)))+(22400*5)]

HCV = [5336.75+345H] Cal/gm…… (1)

NCV = HCV – (0.09 * H*587)

Where 587 Cal/gm is latent heat of steam

9210 = HCV – (0.09*H*587)

HCV = 9210 + 52.83 H Cal/ gm……. (2)

On solving eq (1) & eq (2)

H = 13.25 %

Substitute H value in eq (1)

HCV = 9908 Cal / gm

4. Calculate the HCV of the coal from the given data:

Weight of the fuel burnt=0.92 g

Initial temperature=12^{0}C

Final temperature=19.2^{0}C

Weight of the water in calorimeter is 1458 g

Water equivalent of calorimeter = 14g

Solution: Weight of the fuel burnt=0.92 g

Initial temperature=12^{0}C

Final temperature=19.2^{0}C

Weight of the water in calorimeter is 1458 g

Water equivalent of calorimeter = 14g

Use the formula HCV = [(W + w)(t_2-t_1)]/x

where W = Weight of the water in calorimeter, w = Water equivalent of calorimeter, t_2 = Final temperature, t_1 = Initial temperature, x = Weight of the fuel burnt

Substitute the values in above formula

HCV = [(1458+14) (19.2-12)]/0.92

= 11520 k Cal/ m^3

5. Calculate the GCV and NCV of a fuel from the following data. Volume of the fuel burnt at STP is 0.08m^3.Weight of the water used for cooling is 24kg.Temperature of inlet and outlet water is 25°C and 30°C respectively. Weight of water obtained by steam condensation is 0.02 kg.

Solution:

For calculating the GCV of the coal use the formula GCV = w (T_2-T_1)/V

where w = weight of water and v = volume of fuel burnt.

GCV = 24000(30-25)/0.08

= 1500 Cal /g

NCV = GCV – ((M/V)*587)

M is the weight of the water obtained by steam.

= 1500 – ((0.02/0.08)* 587)

= 1353.25 Cal/g

6. Calculate the HCV and NCV of the coal for the following data.

Weight of the coal = 0.8g, Water equivalent of calorimeter = 460g, Weight of the water = 2600 g, Rise in temperature = 2.42°C, Cooling corrections = 0.052°C, Fuse wire corrections = 10 calories H=6% and assume latent heat of steam=600cal/g.

Solution:

We know the formula for HCV and that is

HCV = [(W + w) (t_2-t_1+cooling corrections)/x] - fuse wire corrections

W is the weight of the water and w is the water equivalent and t2 and t1 are the final and initial temperatures.

HCV = [(2600+460) (2.42+0.052)/0.8] – 10

= 9445.4 Cal/g

NCV=HCV – (0.09*H*latent heat of steam)

=9445.4 – (0.09*6*600)

= 9121.4 Cal/g

7. A sample of coal containing the following elements:

C=90%; H=5%; ash=4%; weight of the coal burnt=0.90g; weight of the water taken=600g; water equivalent of bomb calorimeter=2000g; rise in temperature=2.48°C; fuse wire correction=10 Cal; cooling correction=0.02°C; acid correction = 50 cal. Calculate net and GCV of coal in Cal/g. Assume latent heat of steam is 580 Cal/g.

Solutions:

For HCV we can use the formula

HCV = (Wow) (t_2-t_1+cooling corrections) - (acid + fuse corrections) / weight of the fuel

= (600+2000) (2.48+0.02)-(50+10)/0.9

= 7155.5 Cal / g

NCV=HCV – (0.09*H*latent heat of steam)

=7155.5 – (0.09*5*580)

= 6894.5 Cal / g

8. Calculate the weight and volume of air required for the complete combustion of 2kg of carbon.

Solution:

$C+O_2 \rightarrow CO_2$

12 kg of C requires 32kg of oxygen for combustion

Weight of oxygen for required = 2*32/12 = 5.33 kg

Air contains 23% of O_2 by weight

Weight of air = 5.33*100/23 = 23.17 kg

At NTP 1 mole of oxygen occupies 22.4 liters

5330 g mole of oxygen occupies = 22.4/32 * 5330 = 3731 liters

Then calculates for the weight of carbon that has calculated before, and then find volume of air by knowing that 100 parts of air contains 21 parts of oxygen.

Volume of air = 3731*100 / 21 = 17766.66 liters

9. The coal sample contains C = 80%; H = 4%; O = 2%; N = 12%; S = 2%, the remainder is ash and then calculate the volume of air required for perfect combustion of 1kg of fuel assuming STP conditions.

Solution:

$C+O_2 \rightarrow CO_2$

1 kg of fuel contains 80% of Carbon i.e., 0.8 kg of C

12 kg of C requires 32 kg of O for combustion

Weight of oxygen required = 0.8*32/12 = 2.133 kg

At NTP STP 1 mole of oxygen occupies 22.4 liters

2133 g mole of oxygen occupies = 2133*22.4/32 = 1493.1 liters

We know that 100 parts of air contains 21 parts of oxygen by volume.

Volume of air = 1493.1*100/21 = 7110 liters

10. 2.5 gm of air-dried coal sample was taken in a silica crucible, after heating it in an electric oven at 1050°C−1100°C for 1 hour; the residue was weighed 2.410 gm. The residue was heated in silica crucible covered with vented lid at temperature 950−9700°C for exactly 7 minutes. After cooling the weight of residue was found to be 1.78 gm. The residue was ignited at 7500°C to a constant weight of 0.246 gm. Calculate results of proximate analysis.

Solution:

Weight of coal taken = 2.5 gm

Mass of moisture in coal = 2.5 – 2.415 = 0.085 gm

% of moisture = loss in wt. of coal/weight of coal taken x 100

= 0.085 / 2.5 x100 = 3.4%

% Mass of volatile matter = 2.415 – 1.78 = 0.635gm

% of volatile matter = 0.635 /2.5 x100 = 25.4%

Mass of residue after ignition = 0.246 gm

% of Ash = weight of residue left/weight of coal taken x100

= 0.246/2.5 x 100 = 9.8 %

% of fixed Carbon = 100- (% moisture +% volatile matter+ %Ash)

= 100- (3.4 + 25.4 + 9.8)

= 100- 38.6

= 61.4%

% Moisture = 3.4%

% Volatile matter = 25.4%

% Ash = 9.8 % %

Fixed Carbon = 61.4 %

11. 1.95 gm of coal sample was taken for nitrogen estimation by Kjeldahl's method. The ammonia liberated required 9.5 ml of 0.4 N H_2SO_4 for neutralization. Calculate the percentage of nitrogen in coal sample.

Solution: Weight of coal = 1.95 gm

Normality of H_2SO_4 = 0.4 N

Volume of H_2SO_4 = 9.5 ml

Required % N =?

% N = (Volume of H_2SO_4 x Normality x 1.4) / wt. of coal

= 9.5*0.4*1.4 / 1.95

=2.72 %

12. 0.5 gm of coal sample was burnt in Bomb calorimeter experiment produced 0.06 gm of $BaSO_4$. Calculate percentage of Sulphur.

Solution:

% S = (weight of $BaSO_4$ x 32 x100) / (weight of coal x 233)

= (0.06x32x100) / (0.5x233)

= 1.64 %

13. Find the volume of air required for complete combustion of 1 m^3 of acetylene and the weight of air necessary for the combustion of 1 kg of fuel?

Solution:

The reaction involved in this reaction will be-

$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$

Therefore, air required per m^3 of C_2H_2 = 2.5 × (100/21) = 11.9 m^3

Now oxygen supplied per kg of fuel = (162/52)

Therefore, air supplied per kg of C_2H_2 = (160/52) × (100/23) = 13.378 kg.

14. The % analysis by volume of product gas is H_2 – 18.3%, CH_2 – 3.4%, CO – 25.4%, CO_2 – 5.1% and N_2 – 47.8%. Calculate the volume of air required m^3 of the gas?

Solution:

Total moles of O_2 required = 28.65

Therefore, air required for 100 moles of fuel gas = 28.65 × (100/21) = 136.43 mols

Therefore, air required per m^3 of gas = (136.43/100) = 1.3643 m^3.

FAQ

1. Define fuel.
2. What is meant by gross calorific value?
3. Define octane number and cetane number.
4. What are the advantages and disadvantages of a gaseous fuel?
5. Write a note on proximate analysis includes the determination of moisture, volatile, ash and fixed carbon content.
6. Write notes on ultimate analysis of coal.
7. Write notes on knocking.
8. Write notes on knocking.
9. Explain the cracking process?
10. Explain flue gas analysis by Orsat apparatus.
11. Discuss Bergius method of manufacture of Synthetic Petrol?
12. Discuss Fischer – Tropsch method of manufacture of Synthetic Petrol?
13. Write short notes on a) Natural gas, b) LPG gas, c) CNG gas.
14. Write about Bomb calorimetry?

Unit 5

Water Technology

Introduction

Water is required for the survival of humans, animals, and plants. Despite the fact that water covers 80% of the earth's surface, less than 1% of it is available for immediate use. Rain rivers and lakes (surface water), wells and springs (ground water), and sea water are the primary sources of water. Rain water is the purest form of water, but it is extremely difficult to collect, whereas sea water is the most impure form. As a result, both surface and ground water are typically used for industrial and domestic purposes. Such water must be free of unwanted impurities.

"***Water treatment refers to the process of removing all types of impurities from water and making it suitable for domestic or industrial use.***"

5. 1. Hardness of Water

Hard water is water that does not lather or produces very little lather with soap. Soft water, on the other hand, readily produces a large amount of lather when combined with a small amount of soap. The presence of dissolved salts such as bicarbonates, carbonates, sulphates, chlorides, and nitrates of bivalent metal ions such as calcium and magnesium cause water hardness.

Soap is sodium or potassium salt of higher fatty acids like stearic, oleic and palmitic acids.

Na-Stearate + H_2O → NaOH + stearic acid (stearate = $C_{17}H_{35}COO^-$)

Stearic acid + Na-Stearate → formation of lather

When soap comes into contact with hard water, sodium stearate reacts with the dissolved calcium and magnesium salts to form calcium stearate or magnesium stearate, both of which are white precipitates.

2 Na-Stearate + Ca^{2+} → Ca-Stearate +2 Na^+

The different types of water are commercially classified on the basis of degree hardness as follows:

Hardness	**Types of water**
0-70 mg/ltr	Soft water
70-150 mg/ltr	Moderate hard water
150-300 mg/ltr	Hard water
300 mg/ltr & above	Very hard water

Units of hardness:

Unit	**Unit name**	**conversion**
ppm	Parts perm million	1 ppm = 1mg/L
^{0}Fr	Degree French	1 ppm = 0.1 ^{0}Fr

^{0}Cl	Degree Clark	1 ppm = 0.07^{0}Cl

5.2 Characteristics of potable water

Potable water is water that is free of contaminants or water that is safe for human consumption.

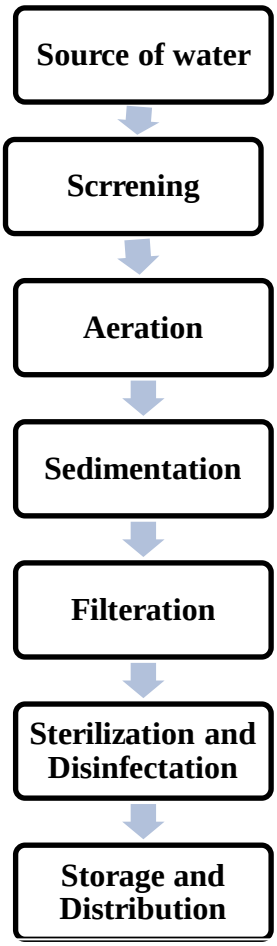
The following is the common standard for drinking water:

1. Water should be colorless, odorless, and transparent (no turbidity).
2. Turbidity should not exceed 10ppm. pH should be between 7.0 and 8.5.
3. The total hardness should not exceed 125ppm.
4. Pathogenic bacteria are not present.
5. Chloride, fluoride, and sulphate levels in water should be less than 250ppm, 1.5ppm, and 250ppm, respectively.

5.3 Method of purification for municipal water supply

The flow diagram of municipal water treatment for domestic use and the various stages involved in purification is as follows:

1) **Screening:** Water is passed through screens with a large number of holes to remove impurities such as wood pieces, leaves, and so on.
2) **Aeration:** Aeration facilitates the exchange of gases between air and water while also increasing the oxygen content of the water.
3) **Sedimentation:** The suspended and colloidal impurities are allowed to settle naturally through gravity. The basic idea behind this treatment is to allow water to flow at a very slow velocity, allowing heavier particles to settle due to gravity.
4) **Filtration**: Filtration aids in the removal of colloidal and suspended impurities that are not removed by sedimentation. Typically, filters are used. Slow sand filters and rapid sand filters, also known as pressure filters, are the two types of sand filters.
5) **Sterilization or Disinfection:** The removal of harmful pathogenic bacteria from drinking water is accomplished through orientation or disinfection.



5.4 Various sterilization and disinfection methods:

1. **Boiling:** Harmful bacteria are killed by boiling water for 15-20 minutes. This is not possible for the municipal water supply. This method of sterilisation is used in the home.
2. **Passing ozone:** When ozone is introduced into water, it acts as a disinfectant. Ozone (an unstable isotope of oxygen) generates nascent oxygen, a powerful disinfectant.

$O_3 \rightarrow O_2 + [O]$ nascent oxygen

The treatment is expensive and cannot store ozone for an extended period of time.

3. **Chlorination:** Chlorination is the process of using chlorine as a powerful disinfectant. Chlorinating agents are classified into three types.
   i) *By passing chloramines*: Chlorine is combined with ammonia in a ratio of 2:1 by volume to form a stable chloramine that produces hypochlorous acid, a powerful disinfectant.

   $Cl_2 + NH_3 \rightarrow ClNH_2 + HCl$

   $ClNH_2 + H_2O \rightarrow HOCl + NH_3$

ii) *By bleaching powder:* The chlorine content of bleaching powder is 80%. When using bleaching powder as a disinfectant. Because hypochlorous acid is used to disinfect, this method is also known as hypo-chlorination.

$CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$

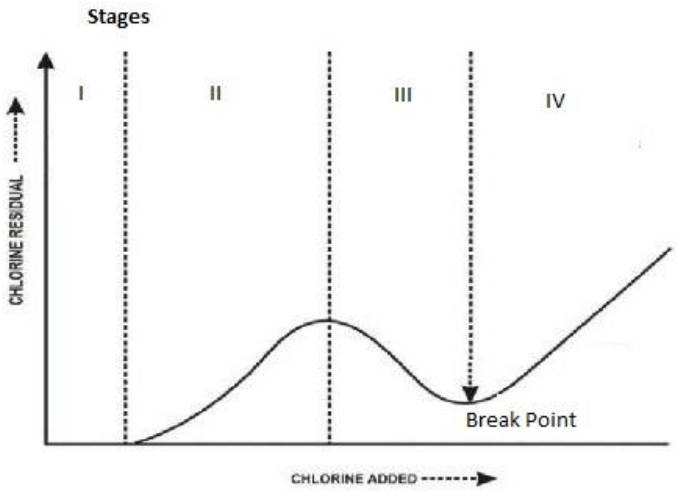
$Cl_2 + H_2O \rightarrow HOCl + HCl$

iii) *Chlorination:* Chlorination is the process of adding a calculated amount of chlorine to water in order to kill pathogenic bacteria.

$Cl_2 + H_2O \rightarrow HOCl + HCl$

5.5 Break Point Chlorination

Another name for it is free residual chlorination. This method of chlorination is controlled. Only the necessary dose of chlorine is added. Four stages make up the break point chlorination curve.



Stage I. No residual chlorine in the sample. All chlorine added is consumed for oxidation and destruction.

Stage II. With a higher chlorine dose, there is more residual chlorine present. Even at lower chlorine concentrations, some organic compounds resist oxidation. Chloramines and some chloro-organics are produced.

Stage III. As the applied chlorine dose is increased, the amount of residual chlorine decreases. Here, the chlorine that was added is used to oxidize the remaining organic compounds, such as ammonia. Complete oxidation and disinfection occur at break point.

Stage IV. The applied chlorine dose is not consumed so left us residual chlorine.

Advantages:

It destroys all pathogens.

It oxidizes all organic compounds, ammonia and other reducing substances.

Remove color, bad taste and odor from water.

Prevents the growth of algae, weed etc.

Dichlorination:

Over chlorination after "break point" produces unpleasant, taste, odour, and toxicity to water. The over chlorination is removed by passing water through a bed of granular carbon and also by the addition of SO_2 and sodium thiosulphate.

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

$Na_2SO_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl$

$Na_2S_2O_3 + 4Cl_2 + 5H_2O \rightarrow 2NaHSO_4 + 8HCl$

5.6 Characteristics of boiler feed water

1. Components that cause hardness, such as dissolved calcium and magnesium salts, produce sludges and scales that waste fuel and increase the risk of a boiler explosion. Therefore, the boiler feed water's hardness should be less than 0.2 ppm.
2. Caustic embrittlement results from the presence of caustic alkalis. Therefore, the alkalinity of boiler feed water (caused by OH^-) should range from 0.15 ppm to 0.45 ppm.
3. To avoid boiler corrosion, boiler feed water needs to be free of dissolved gases like CO_2, O_2, and others.
4. Turbidity, oil, and other substances that cause foaming and priming in boilers shouldn't be present in boiler feed water.

5.7. Boiler Troubles and prevention methods

Boilers are the equipment used in industrial settings to generate steam. Steam is created by heating the water that is fed into the boiler. "Boiler feed water" refers to the water that is fed into the boiler.

Boiler water must meet the following requirements:

(1) Be free of hardness-causing salts; otherwise, sludge and scale will result. (2) Free of oil and greases; if not, foaming will result. (3) Free of dissolved salts, suspended impurities, or boiler corrosion would result; (4) Free of dissolved gases, suspended salts, or caustic embrittlement.

Boiler Troubles

***1. Caustic Embrittlement*:** (Inter crystalline cracking of boiler metal) Boiler intercrystallite cracking brought on by Na_2CO_3 is the cause. Na_2CO_3 hydrolyzes under high pressure to produce NaOH. As a result, water is caustic. The tiny hair cracks are filled with the NaOH-satisfying water.

$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$

Boiler metal's hairline cracks are occupied by this NaOH, which also transforms the insoluble Fe into soluble Sodium Ferroate. As a result, it enlarges cracks in bends, joints, and crevices.

$$Fe+2NaOH \rightarrow Na_2FeO_2+ H_2 \uparrow$$

(Insoluble) (Soluble)

Prevention of caustic embrittlement:

1. As softening agent, we can use sodium phosphate instead of sodium carbonate.

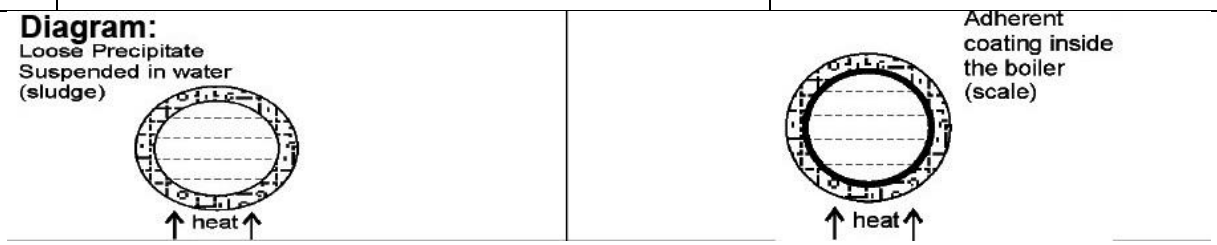
2. The hair line cracks can be sealed by waxy materials like Tannin and Lignin.

2. Sludge and scale: If the water contains hardness causing salts like $MgSO_4$, $MgCl_2$, $CaSO_4$, $Ca(HCO_3)_2$ on evaporation, the salts are precipitated to produce scale and sludge.

Differences between scales and sludges:

No	Scales	Sludges
1	Hard, thick, strong adherent precipitate	Loose, slim, non-adherent precipitate
2	Due to salts like $CaSO_4$, $Ca(HCO_3)_2$	Due to salts like $MgSO_4$, $MgCl_2$
3	Due to poor conductance, they decrease the boiler efficiency to maximum extent, cause reduced fuel economy, improper boiling, boiler explosion etc.,	Due to poor conductance, they decrease the boiler efficiency to lesser extent and causing chocking in the pipelines.
4	It can be prevented by special methods like i) external treatment of ion exchange, ii) Internal carbonate, phosphate, Calgon conditioning iii) Mechanical hard scrubbing methods.	It can be prevented by periodical replacement of concentrated hard water by fresh water. This process is known as "blow down" method.

Diagram:



3. Priming and Foaming:

1) Due to rapid boiling, the steam may carry some water droplets along with it. This is called wet steam. The process of wet steam production is called Priming. It can reduce the heat of the steam and cause corrosion in the pipelines.

Priming is due to:

a) Improper design of boiler

b) High water level

c) High velocity of steam

d) uneven boiling

Priming can be controlled by

i) Proper boiler design ii) Maintaining proper water level iii) Proper boiling

2) If oils and greases are present, they produce stable bubbles on the water surface. This will increase the wet steam production. This is known as "Foaming".

Foaming is prevented by adding i) Anti foaming agents (e.g.) synthetic poly amides, castor oil ii) Coagulants (e.g.) Aluminum hydroxide

3) Foaming and priming are collectively known as 'Carry over".

4. Boiler corrosion

Boiler corrosion It may be due to three major reasons:

i) Dissolved Oxygen

ii) Dissolved CO_2

iii) Dissolved salts like $MgCl_2$

Corrosion Due to dissolved oxygen:

Dissolved oxygen in presence of water, causes corrosion.

$$4Fe + 6H_2O + 3O_2 \rightarrow 4Fe(OH)_3$$

(Rust)

Prevention from oxygen:

a) Chemical method -

i) Adding Sodium Sulphite:

$$2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$$

This method results in other precipitates which can have some side effects. So, this method is less preferred.

ii) Adding Hydrazine:

$$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$$

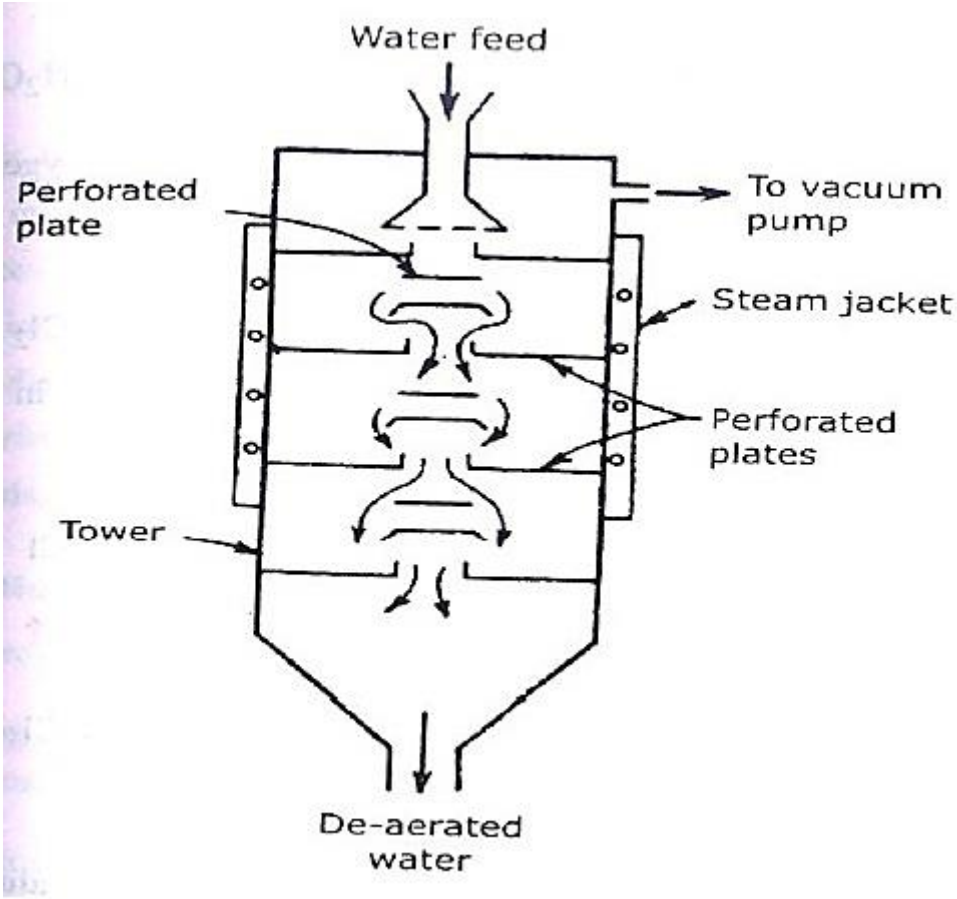
This method results in inert gas and pure water, and has no side effects. So, it is preferred.

b) Mechanical deaeration method:

1. This is based on the principle that at high temperature, low pressure and high exposed area, the solubility of gases in water is decreased. So, the gases can be expelled easily.

2. Here, the water is fed into the mechanical deaerator which is provided with vacuum pump, heaters and perforated plates.

3. The out coming water will be free from dissolved gases.



Corrosion due to CO_2

Salts like Calcium bicarbonate on heating produces CO_2. CO_2 dissolves in water to form carbonic acid which corrodes the boiler metal.

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$$

$$H_2O + CO_2 \rightarrow H_2CO_3$$

Prevention from CO_2

Chemical method: By adding calculated amount of ammonium hydroxide

$$2NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$$

Mechanical deaeration method (similar to oxygen method)

Corrosion due to Dissolved salts like $MgCl_2$:

Dissolved salts like $MgCl_2$ cause acid formation. This will be prevented by alkali neutralisation.

$$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HCl \text{ (Corrosive acid)}$$

Neutralisation:

Excess acidic nature is neutralized by adding alkalis and vice versa.

$$HCl + NaOH \rightarrow NaCl + H_2O$$

5.8. Turbine deposits and prevention methods

Turbine is a rotatory mechanical device that extracts from a fluid flow and converts into useful work i.e., electricity.

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it do mechanical work on a rotating output shaft. About 90% of all electricity is produced by using steam turbines.

When steam is carried over the turbine plates (blades) due to the boiler trouble carry over some droplets of water etc. are deposited on the plates which are called turbine deposits.

Disadvantages of turbine deposits:

1. Turbine deposits cause corrosion blades, rotors and discs by electrochemical corrosion.
2. Turbine deposits reduce the turbine efficiency.
3. The megawatt (MW) generating capacity of the turbine is reduced by turbine deposits.

Composition of turbine deposits: Turbine deposits when analysed has the following composition. Most deposits are silicates and amorphous silica (SiO_2). They are sodium silicate (Na_2SiO_3), sodium meta silicate hydrates ($NaFeSi_2O_6$), potassium aluminum silicate ($KAlSi_3O_8$), magnesium silicates, calcium silicates and sodium molybdenum oxide silicate ($Na_8Al_6Si_6MoO_4$) etc.

Estimation of turbine deposits:

1. A deposit collector/simulator is a tool that can be installed anywhere in a turbine and simulates the turbine conditions governing the deposition of superheated steam. The deposits are gathered from steam that has been overheated and is passing through a collector, allowing the concentration of impurities in the steam to be calculated.

2. Deposit analysis: Wet chemical methods, optical and SEM microscopy (SEM determines particle size and shape), X-ray diffraction, and elemental analysis are used to chemically analyses the deposit from the collector to determine its composition.

3. An optional flow totalizer measures the concentration of impurities in the steam as well as the average flow rate of the total flow over the course of the test.

Causes of turbine deposits:

1. *Entertainment:* It is the cause of turbine deposition due to carry over where minute drops of boiler water in the steam along with the same solid produce and dissolved salt deposit on the turbine. Priming and foaming are responsible for high level deposits leading to super heater tube failures.
2. *Attemparting water impurity:* Controlled cooling in a steam boiler is called attemperating. Turbine deposits are also formed due to impurities of water used for steam attemperator and leakage in closed heat exchangers used for attemperator.
3. *Vaporization of boiler water salts:* The vaporization of salts presents in water at operating pressures below 2,400 psi cause deposition problem.

4. *Localized silica saturation:* Turbine deposits are formed when localized silica saturation increases and silica condenses from steam in those areas of the turbine causing turbine deposits.
5. *Turbine velocity:* The velocity in the turbine is another reason for turbine deposits. The steam flows from inlet to outlet of the turbine with in the fraction of a second. As a result, the deposited is shifted downstream from saturation point by steam velocities.

Prevention of turbine deposits:

1. To maintain low silica concentration.
2. To monitor the condensate.
3. To make use of external treatment equipment like steam purifier.
4. To increase the number of blow down operations.

Removal of turbine deposits:

1. Water cooling under the supervision and recommendation of the turbine vendor should be carried out.
2. To remove water insoluble deposits blasting of turbines with aluminum oxides or another soft grit materials is carried out.

5.9. Treatment methods

5.9.1. External treatment methods:

Ion exchange method (Demineralization method):

In this method cations and anions are separately removed. It uses two column of cation exchange column and anion exchange column filled with resins. Resins are long chain insoluble cross linked organic polymers.

There are two types.

(i) Cationic exchange resins – RH^+

e.g., sulphonated coals, RSO_3H,

(ii) Anion exchange resins $R'OH^-$,

e.g., urea formaldehyde, amines $R\text{-}NH_2$.

Cation exchangers: These are capable of exchanging their H^+ ions with cations of the dissolved salts, which comes in their contact. General formula is R-H, where R is general structure for resin and H is exchangeable with cation.

Process: Raw water is first passed through cation exchanger and the removal of cation takes place in the following way.

$2RH + Ca(HCO_3)_2 \rightarrow R_2Ca + 2H_2CO_3$

$2RH + CaCl_2 \rightarrow R_2Ca + 2HCl$

$2RH + MgSO_4 \rightarrow R_2Mg + H_2SO_4$

The Ca^{2+}, Mg^{2+} are retained by the cation exchangers as CaR and MgR releasing H^+ into water.

Regeneration: After the deionization of certain amount of water the cation exchanger will be exhausted. Regeneration of cation exchanger is carried out by passing dil.HCl or H_2SO_4 solution into the cation exchanger bed. The H^+ ions of the acid are exchanged with the cations present in the cation exchanger regenerating it in the following way.

$R_2Ca + 2HCl \rightarrow 2RH + CaCl_2$ (washings)

$R_2Mg + 2H_2SO_4Cl \rightarrow 2RH + MgSO_4$ (washings)

Anion exchangers: These are capable of exchanging their OH^- ions with anions of the dissolved salts, which comes in their contact. General formula is R'-OH, where R' is general structure for resin and OH is exchangeable with anion.

Process: Raw water is first passed through anion exchanger and the removal of anion takes place in the following way.

$R'OH + HCl \rightarrow R'Cl + H_2O$

$R'OH + H_2CO_3 \rightarrow R'HCO_3 + H_2O$

Regeneration: After the deionization of certain amount of water the anion exchanger will be exhausted. Regeneration of anion exchanger is carried out by passing dil.NaOH solution into the anion exchanger bed. The OH^- ions of the base are exchanged with the anions present in the anion exchanger regenerating it in the following way.

$R'SO_4 + 2NaOH \rightarrow 2R'OH + N_a2SO_4$ (washings)

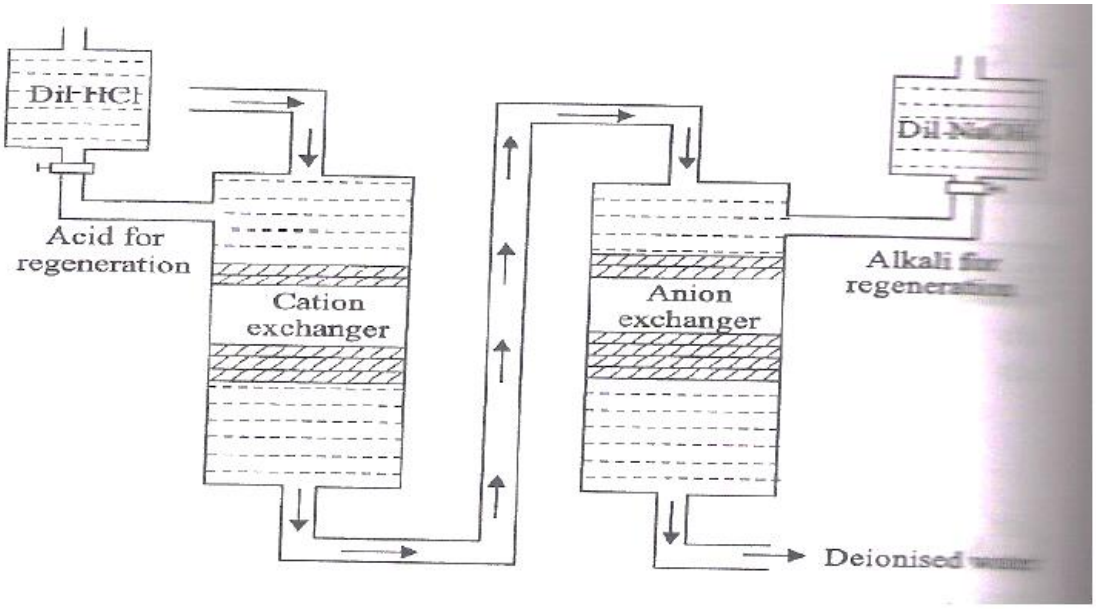
$R'Cl + NaOH \rightarrow 2R'OH + NaCl$ (washings)

Advantages:

1. Highly acidic or alkali water sample can be purified by this process.
2. The hardness possessed by the deionised water is **2 ppm**.
3. The deionised water is most suitable for high pressure boilers.

Disadvantages:

1. The ion exchanging resins are expensive; hence the cost of purification is high.
2. Raw water should contain turbidity below **10 ppm**



Mixed bed deionizer: It consists of a single cylinder containing an intimate mixture of cation exchanger and anion exchanger. When raw water is passed through the bed it comes in contact for a number of times with two kinds of ion exchangers alternatively. The net result is equivalent to passing the raw water through several series of cation and anion exchangers. The hardness of this deionized water is less than **1 ppm**.

Regeneration: The mixed bed is back washed (by forcing water in the upward direction). The anion exchanger which is lighter (low specific gravity) gets displaced to from upper layer and heavier cation exchanger remains as lower layer. The cation exchanger separated and treated with dil.H_2SO_4 and the anion exchangers with dil. NaOH solutions and then washed with water. The two beds were mixed again by forcing compressed air.

Zeolite method (Permutit process):

Basis of softening is cation exchange using Zeolites. Can be remove temporary and permanent hardness.

Zeolites are Sodium alumina silicates. Chemical formula is $Na_2O\ Al_2O_3\ xSiO_2\ yH_2O$. Conveniently represented as, Na_2Ze.

Two types of Zeolites:

Natural Zeolites	Synthetic Zeolites
Obtained from natural sources	Prepared by chemical synthesis
Less porous in nature	More porous
Less ion exchange capacity	More ion exchange capacity

Process:

Zeolites are cation exchangers; they take the hardness causing cations Ca^{2+} and Mg^{2+} from water and replace with Na^+.

$Na_2Ze + Ca^{2+} \rightarrow CaZe + 2Na^+$

$Na_2Ze + Mg^{2+} \rightarrow MgZe + 2Na^+$

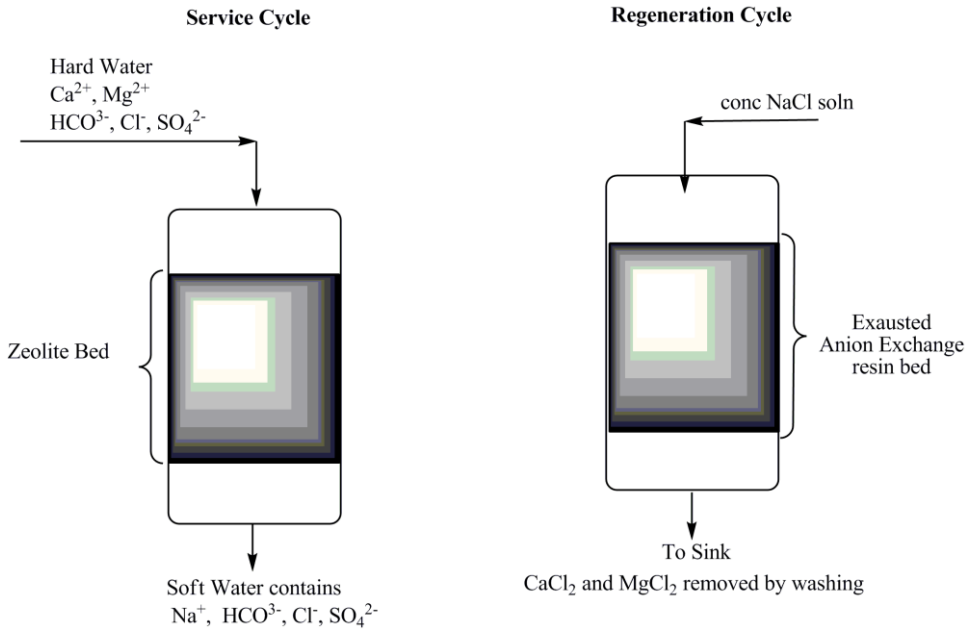
Water become soft but will be richer in sodium ions.

Service Cycle: Water is passed through a cylindrical tank packed with zeolite bed. Zeolite remove cations like Ca^{2+} and Mg^{2+} from water and equivalent amount of Na^+ ions are released to water.

Regeneration Cycle: When the zeolite bed is exhausted, it is regenerated by washing zeolite bed with conc. NaCl solution.

$CaZe + 2NaCl \rightarrow Na_2Ze + CaCl_2$

$MgZe + 2NaCl \rightarrow Na_2Ze + MgCl_2$



Advantages:

It removed hardness up to **10 ppm**.

Apparatus is compact and process is economic.

Disadvantages:

Softened water contains more sodium salts.

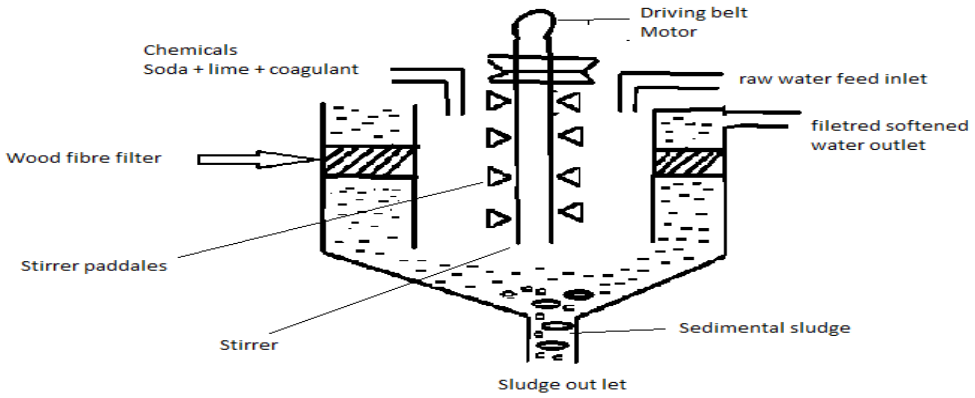
Acidic ions are left in the solution.

Cold lime soda process:

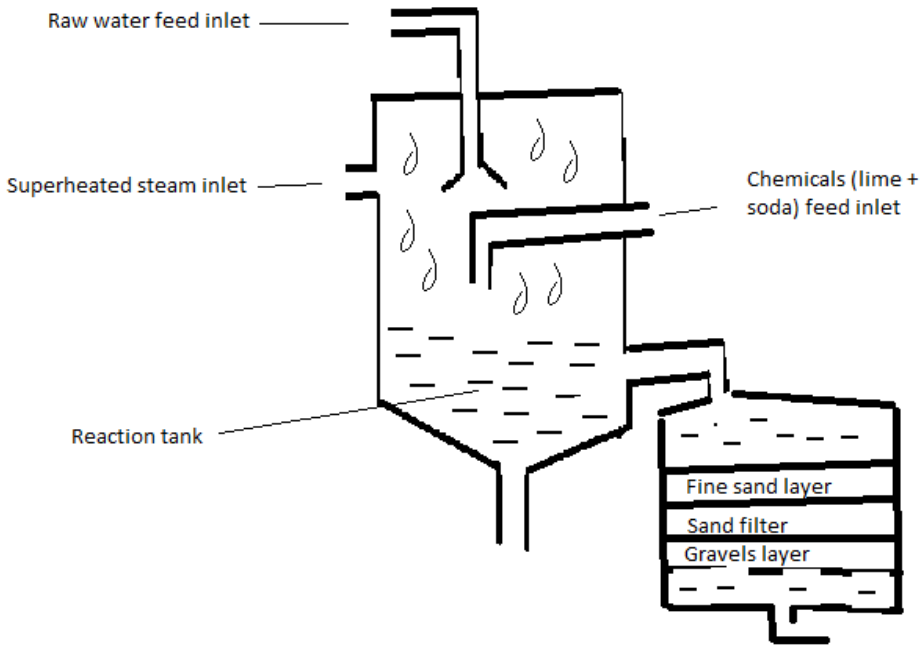
In this method, the calculated quantities of lime are mixed with water at room temperature, and precipitates formed are finely divided. They can now settle down easily. Filtration cannot be easily. Hence

small amounts of alum were added. The addition of sodium aluminate ($NaAlO_2$) can act as coagulant and also helps in the removal of silica and oil.

Raw water along with calculated amounts of chemicals are fed from top into the inner chamber, fitted with a vertical rotating shaft carrying a number of paddles. As the water and chemicals flow down the chamber vigorous mixing and softening of water takes place. The sludge settling at the bottom of the outer chamber is drawn off occasionally. The water sample contains residual hardness 50-60 ppm.



<u>Hot lime soda process:</u>



In this process the raw water is treated with softening chemicals at 80-150^{0}C. Because the hot lime soda process is carried out at a temperature of boiling point. (1) The reaction proceeds faster, (2) The precipitates formed settle down rapidly, (3) No coagulant rapidly, (4) Filtration is fast, (5) Dissolved gases like CO_2 driven out. The residual hardness of water will be 15-30 ppm.

Hot lime soda process contains essentially 3 parts.

a. A reaction tank in which water + chemical + steam is thoroughly mixed.
b. A conical sedimentation vessel in which sludge settle down.
c. A sand filter, which ensures complete removal of sludge removal from the softened water.

Advantages of lime-soda process:

1. The process decreases the pH value of treated water there by corrosion of the distribution pipes reduced.
2. Due to alkaline nature of water the pathogenic bacteria in water are reduced.
3. Besides removal of hardness causing salts, minerals and manganese present in water are removed.
4. This process is economical.

Disadvantages of lime-soda process:

1. Skilled supervision required.
2. Disposal of large amounts of sludge is a problem.
3. The water softened by this process contains 15 ppm of residual hardness which is not good for boilers.

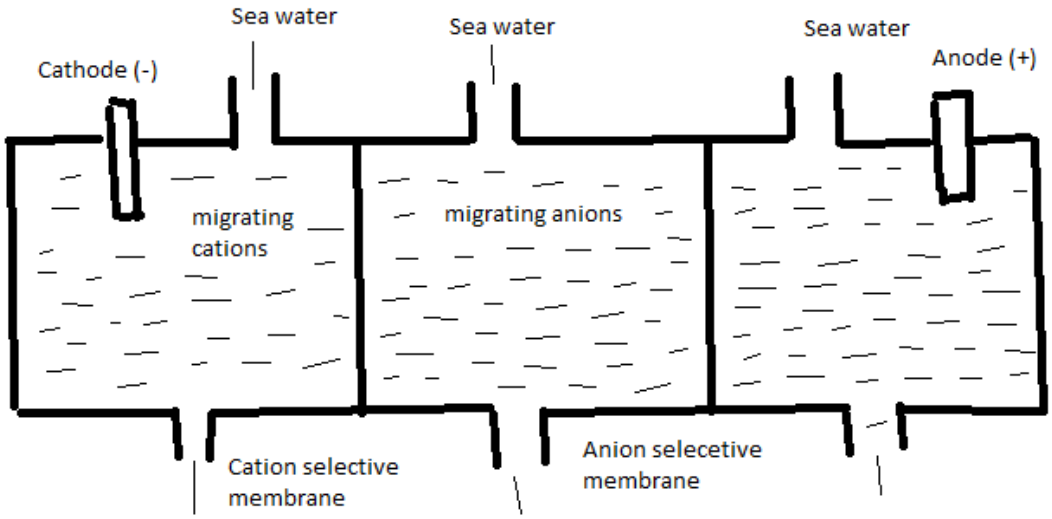
5.9.2. Desalination of brackish water

Water containing high concentrations of dissolved solids with a peculiar salty or brackish taste is called brackish water. Sea water is an example of brackish water containing 3.5 % dissolved salts.

5.9.3 Electro dialysis

Electro dialysis is based on the principle that the ions present in saline water migrates towards their respective electrodes through ion selective membranes under the influence of applied emf.

The unit consists of a chamber with two electrodes, the cathode and anode. The chamber divided into 3 components with the help of thin rigid, ion selective membranes which are permeable to either cation or anion. The anode is placed is near anion selective membrane while the cathode is near cation selective membrane. The anion selective membrane is containing positively charged functional group such as R_4N^+ and is permeable to anions only. The cation selective membrane consists of negatively charged functional groups. Such as RSO_3^- and is permeable to cations only. Under the influence of the applied emf across the electrodes the cations move towards cathode through the membrane and the anion moves towards anode through membrane. The net result is depletion of ions in the central compartment while it increases in the cathodic and anodic compartments. Desalinated water is periodically drawn from the central compartment while concentrated brackish water is replaced with fresh sample.



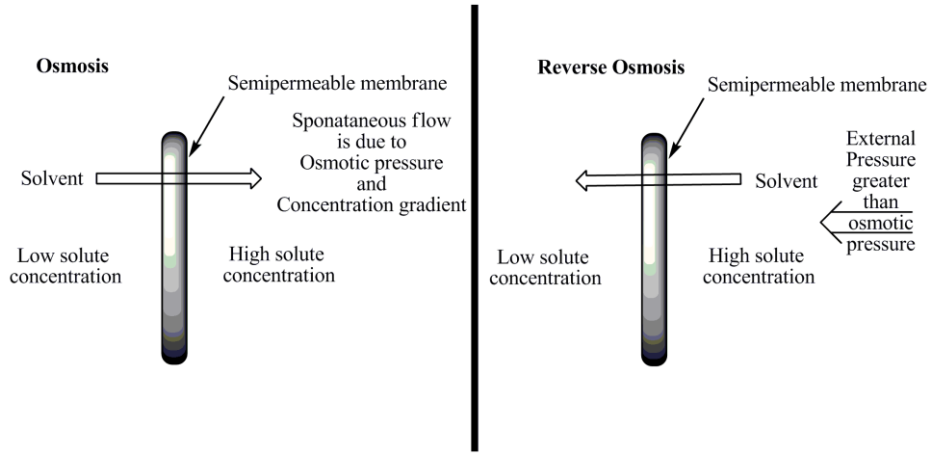
Advantages:

1. The unit is compact.
2. The process is economical as far as capital cost and operational expenses are concerned.

5.9.4. Reverse osmosis

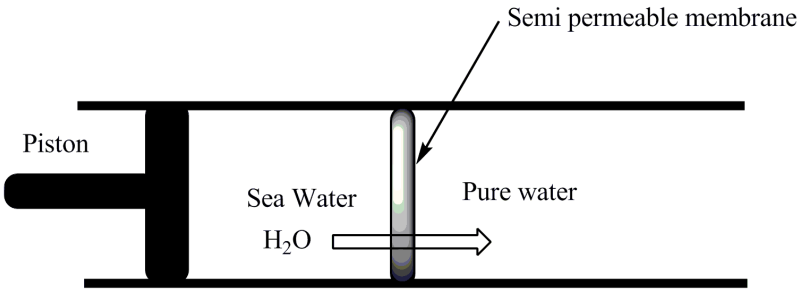
Brackish water is the water which contains high concentrations of dissolved salts, example sea water. The process of removal common salt from water is called desalination.

Osmosis: In a system where two solutions which differ in their concentration are separated by a semi-permeable membrane, solvent molecules pass through the membrane from the side having lower solute concentration to the side having higher solute concentrations. This spontaneous flow depends on osmotic pressure and concentration gradient.



Reverse Osmosis: If we apply an external pressure in excess of the osmotic pressure on the concentrated side, the solvent flow is reversed. That is solvent is forced to move from the side having higher solute concentration to the side having lower solute concentrations.

This principle can be used for desalination of sea water.



Pure water is separated from dissolved minerals. This membrane filtration is called super filtration.

Method of purification: The reverse osmosis cell consists of a chamber fitted with a semipermeable membrane, sea water/ impure water is taken and a pressure of 15 to 40 kg/cm^2 is applied on the sea water/ impure water. Semipermeable membrane is made of very thin films of cellulose acetate. Superior membrane made of polymethacrylate and polyamide polymers have come to use.

Advantages:

1. Both ionic and non-ionic, colloidal and high molecular weight organic matter also removed from the water sample.
2. Cost of purification of water is less and maintenance cost is less.
3. This water can be used for high pressure boilers.

5.9.5. Internal treatments

a. **Colloidal Conditioning**: This is done in low pressure boilers. Organic substances like kerosene, tannin etc. are added as sequestering agents. They get coated over minute scale forming particles and prevent their coagulation. Hence the salts remain as non-sticking loose sludge. It can be removed by blow-down operation.

b. **Carbonate conditioning**: This is done in low pressure boilers. Scale formation due to CaSO4 can be avoided by adding sodium carbonate to boiler water.

$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$

The $CaCO_3$ precipitated is a non-sticking loose sludge. It can be removed by blow-down operation. If used in high pressure boilers it may result in caustic embrittlement.

c. **Calgon Condition (Calgon treatment)**: Calgon is sodium-hexa-metaphosphate . Calgon when added to boiler water forms a soluble complex with Ca^{2+}. Thus, scale formation is prohibited.

$Na_2[Na_4(PO_3)_6] + CaSO_4 \rightarrow Na_2[Ca_2(PO_3)_6] + Na_2SO_4$

d. **Phosphate Conditioning**: This is done in high pressure boilers. Here scale formation is avoided by adding sodium phosphate. Ca^{2+} and Mg^{2+} ions react with phosphate anion to form non-sticking loose sludge of calcium and magnesium phosphate. The sludge can be removed by blow down operation.

$2Na_3PO_4 + Ca^{2+} \rightarrow Ca_3(PO_4)_2\downarrow + 6Na^+$

FAQ

1. Write a note on different types of hard water.
2. What are the characteristics of potable water?
3. Explain the treatment of water for municipal supply. (or) How the domestic water can be purified? (or) Write the process of purification of drinking water.
4. Write a note on break point chlorination. (or) What is break point chlorination and what is its significance? (or) What is the disadvantage if the chlorine present in water crosses or exceed the break point chlorination point along with explanation of break point chlorination.
5. What are the characteristics of boiler feed water?
6. What are boiler troubles? How can you prevent boiler troubles?
7. What is priming and foaming? How can you prevent priming and foaming in boiler water? (Look priming and foaming)
8. Define scales and sludges. How can you control scales and sludges formation in boiler? (Go to scales and sludges)
9. Differentiate scales and sludges.
10. What is caustic embrittlement and how can prevent it? (or) write a note on caustic embrittlement. (Caustic embrittlement topic)
11. Explain boiler corrosion. (or) What is boiler corrosion and what are the prevention methods for that? (Boiler corrosion concept)
12. Write a note on turbine deposits. (or) What are turbine deposits? Mention the prevention methods.
13. Write a note on ion exchange method. (or) How can you softening the water by using ion exchange method?
14. Explain the water softening process of anion exchange method. (See anion exchange method)
15. Write a note on cation exchange method. (See cation exchange method)
16. What is mixed ion exchange method? Explain the process of it. (See mixed ion exchange process)
17. What is permutit process? (or) Explain permutit process.
18. Explain lime soda process for softening of water.
19. What is cold lime soda process? How it softening the water? (See cold lime soda process)
20. Write the differences between cold lime soda and hot lime soda process.
21. Write a note on hot lime soda process. (See hot lime soda process)
22. What is brackish water? Explain any one method for the treatment of sea water or brackish water.
23. Write a note on electrodialysis for the softening of brackish water or sea water.
24. Define reverse osmosis. How is the reverse osmosis method used for the softening of sea water or brackish water? Mention its advantages.
25. 25. Write a note on a) Colloidal conditioning b) Carbonate conditioning, c) Calgon treatment and d) Phosphate conditioning.